

# SLE<sub>3</sub> universality for weak finite-range Ising interactions

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## Abstract

We prove that every sufficiently small, square-symmetric finite-range perturbation of the planar Ising contour energy has a domain-independent inverse temperature at which its spin interface converges to chordal SLE<sub>3</sub>. The result allows interactions of either sign, arbitrary admissible exterior contours, and uniformly approximated Jordan domains. Convergence holds for the full oriented curve law in uniform distance modulo increasing reparametrization.

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# 1 Introduction

Critical nearest-neighbor Ising spin interfaces have a conformally invariant scaling limit: with opposite boundary signs on two arcs, the limiting interface is chordal  $\text{SLE}_3$ . We prove that this conclusion persists under every sufficiently small finite-range perturbation of the contour energy that preserves the symmetries of the square lattice. The perturbation may contain arbitrary many-edge interactions and may have either sign. The boundary specification is part of the result: interactions across the boundary remain present, and the exterior may contain additional closed contours.

## 1.1 The model and its boundary data

Fix  $J > 0$ . Let  $\mathcal{E}^\bullet$  be the nearest-neighbor edges of  $\mathbb{Z}^2$ , and let  $\mathcal{E}$  be those of the dual lattice  $\mathcal{F} = \mathbb{Z}^2 + (1/2, 1/2)$ . Write  $e^\bullet \leftrightarrow e$  when the primal and dual edges cross. A potential  $U$  assigns a real number to each finite subset of  $\mathcal{E}$ , with  $U(\emptyset) = 0$ . We assume that  $U$  is invariant under lattice translations and the rotations and reflections of the square lattice. Its range is finite: there is  $R < \infty$  such that

$$U(S) = 0 \quad \text{if the Euclidean diameter of the union of the edges in } S \text{ exceeds } R.$$

All values of  $U$  are fixed before the mesh is rescaled.

For a finite  $\Omega \subset \mathbb{Z}^2$ , put

$$Q_\Omega = \bigcup_{v \in \Omega} (v + [-1/2, 1/2]^2), \quad D_\Omega = \text{int } Q_\Omega.$$

We require  $Q_\Omega$  to be the closure of a Jordan domain. Let  $E_\Omega \subset \mathcal{E}$  consist of the edges crossing primal bonds with at least one endpoint in  $\Omega$ , and let  $F_\Omega$  be their endpoints. Choose distinct  $f_{\text{in}}, f_{\text{out}} \in \partial D_\Omega \cap F_\Omega$ . For each mark choose a dual edge outside  $E_\Omega$  joining that mark to a vertex outside  $Q_\Omega$ ; denote these edges by  $e_{\text{in}}, e_{\text{out}}$ . They are the exterior stubs, and  $\xi = ((f_{\text{in}}, e_{\text{in}}), (f_{\text{out}}, e_{\text{out}}))$ .

For a dual-edge set  $A$ , let  $d_f(A)$  be its degree at  $f$ . The permitted interior contours are

$$\mathcal{P}_\Omega^\xi = \left\{ P \subset E_\Omega : d_f(P) + \mathbf{1}_{\{f_{\text{in}}\}}(f) + \mathbf{1}_{\{f_{\text{out}}\}}(f) \text{ is even for every } f \in \mathcal{F} \right\}.$$

We require this set to be nonempty. An admissible exterior contour is a finite set  $P_o \subset \mathcal{E} \setminus E_\Omega$  containing the two stubs and satisfying

$$d_f(P_o) + \mathbf{1}_{\{f_{\text{in}}\}}(f) + \mathbf{1}_{\{f_{\text{out}}\}}(f) \equiv 0 \pmod{2} \quad (f \in \mathcal{F}).$$

Thus  $P \cup P_o$  is even for every  $P \in \mathcal{P}_\Omega^\xi$ .

For inverse temperature  $\beta > 0$  and coupling  $\lambda \in \mathbb{R}$ , the law is

$$\mu_{\Omega, \beta, \lambda}^{\xi, P_o}(P) = \frac{1}{Z_{\Omega, \beta, \lambda}^{\xi, P_o}} \exp \left\{ -2\beta J |P| - \beta \lambda \sum_{\emptyset \neq X \subseteq P} \sum_{Y \subseteq P_o} U(X \cup Y) \right\}. \quad (1.1)$$

The subset sums include equality, and the second includes the empty set. The normalizing constant is the sum over  $\mathcal{P}_\Omega^\xi$ . In particular, the boundary data retain every interaction joining an occupied interior edge to occupied exterior edges.

The curve  $\gamma(P)$  uses  $P$  and the half of each exterior stub between its mark and its midpoint. At degree two, join the two occupied edges. At degree four, join north to east and south to west, also at a mark if its half-stub raises the degree to four. Draw these connections disjointly in disks of radius less than  $1/10$  around the dual vertices. There is a unique component between the two stub midpoints; orient it from the incoming midpoint to the outgoing midpoint. This is  $\gamma(P)$ . Its edges are each traversed at most once, and the resolved curve is simple. Any fixed local drawing with these connections is allowed.

## 1.2 The main theorem

An oriented curve is a continuous map from  $[0, 1]$  into the plane, considered modulo increasing reparametrization and zero distance for

$$d_{\text{curv}}(\gamma, \eta) = \inf_{\alpha, \rho} \sup_{t \in [0, 1]} |\gamma(\alpha(t)) - \eta(\rho(t))|, \quad (1.2)$$

where  $\alpha, \rho$  range over increasing homeomorphisms of  $[0, 1]$ . The orientation distinguishes the two endpoints.

For a bounded Jordan domain  $D$  and distinct  $a, b \in \partial D$ , chordal  $\text{SLE}_3$  from  $a$  to  $b$  is the conformal image of the upper-half-plane Loewner trace with

$$\partial_t g_t(z) = \frac{2}{g_t(z) - \sqrt{3}B_t}, \quad g_0(z) = z,$$

where  $B$  is standard real Brownian motion; the conformal map sends 0 to  $a$  and  $\infty$  to  $b$ . This law is independent of the choice of such map. The trace is simple and meets the Jordan boundary only at its endpoints; see Rohde and Schramm [22], Schramm [23].

**Theorem 1.1.** *Let  $J > 0$ , and let  $U$  satisfy the finite-range and symmetry assumptions above. There is  $\lambda_0 = \lambda_0(J, U) > 0$  such that for every  $|\lambda| < \lambda_0$  there is a positive inverse temperature  $\beta_c = \beta_c(J, U, \lambda)$ , independent of domains and boundary data, with*

$$\beta_c(J, U, 0) = \beta_0 := \frac{\log(1 + \sqrt{2})}{2J},$$

for which the following holds.

Let  $D$  be any bounded Jordan domain, let  $a, b$  be distinct boundary points, and let  $\delta_n \downarrow 0$ . For each  $n$ , choose a domain  $\Omega_n$ , marks and stubs  $\xi_n$ , and an admissible exterior contour  $P_{o,n}$  as above. Suppose there are boundary homeomorphisms

$$\phi_n : \partial D \longrightarrow \partial(\delta_n D_{\Omega_n})$$

such that

$$\sup_{z \in \partial D} |\phi_n(z) - z| \longrightarrow 0, \quad \phi_n(a) = \delta_n f_{\text{in},n}, \quad \phi_n(b) = \delta_n f_{\text{out},n}.$$

If  $P_n$  has law  $\mu_{\Omega_n, \beta_c, \lambda}^{\xi_n, P_{o,n}}$ , then  $\delta_n \gamma(P_n)$  converges in distribution, in the metric (1.2), to chordal  $\text{SLE}_3$  in  $D$  from  $a$  to  $b$ .

No smoothness of  $\partial D$  is assumed. The same unit-lattice potential  $U$  is used for every  $n$ , and the exterior contours may vary arbitrarily with  $n$ . The theorem concerns the full curve law; it does not assume convergence of an observable or a tightness conjecture for the perturbed model.

## 1.3 Historical context

Schramm's Loewner description identifies conformally invariant random interfaces through a one-dimensional driving process [23]. Smirnov proved conformal covariance of the critical FK-Ising fermionic observable on the square lattice [25, Theorem 2.2]; Chelkak and Smirnov developed the spin and FK observables on isoradial graphs [9, Theorems A–B]. Together with geometric crossing control, these observables lead to convergence of the critical nearest-neighbor spin interface to  $\text{SLE}_3$  [10, Theorem 1]. Conformal covariance of spin correlations and convergence of nested loop collections

to  $\text{CLE}_3$  are established in Benoist and Hongler [4], Chelkak et al. [11]. The spin alternating-arm exponents, including the four-arm value  $21/8$ , are given by Wu [26].

For non-integrable perturbations, constructive renormalization has established universality for several correlation observables. Giuliani, Greenblatt, and Mastropietro prove full-plane energy-correlation scaling limits for weak, rotation-invariant finite-range pair perturbations [15]. Antinucci, Giuliani, and Greenblatt extend energy universality to finite cylinders with periodic boundary conditions in one direction and free boundary conditions in the other, allowing translation-invariant finite-range even multispin interactions [3, Theorem 1.1 and Corollary 1.2]. Cava, Giuliani, and Greenblatt prove boundary-spin scaling limits in the half-plane with free boundary conditions, including the renormalized Pfaffian limit under weak translation-invariant finite-range even perturbations [8, Theorem 1.1]. These results treat either sign of a sufficiently small perturbation and give precise information in the stated plane, cylinder, and half-plane geometries.

A different approach uses random currents. For ferromagnetic, translation-invariant, square-symmetric finite-range pair interactions whose interaction graph is connected, Aizenman et al. [2, Theorem 1.2] prove an asymptotic Pfaffian structure of critical boundary-spin correlations in the half-plane, without a small-perturbation assumption. Its relative Pfaffian relation and the preceding correlation scaling limits concern observables and hypotheses different from the full signed contour-potential interface law considered here.

A close predecessor for the interface problem is the work of Greenblatt and Peltola [16] on non-integrable finite-range interactions. Their Theorem 4.4 expresses an exploration local martingale as a ratio of Grassmann correlations, and their Conjecture 4.5 formulates fermionic asymptotics, local renormalization, and boundary stability that would yield an  $\text{SLE}_3$  limit. Their Section 5 explains why estimates in fixed geometries must be extended to the irregular boundaries created by exploration. We establish the interface limit for the symmetric class in Theorem 1.1 through bounded spin observables, sparse changes, and restricted boundary comparisons. The specific fermionic asymptotics in their conjecture are not a conclusion of this argument.

There is a complementary geometric lineage. Multi-crossing estimates control random-curve regularity in Aizenman and Burchard [1, Hypothesis H1 and Theorem 1.1], and conditional bounds on unforced crossings yield tightness and Loewner limits in Kemppainen and Smirnov [18, Condition G2 and Theorem 1.5]. The distinction between driving-function convergence and strong curve convergence is developed in Sheffield and Sun [24, Theorem 1.2 and Example 2.1]. Our geometric input is a mesh-first interior four-passage estimate. We prove the trace-to-curve implication needed here directly by excluding collapsing reverse traversals.

## 1.4 Proof strategy

We work with spins by assigning plus at infinity and changing sign each time a primal path crosses an edge of the even contour set  $P \cup P_o$ . This assigns a spin  $\sigma_v \in \{-1, +1\}$  to each lattice vertex; the spins outside  $\Omega$  are fixed by  $P_o$ . A prescribed spin is called a *pin*. Expanding each contour interaction in these spins gives a finite-range even spin interaction. Section 1.5 proves the exact identity of the two conditional laws, including all boundary interactions. Our first comparison is between the periodic version of this bulk interaction and the critical nearest-neighbor law on a large periodic square.

Section 2 specifies the pure-model estimates used in that comparison. Besides the public results above, we use the buffered likelihood comparisons, ordered-source transmission, nearby-line prediction, and complete-resolution results of the companion article *Buffered comparison and stopping-band resolution in critical Ising* [19]. These are equilibrium statements about the ordinary critical model. Their hypotheses distinguish arbitrary common pins in likelihood comparisons from

the free neighborhoods needed to transmit a magnetic response.

The linear transfer takes a bounded spin function supported in a square, conditions it under the pure law on the boundary spins of a concentric square larger by a fixed ratio  $L$ , and multiplies by  $L^2$  to normalize an interaction density. The first obstacle is that repeated transfer creates functions of more and more microscopic spins. Formulas for a fixed number of field insertions do not control all such functions. Camia, Garban, and Newman construct smeared magnetization-field limits from spin correlations [7, Section 3]; Section 3 proves the further uniform approximation needed here. We observe each spin through a weak Gaussian signal. Conditional on these observations the pure law remains ferromagnetic. Ordered-source transmission and prediction from nearby spin sums show that the remaining uncertainty has little effect across a positive gap. Only finitely many spatial averages of the observation are then needed. This reduces the transfer of arbitrary bounded spin functions to the explicit continuum spin formulas. Reflection positivity and a finite determinant identity show that, among even square-invariant interaction densities, exactly one scalar direction can grow under this transfer.

Section 4 turns the linear transfer into an exact flow of local interactions. A finite-state identity explains the construction: moving a potential toward its conditional expectation can be reversed by random local refreshes at rates independent of the spins. We control the interactions produced at successive scales. Positive long-distance energy covariance shows that changing temperature moves transversely to the set of initial interactions whose trajectories decay. We can therefore choose the inverse temperature to obtain such a trajectory. This choice uses only bulk data and is therefore independent of the domain. Reversing the flow couples the resulting torus spin law to the pure law by changes inside independently marked squares. The marking probabilities decay with scale; the spins assigned inside marked squares remain unrestricted.

Section 5 proves that these possible changes preserve every fixed finite collection of polygonal crossing, circuit, and four-passage tests with probability tending to one. The geometric reason has three forms. An edit in the interior of a test can change its answer only if four long contour portions reach the edit. Near a straight test edge two portions must remain on one side, and near a corner one portion must extend to a fixed distance. The corresponding probabilities decay faster than the numbers of possible locations grow. The strict bulk threshold comes from the spin four-arm exponent  $21/8 > 2$ ; the half-plane estimate is derived here. A bootstrap over the edit scales makes the conclusion valid for every assignment in all marked squares, including assignments chosen after examining the pure configuration.

To pass from periodic squares to the prescribed domain, we need comparison under boundary conditions. Signed perturbations need not preserve ferromagnetic comparison for the physical spins. Section 6 represents each small interaction by additional binary variables on nearest-neighbor bonds. Their local conditional probabilities have strict margins under specified changes from minus pins to free spins or from free spins to plus pins. These margins provide the boundary comparisons used here. Combined with the bulk crossing tests, they also bound the pointwise density ratio between a target-domain spin marginal on a square and the perturbed torus marginal on that square, when a fixed free buffer separates the square from all pins.

Finally, Section 7 constructs two displaced pure comparison domains, favoring the opposite colors. Finite collections of colored strip crossings transfer barriers near their interfaces to the target. In any subsequential limit, the barriers and connectedness of the target trace order the two limiting simple chords. Both chords have the same SLE<sub>3</sub> law, so they coincide and identify the trace. A persistent backward traversal near a simple chord would create six passages through a small interior annulus. The four-passage estimate excludes this event and yields convergence in the full oriented curve metric (1.2).

The inverse temperature is selected by the decaying trajectories of the conditional flow in

Section 4. Its local uniqueness refers to that construction. No identification with a temperature branch selected by a different scale map, or joint field-and-interface scaling limit, is required for Theorem 1.1.

## 1.5 Exact spin formulation

The proof uses spin conditional expectations in the bulk. We record the exact passage from (1.1) to spins, including the exterior data. The contour-potential formulation with exterior interactions is also used by Greenblatt and Peltola [16, Section 4]. For a primal bond crossing  $e$ , write

$$d_e(\sigma) = \frac{1 - \sigma_v \sigma_w}{2}, \quad e^\bullet = \{v, w\}, \quad \sigma \in \{-1, 1\}^{\mathbb{Z}^2}.$$

The formal spin Hamiltonian is

$$\mathcal{H}_{\beta, \lambda}(\sigma) = -\beta J \sum_{\{v, w\} \in \mathcal{E}^\bullet} \sigma_v \sigma_w + \beta \lambda \sum_{\emptyset \neq S \subset \mathcal{E}} U(S) \prod_{e \in S} d_e(\sigma). \quad (1.3)$$

Only terms depending on the spins in the chosen finite domain enter its conditional law. All resulting sums are finite.

**Lemma 1.2.** *For the boundary data above, every even completion  $P \cup P_o$  determines a unique spin configuration that is plus sufficiently far away and has disagreement set  $P \cup P_o$ . Its restriction  $\tau$  to  $\mathbb{Z}^2 \setminus \Omega$  is independent of  $P$ . The map  $P \mapsto \sigma|_\Omega$  is a bijection between  $\mathcal{P}_\Omega^\xi$  and all spin assignments in  $\Omega$ . Under this bijection, (1.1) is precisely the Gibbs conditional law for (1.3) with exterior spins  $\tau$ . Denote this law by  $\nu_{\Omega, \beta, \lambda}^\tau$ .*

*The exterior nearest neighbors along each of the two boundary arcs between the marks have a constant sign, and the signs of the two arcs are opposite. The interaction in (1.3) is finite-range, invariant under global spin flip and the square symmetries, and has finitely many local interaction types up to translation.*

*Proof.* A finite even dual subgraph has even intersection parity with every closed primal walk. One way to see this is to reduce a closed walk modulo two to elementary primal faces; the intersection parity around such a face is the degree of the dual vertex it encloses. Starting with a plus spin at infinity, flip signs on crossing an occupied dual edge. Path independence follows from this parity statement. Finiteness fixes the plus sign in the unbounded region and gives uniqueness.

The exterior cell centers  $\mathbb{Z}^2 \setminus \Omega$  form a connected nearest-neighbor graph. Indeed the complement of a Jordan polyomino has no bounded component, and a path in its exterior can be perturbed to cross cell sides away from vertices, giving a path of exterior cells. The Jordan condition excludes a diagonal pinch. A primal path using only exterior vertices crosses only dual edges outside  $E_\Omega$ . Their occupied set is  $P_o$ , so all exterior signs are fixed independently of  $P$ .

Conversely, extend any assignment in  $\Omega$  by these fixed exterior spins. Its disagreement set outside  $E_\Omega$  is  $P_o$ , and its complete dual disagreement graph is even. The interior part therefore belongs to  $\mathcal{P}_\Omega^\xi$ . These two constructions are inverse.

For the signed-arc assertion, traverse the polygonal boundary. At a flat boundary vertex, the two successive exterior neighboring cells are separated by one exterior primal bond. At an unmarked vertex the crossing exterior dual edge is absent by the parity condition, so the two exterior signs agree. At a convex corner, the corresponding exterior path has two bonds; its crossing parity is again even at an unmarked vertex. At a reflex corner, the two boundary edges have the same adjacent exterior cell. At either mark the exterior parity is odd, and the sign changes. Reflex corners cannot

support an exterior stub of the required kind. Thus there are exactly the two asserted constant, opposite arc signs.

Now fix an interior spin configuration and its contour  $P$ . A nonconstant interaction term in (1.3) has a support  $S$  meeting  $E_\Omega$ . Write

$$X = S \cap E_\Omega, \quad Y = S \setminus E_\Omega.$$

The product of disagreement indicators is one exactly when  $X \subseteq P$  and  $Y \subseteq P_o$ . Summing over all such supports gives precisely the double sum in (1.1). Terms supported entirely outside  $E_\Omega$  are constant in the interior spins. The nearest-neighbor energy is  $2\beta J|P|$  up to an exterior-dependent constant. This proves equality of the conditional laws.

Finally, choose an edge in a nonempty support of  $U$ . Finite range leaves only finitely many possible other edges relative to it, hence only finitely many support types up to translation. Expanding the products of  $d_e$  gives finite local spin polynomials, invariant under spin flip and under the prescribed square symmetries.  $\square$

In particular, exterior contours do not give an additional bulk parameter. Their only effect is through the actual exterior spins in the conditional law. This observation does not permit deleting interactions across a narrow fjord or replacing distant exterior spins by their adjacent arc signs.

## 2 Pure critical Ising inputs and contour conventions

The comparison and regularity estimates in this section concern only the nearest-neighbor Ising model at coupling  $K_0 = \beta_0 J = \frac{1}{2} \log(1 + \sqrt{2})$ . Their role is to supply uniform control of bounded spin observables across a free region, and geometric estimates for the contours of the pure model. Section 3 will derive the uniform transfer theorem from these inputs.

We write  $\mu_0$  for the critical full-plane law and  $\mathbb{E}_0$  for its expectation. A *pin* is a prescribed spin. Common pins have the same locations and values in the laws being compared; they may have either sign. A free lattice chart is a region isometric to part of the square lattice in which no spins have been pinned. A collar always includes the edges of that chart, as well as its vertices. Write  $B_s(x) = x + [-s, s]^2 \cap \mathbb{Z}^2$ , and  $B_s = B_s(0)$ . For integer  $s$ , the outer layer is  $\partial B_s = B_s \setminus B_{s-1}$ ; integrating the open square means integrating  $B_{s-1}$  while keeping this layer fixed. Integer rounding and a bounded displacement of these layers can be absorbed by a fixed fraction of any stated positive collar once the lattice scale tends to infinity.

We cite three equilibrium estimates and one complete-resolution result from the companion article *Buffered comparison and stopping-band resolution in critical Ising* [19]. The citations below specify its named results and section headings. We use these results as pure-model inputs; the transfer, scale flow, and perturbed-interface arguments are proved in this article.

### 2.1 Pointwise comparisons through collars

Pointwise comparison is stronger than comparison of a fixed collection of observables. It will permit arbitrary boundary assignments to enter the transfer argument through uniformly bounded likelihoods.

**Proposition 2.1** (Buffered and small-patch comparisons). *Consider a finite critical nearest-neighbor ferromagnetic Ising graph, disjoint vertex sets  $I, J, F$ , and common pins on  $F$ . Let  $\mu_b$  be the marginal law on  $I$  obtained by prescribing an arbitrary configuration  $b$  on  $J$ . Whenever the following sets lie in the indicated square-lattice charts, the bounds hold for all configurations  $a$  on  $I$  and all  $b, b'$ .*

1. For every  $c > 0$  there is  $K_c < \infty$  such that  $I \subset B_R(z)$ ,  $J \subset B_{(1+c)R}(z)^c$  imply

$$K_c^{-1} \leq \frac{\mu_b(a)}{\mu_{b'}(a)} \leq K_c.$$

2. There are absolute  $K < \infty$ ,  $\zeta > 0$  such that  $I \subset B_w(z)$ ,  $J \subset B_R(z)^c$ , and  $R \geq 8w \geq 8$  imply

$$\left| \log \frac{\mu_b(a)}{\mu_{b'}(a)} \right| \leq K(w/R)^\zeta.$$

No free neighborhood of  $I$  is required in these two assertions: additional common pins may lie anywhere outside  $I \cup J$ .

3. Either law can be replaced by any mixture of the indicated conditional laws. The fixed-ratio comparison also holds for laws from different domains, including the plane and a torus, if there is a common isometric lattice collar separating the observed spins from every difference in the data. In this cross-domain assertion the collar is unpinned in both laws. Applying a common randomization kernel to the observed spins preserves each density bound.

All constants are independent of the number, positions, and values of the common pins and of the mesh. In the first two assertions the same ratio bounds hold when the roles of the two terminal sets are reversed.

*Source and the terminal symmetry.* The first three assertions are Corollary 2.5, “Buffered and small-patch comparisons”, in Section 2, “Pointwise comparison through buffers”, of [19]. We record the symmetry needed below. For the positive joint marginal  $p(a, b)$ , put  $f = \log p$ . The source’s Lemmas 2.1 and 2.2, “Extreme rectangles in a ferromagnetic marginal” and “Pointwise ratios with arbitrary common pins”, bound

$$|f(a, b) + f(a', b') - f(a, b') - f(a', b)| \leq 4 \operatorname{arctanh} q_0,$$

where  $q_0$  is the zero-field FK connection probability after deleting the common pinned vertices and separately contracting the two terminal sets. Deletion retains the fields induced by the pinned spins in the original law; all these fields are then set to zero in the comparator. The right side is symmetric in the terminals. A dual circuit in the separating collar bounds  $q_0$  away from one; several separated dual circuits give its power decay in the small-patch case. Normalizing either family of conditional laws turns the oscillation bound into the displayed pointwise ratio bounds. Thus reversing which terminal is observed requires no new geometric estimate; this is also Remark 2.3 of the companion.  $\square$

**Corollary 2.2** (Two collars around an annulus). *Let  $2 \leq r < R/4$ ,  $A = B_R \setminus B_r$ , and  $W = B_{2R} \setminus B_{r/2}$ , with integer rounding of the boundaries. Suppose  $W$  is an unpinned square-lattice chart. The law on  $A$ , conditional on arbitrary spins outside  $W$ , has pointwise density relative to  $\mu_0|_A$  between  $K^{-1}$  and  $K$ , for an absolute  $K$  independent of  $R/r$ . The assertion applies in a torus chart and to mixtures of exterior conditions. Fixed changes of the two collar ratios only change  $K$ .*

*Proof.* First change the spins outside  $B_{2R}$ , keeping all inner pins common. Proposition 2.1 across the outer collar costs a fixed factor. Next change the spins in  $B_{r/2}$ , keeping the outer data common. Use the terminal-symmetric form of the same proposition across the collar from  $r/2$  to  $r$ : the small terminal is now the source and  $A$  lies beyond its fixed-ratio collar. This costs another fixed factor. The product is independent of the distance between the collars. Finally mix the exterior data according to the full-plane law. Cross-domain comparison is legitimate because both collars are common lattice charts.  $\square$

## 2.2 Signed sources and line magnetizations

The transfer proof will produce a signed density that has a positivity property under increasing tests, but is not necessarily increasing. The following definition states precisely the property required of it.

**Definition 2.3** (Order-positive source). For a probability measure  $\mu_S$  on spins in a finite set  $S$ , a real function  $q$  is an order-positive centered density if

$$\mathbb{E}_{\mu_S} q = 0, \quad \mathbb{E}_{\mu_S} [qg] \geq 0 \quad \text{for every increasing } g.$$

Here density means the density of a signed measure.

We must also specify the small free neighborhoods used by the transmission estimate. Let  $S, T$  be disjoint source and target sets. A cut  $J$ , disjoint from  $S, T$  and from all common pins, is *separating* if every free-graph path from  $S$  to  $T$  meets  $J$ . Its target-side Gibbs continuation is then independent of the source after the cut spins have been prescribed.

Fix  $A, K \geq 1$  and  $a \in (0, 1/(16A))$ . The cut is *admissible at scales  $w, s$* , where  $0 < w \leq s$ , if it has a partition into patches  $b_i$ , straight counting segments  $L_i \subset J$ , centers  $z_i$ , and scales  $\ell_i \in [aw, Aw]$  such that

1.  $b_i \cup L_i \subset B_{\ell_i}(z_i)$  and  $a\ell_i \leq |L_i| \leq 2\ell_i$ ;
2.  $B_{16\ell_i}(z_i)$  is a pin-free planar lattice chart, including freedom from source pins;
3. each vertex belongs to at most  $K$  counting segments;
4.  $\text{dist}(T, z_i) \geq s/A$ ,  $16\ell_i \leq s/A$ , and the small-patch comparison applies from  $b_i$  toward  $T$  in a chart extending to a fixed positive multiple of  $s$ .

The constants  $a, A, K$  and the fixed chart geometry are part of this definition; the number of patches is not. A square contour in a free collar of width comparable to  $w$ , at distance comparable to  $s$  from the target, is admissible. Partition its sides into intervals of length a sufficiently small fixed multiple of  $w$ . At corners use counting intervals on an adjacent side. Their overlap remains bounded. The same construction works for a fixed finite union of square contours, including the two boundaries of a square band.

**Proposition 2.4** (Transmission of a signed source). *There is a fixed exponent  $\eta \in (0, 1/8)$  with the following property. Let  $\mu$  be a pure critical Gibbs law with common pins, let  $q = q(\sigma_S)$  be order-positive and centered for its source marginal, and let  $J$  be an admissible cut at scales  $w, s$ . For every bounded target observable  $h = h(\sigma_T)$ , when  $w$  is sufficiently large,*

$$|\mathbb{E}_{\mu}[qh]| \leq C \|h\|_{\infty} (w/s)^{-7/8+\eta} \mathbb{E}_{\mu}[qM_J], \quad M_J = s^{-7/8} \sum_{x \in J} \sigma_x. \quad (2.1)$$

*The constant depends only on the admissibility constants and the fixed chart geometry. It is independent of the mass of  $q_+$ , the mesh, the number and values of source pins, and the ratio  $w/s$ . The same inequality holds for expectation differences of two ordered source configurations, and after averaging an ordered coupling of random source configurations, with the common Gibbs continuation. All changed pins and fields lie on the source side of the cut; the target-side continuation is identical in the compared laws. One may decrease  $\eta$ , keeping it positive.*

*Source and signed-measure interpretation.* These are Proposition 4.2, “Transmission”, and Corollary 4.11, “Random and signed sources”, in Section 4, “Ordered transmission across a cut”, of [19]. To clarify the signed assertion, put  $m = \mathbb{E}q_+ = \mathbb{E}q_-$ . If  $m = 0$ , both sides vanish. Otherwise the

source probability measures  $m^{-1}q_+\mu_S$  and  $m^{-1}q_-\mu_S$  are stochastically ordered by Definition 2.3. Couple them in that order, apply the deterministic-source transmission inequality, and multiply by  $m$ . Its nonnegative magnetic difference permits averaging and gives (2.1). No division by a mesh-dependent lower bound on  $m$  occurs.  $\square$

**Proposition 2.5** (Nearby line sums). *Let  $J_s, J'_s$  be horizontal or vertical lattice segments of length  $O(s)$ , with corresponding portions displaced by at most  $Cw$  and unmatched end portions of total length  $O(w)$ . The assertion also applies to concentric square contours of radii comparable to  $s$ , differing in radius by at most  $Cw$ , and to a fixed finite union of these pairs. Assume their union has a free neighborhood of width  $cs$ , covered by a fixed number of buffered lattice charts. For each fixed  $h = w/s \in (0, h_0]$ ,*

$$\limsup_{s \rightarrow \infty} \left\| s^{-7/8} \sum_{J_s} \sigma_x - s^{-7/8} \sum_{J'_s} \sigma_x \right\|_{L^2(\mu)} \leq C_0 h^{3/8}. \quad (2.2)$$

*The bound is uniform over arbitrary pins beyond the stated free neighborhood and mixtures of their laws. The constant depends only on the fixed geometry and buffer.*

This is Proposition 5.1, “Differences of nearby line sums”, in Section 5, “Nearby contours and prediction”, of [19]. Its exponent follows from integrating the full-plane covariance kernel  $c|x-y|^{-1/4}$  along segments: separations below  $h$  contribute  $O(h^{3/4})$  to the squared norm, while above  $h$  the first-derivative bound gives  $Ch \int_h^1 t^{-5/4} dt = O(h^{3/4})$ . Short unmatched end pieces contribute  $O(h^{7/4})$  in squared norm. Buffered pointwise comparison transfers the nonnegative squared error to the allowed laws. This explanation uses the continuum kernel after taking the mesh limit and asserts no uniform discrete derivative bound.

**Corollary 2.6** (Prediction from a nearby contour). *In Proposition 2.5, let  $\mathcal{F}$  be any spin information containing  $\sigma_{J'_s}$ , and let  $P_{\mathcal{F}}$  denote conditional expectation. Then, for each fixed  $w/s$ ,*

$$\limsup_{s \rightarrow \infty} \|M_{J_s} - P_{\mathcal{F}} M_{J_s}\|_2 \leq C_0 (w/s)^{3/8}.$$

*Both line sums use the same normalization  $s^{-7/8}$ .*

*Proof.* The second line sum is an  $\mathcal{F}$ -measurable competitor for the orthogonal projection. Apply (2.2). This is also the Corollary 5.2, “Prediction from a nearby contour”, in Section 5 of [19].  $\square$

In particular, when  $w \asymp s/n$  for  $n$  concentric cuts, Proposition 2.4 has a constant independent of  $n$ . In Proposition 2.5 one first fixes  $n$  and then takes  $s$  large; the threshold may depend on  $n$ . The fixed exponent  $\eta > 0$ , geometric constants, and normalization are chosen before either operation.

## 2.3 Spin correlations and the thermal direction

**Proposition 2.7** (Spin normalization and moment bounds). *There is  $a_\sigma > 0$ , determined by the square-lattice convention, such that, for distinct physical points  $z_1, \dots, z_n$ ,*

$$a_\sigma^{-n} \delta^{-n/8} \mathbb{E}_0 \prod_{i=1}^n \sigma_{x_i^\delta} \longrightarrow S(z_1, \dots, z_n), \quad \delta x_i^\delta \longrightarrow z_i.$$

*For odd  $n$ ,  $S = 0$ ; for even  $n$ ,  $S$  is the nonnegative square root of*

$$S(z_1, \dots, z_n)^2 = 2^{-n/2} \sum_{\substack{u_i \in \{-1, 1\} \\ \sum_i u_i = 0}} \prod_{i < j} |z_i - z_j|^{u_i u_j / 2}. \quad (2.3)$$

Thus  $S(z, w) = |z - w|^{-1/4}$ . The convergence is uniform when the points lie in compact sets with fixed mutual separation. Also

$$0 \leq \mathbb{E}_0[\sigma_x \sigma_y] \leq C(1 + |x - y|)^{-1/4}, \quad (2.4)$$

$$0 \leq \mathbb{E}_{B_r}^+[\sigma_0] \leq C(1 + r)^{-1/8}. \quad (2.5)$$

For  $n \geq 2$ , repeated sites allowed,

$$\left| \mathbb{E}_0 \prod_{i=1}^n \sigma_{x_i} \right| \leq C^n \prod_{i=1}^n \left( 1 + \min_{j \neq i} |x_i - x_j| \right)^{-1/8}. \quad (2.6)$$

For every fixed finite collection of bounded compactly supported Riemann-integrable spatial test functions  $\varphi$ , the smeared variables

$$\Phi_\delta(\varphi) = a_\sigma^{-1} \delta^{15/8} \sum_{x \in \mathbb{Z}^2} \varphi(\delta x) \sigma_x$$

have joint moment limits obtained by integrating  $S$  against the test functions. Here Riemann integrability is equivalent to having a set of discontinuities of two-dimensional Lebesgue measure zero. Spatial smearings are chosen in this class and fixed before taking the mesh limit; polygonal cell indicators are included. The limiting correlation functions are covariant under Möbius transformations, including inversion, with weight  $1/8$  at each spin insertion.

*Sources and the domination needed for smearing.* The construction of smeared magnetization fields from spin correlations is developed in [7, Section 3]. We give the domination argument for the spatial test functions used here. The spin limit and its full-plane normalization follow from [11, Theorems 1.1–1.2, Remark 1.4, and Equation (1.6)]. That paper uses a rotated lattice with nearest-neighbor spacing  $\sqrt{2}$  times its mesh parameter; its positive normalization constant is absorbed into  $a_\sigma$ . The plus-boundary one-point convergence in a fixed square and the full-plane two-point asymptotic give (2.5) and (2.4) for large distances. Enlarging  $C$  covers the finitely many smaller distances.

For (2.6), put disjoint squares of radius a small fixed fraction of the nearest other insertion distance around the distinct insertion sites. Conditional on the spins outside these squares, the squares are independent. Boundary monotonicity bounds the absolute conditional mean at each insertion by the corresponding plus-square mean. A site appearing an even number of times contributes one; for a repeated site the corresponding factors on the right of (2.6) are one. Multiplication gives the bound, after harmless bounded-distance changes of the boxes.

For each fixed  $n \geq 2$ , the rescaled correlation kernel

$$K_{\delta,n}(z_1, \dots, z_n) = a_\sigma^{-n} \delta^{-n/8} \mathbb{E}_0 \prod_{i=1}^n \sigma_{z_i/\delta}, \quad z_i \in \delta \mathbb{Z}^2,$$

is uniformly bounded in  $L^2$  on bounded products, with normalized lattice counting measure  $\delta^{2n}$ . To see this, square the nearest-distance majorant in (2.6) and bound the product of nearest-neighbor maxima by

$$|K_{\delta,n}(z_1, \dots, z_n)|^2 \leq C_n \sum_{\substack{f: \{1, \dots, n\} \rightarrow \{1, \dots, n\} \\ f(i) \neq i \text{ for all } i}} \prod_{i=1}^n (\delta + |z_i - z_{f(i)}|)^{-1/4}.$$

There are finitely many directed choices  $f$ . Each resulting graph consists of directed cycles with attached trees. Integrate tree leaves first. A two-cycle has a single singularity of power  $1/2$ ; on a

longer cycle integrate one vertex by Cauchy–Schwarz and then the remaining path. Only one-variable singularities of power at most  $1/2 < 2$  occur. Their bounded-domain integrals and mesh-regularized lattice sums are uniformly bounded, including when lattice insertion sites coincide. This proves the  $L^2$  bound; its constants may depend on the fixed  $n$ , with no assertion here about their growth in  $n$ . The same integration argument with  $\delta = 0$  gives the corresponding bound for the continuum correlation kernel. The first moment vanishes by spin symmetry, and the zeroth moment is one.

On a fixed bounded product, the set where any two insertion points are within distance  $\varepsilon$  has normalized counting measure  $O_n(\varepsilon^2 + \delta^2)$ . Cauchy–Schwarz therefore bounds its contribution to a smeared moment by  $C_n(\varepsilon + \delta)$ , with the fixed test-function bounds absorbed in  $C_n$ . Away from these diagonals the correlation convergence is uniform. Multiply by a continuous cutoff that vanishes when a pair is within distance  $\varepsilon$  and equals one when all pairwise distances are at least  $2\varepsilon$ . The remaining continuum integrand is bounded and Riemann integrable: its discontinuities lie in the finite union of the coordinate lifts of the spatial test functions’ null discontinuity sets. Its lattice Riemann sums consequently converge. First take the mesh limit and then let  $\varepsilon \downarrow 0$ , using the preceding uniform bound. This proves the asserted joint smeared moments. Finally the covariance rule follows directly from (2.3) and  $|\psi(z) - \psi(w)| = |z - w| |\psi'(z)\psi'(w)|^{1/2}$  for Möbius maps, with complex conjugation included for an orientation-reversing map.  $\square$

**Lemma 2.8** (Reflection positivity). *The full-plane law  $\mu_0$  is reflection positive across a horizontal or vertical lattice reflection line. The limiting smeared spin correlation algebra is reflection positive across a straight line.*

*Proof.* The finite-bond argument below is a special case of the reflection-positivity method of Fröhlich et al. [14]. In a finite reflection-symmetric box, reflection through a bond line leaves the within-half weights paired. Each bond crossing the line has matrix  $(e^{K_0 st})_{s,t=\pm 1}$ , whose eigenvalues are  $2 \cosh K_0$  and  $2 \sinh K_0$ , both positive. Tensoring its positive decomposition and summing the spins in the two halves writes  $\mathbb{E}[F \overline{F} \circ \vartheta]$  as a sum of nonnegative squares for every half-box function  $F$ . For a reflection through vertices, condition on the spins on the line: the two remaining halves are independent with reflected laws and the same conclusion follows. Exhaust the plane by symmetric boxes, then use the moment convergence of Proposition 2.7 for polynomials in smeared fields whose supports lie strictly on one side. Approximation up to the line, when needed, follows in  $L^2$  from (2.4).  $\square$

**Proposition 2.9** (Positive energy covariance). *Let  $e_1, e_2$  be the coordinate unit vectors and set  $b_j(x) = \sigma_x \sigma_{x+e_j} - 1/\sqrt{2}$ . In the full-plane critical law,*

$$\text{Cov}_0(b_j(0), b_k(x)) = \frac{1 + o(1)}{\pi^2 |x|^2}, \quad |x| \rightarrow \infty, \quad j, k \in \{1, 2\}. \quad (2.7)$$

*For a fixed finite local average  $B(x) = \sum_q c_q b_{j_q}(x + d_q)$ , with  $c_q \geq 0$  and  $c = \sum_q c_q > 0$ ,*

$$\text{Cov}_0(B(0), b_k(x)) \sim \frac{c}{\pi^2 |x|^2}, \quad \text{Cov}_0(B(0), B(x)) \sim \frac{c^2}{\pi^2 |x|^2}.$$

*In particular square symmetrization of the bond energy cannot cancel its leading covariance.*

*Normalization and exact criticality.* Use only the pure specialization of [15, Equations (1.8), (1.11), and Theorem 1.2]. Their scaled energy is  $a^{-1} \sigma_x \sigma_{x+ae_j}$ , and their superscript  $T$  denotes connected expectation. At two insertions their massless formula is  $1/(\pi^2 |z - w|^2)$ , independently of both orientation labels: the two permutations and the two nonzero sign choices each give  $|z - w|^{-2}$ ,

with prefactor  $1/(4\pi^2)$ . The remainder bound in their Equation (1.16), together with the following propagator estimate, is uniform when the two points remain in a fixed annulus of separations. This supplies uniformity in the direction of the displacement.

Their formulation permits a nonzero mass parameter tending to zero. To obtain the exactly critical lattice limit, fix a sequence  $a \rightarrow 0$  and choose a subcritical  $\beta(a)$  so close to  $\beta_0$  that the finitely many spin covariances being tested differ from their critical values by  $o(a^2)$ , and  $|\beta(a) - \beta_0| = o(a)$ . Such a choice is possible by continuity of finite-spin expectations at the unique critical Gibbs state. The cited formula applies along this sequence; the error divided by  $a^2$  vanishes. This proves (2.7). Subtraction of the mean does not change a covariance. The finite-sum assertions then follow term by term, since each fixed displacement  $d_q$  is negligible compared with  $|x|$  and every leading coefficient has the same positive sign.  $\square$

## 2.4 Pure interfaces, complete resolutions, and four passages

A *strong* spin path uses nearest-neighbor edges. A *weak* spin path may also join vertices of the same lattice face. These notions must be distinguished at an alternating four-spin face. The fixed triangulation obtained by joining its northwest and southeast spin vertices realizes the NE/SW pairing when all disagreement edges of the face are included. In a bulk square-lattice chart, affine interpolation on these triangles therefore gives a noncrossing zero set whose local connections match the rounded dual contours within  $O(\delta)$ , with the same traversal order. The added diagonals define connectivity and drawing conventions; they do not change the Ising interaction. At the target boundary, the prescribed open curve uses only the interior edges and its stubs. Its local drawings beside signed boundary strips are treated in Lemma 7.2.

**Theorem 2.10** (Pure Dobrushin interface). *In ordinary lattice approximations of a fixed polygonal Jordan domain with two distinct marked boundary points, the critical nearest-neighbor Ising interface with opposite constant signs on the two marked arcs converges to chordal SLE<sub>3</sub>. Convergence is in uniform distance modulo increasing reparametrization, oriented from the first mark to the second. Either local turn at an alternating face is allowed, including the deterministic rule of the problem. Bounded local rounding and half-stubs have vanishing effect on this convergence.*

This is the specialization of [10, Theorem 1 and Equation (1)] used here. The discussion of ambiguous faces in that paper explicitly allows arbitrary local turns. Its theorem has a broader domain scope; only these fixed polygonal approximations are used as direct inputs. The limiting curve is continuous, simple, and disjoint from the domain boundary except at its endpoints, by [22, Theorems 5.1, 6.1, and 7.1] and conformal extension in a Jordan domain.

**Theorem 2.11** (Every complete loop resolution). *In a plus Jordan domain with lattice boundaries converging uniformly modulo parametrization, every complete noncrossing resolution of all spin-disagreement edges converges to the same oriented nested CLE<sub>3</sub>. The assertion is uniform over choices of the complete resolution. Loops are compared in uniform distance modulo increasing reparametrization and choice of starting point; collections are compared by matching loops and charging the diameters of unmatched loops. Thus the statement includes coverage and multiplicities of all loops above each fixed positive diameter cutoff.*

The published input is [4, Theorem 6, its following common-edge refinement, and Remark 7]. It gives nested convergence for leftmost and rightmost loops and simultaneous approximation of every macroscopic oriented edge-simple Ising circuit. The extension to *every complete resolution* is Lemma 6.1, “Convergence of every complete resolution”, in Section 6, “Complete interface resolutions”, of [19]. For clarity, its extra content is not merely a change of drawing: a pairing

switch can merge or split circuits. The simultaneous circuit approximation and the simplicity and disjointness of the limiting loops imply that two macroscopic circuits affected by a switch would have to converge to the same loop. Their winding numbers would then add to 0 or  $\pm 2$ , contradicting the winding number  $\pm 1$  of the merged simple-loop limit. To fix the multiplicity, include a specified canonical leftmost circuit in an auxiliary complete resolution. A second circuit matched to its limit loop would share an edge with that specified circuit by the common-edge refinement, contradicting edge-disjointness. The pairing-switch argument then transfers the count one to every complete resolution. These facts supply coverage and multiplicity, as required by the stated collection topology.

A contour *passage* across  $B_R(z) \setminus B_r(z)$  is a subpath with one endpoint on each of its boundary curves and interior in the annulus. Several passages use disjoint interiors of parameter intervals; hit endpoints may be shared. Four passages can belong to one loop, and repeated counting of one traversal is forbidden. Denote their existence by  $\mathcal{A}_4^{\text{cont}}(z; r, R)$ .

**Proposition 2.12** (Iterated four-passage bound). *Fix a bulk lattice chart containing  $B_{2R}(z)$ , with all pins and all changes of ambient domain beyond a fixed proportional outer collar. For every  $\alpha < 21/8$ , there are  $C_\alpha < \infty$  and  $u_\alpha > 0$ , depending on that fixed collar but not on the outer conditions, such that, for each fixed  $0 < u < u_\alpha$ ,*

$$\limsup_{\delta \downarrow 0} \mathbb{P}(\mathcal{A}_4^{\text{cont}}(z; uR, R)) \leq C_\alpha u^\alpha. \quad (2.8)$$

*This applies to the fixed diagonal resolution, and uniformly to complete resolutions. The same statement holds for circular annuli and for fixed-factor changes of the radii. In particular one may fix  $\alpha > 2$ . The mesh limit precedes the limit  $u \downarrow 0$ .*

*Conversion from the published spin-arm estimate.* Pointwise comparison reduces the event to a plus-boundary model in a fixed larger square, with a multiplicative constant depending only on the outer collar. Rescale so that  $R = 1$ . For a fixed  $u$ , couple a subsequence of complete loop collections to their limit as in Theorem 2.11. Only loops of diameter at least  $1 - u$ , or a fixed smaller cutoff after relaxing radii, can contain the passages under consideration. There are finitely many such limit loops. Uniform parametrized convergence preserves four passages after replacing the two radii by  $2u$  and  $1/2$ . Indeed choose the passages by last and first hits of the two circles, pass to convergent endpoint parameters, and use uniform continuity of the limiting parametrizations. An interval crossing the relaxed annulus cannot collapse to a point. Disjoint interval interiors remain distinct; injectivity of a limiting loop prevents two nontrivial intervals from having the same image. This also treats passages belonging to one loop and passages traversed in opposite directions.

We next pass from this limiting geometric event to strong alternating spin arms. Crop its through subarcs once more, to the radii  $4u$  and  $1/4$ . The finitely many chosen compact subarcs of the simple, pairwise disjoint limit loops have disjoint thin neighborhoods after this crop. They are crosscuts of the annulus. Reading their sides in cyclic order changes the spin sign at each crosscut. If more than four through crosscuts occur, keep four consecutive ones and the adjacent alternating sides. One can follow each chosen side in its own thin neighborhood from the inner to the outer cropped circle.

Now use leftmost approximations for a plus side and rightmost approximations for a minus side, in the *same* spin configuration. By [4, Section 2.5], a leftmost loop has a strong plus path on its left, and a rightmost loop has a strong minus path on its right. The simultaneous circuit approximation and common-edge refinement in [4, Theorem 6 and Remark 7] make both extremal approximations converge to the same selected loop, with its orientation. Their chosen side paths therefore lie in the disjoint neighborhoods and cross the cropped annulus for all sufficiently fine meshes. They supply

four disjoint strong spin paths, of alternating colors. The cropping allows their endpoints to be moved by the bounded lattice distance needed to follow the colored side.

This is an eventual implication on the coupled limit event, so Fatou's lemma bounds its probability by the lower limit of the four-strong-arm probabilities at the further relaxed radii. Wu's Theorem 1.2 gives the interior alternating *spin* exponent  $(16j^2 - 1)/24$ , hence  $21/8$  for four arms. More precisely, [26, Equation (5.4) and Corollary 5.4] gives the required mesh-first upper bound  $u^{21/8+o(1)}$  for these annular events, uniformly in a larger containing square. Combining the two relaxations yields (2.8); multiplying  $u$  by a fixed constant does not change its limiting exponent. Square and circular annuli contain one another after fixed-factor radius changes.

No microscopic inclusion of arbitrary resolved passages in the strong-arm event was used. Such an inclusion would be false as a justification at an individual alternating face, where a colored side of the fixed resolution may have only weak adjacency. Complete oriented convergence and the extremal side paths are the steps that remove this convention difference.  $\square$

## 2.5 Buffered crossings, circuits, and one passage

**Proposition 2.13** (Pure buffered RSW consequences). *Fix a finite collection of polygonal corridors, each a topological rectangle with designated opposite end gates, and polygonal annular bands. Prescribe one color for each required crossing or circuit. Assume each requirement admits a finite rectangle construction with positive-width unpinned buffers, and that the buffered constructions for opposite colors are disjoint. Their simultaneous occurrence has probability bounded below by a positive constant for all sufficiently fine meshes. The constant may depend on this fixed geometry, and is uniform over pins beyond the buffers and ambient plane or torus laws with common lattice charts.*

*There are  $C < \infty$ ,  $\rho > 0$  such that, in a buffered bulk annulus and for  $1 \leq r < R/C$ ,*

$$\mathbb{P}(\text{a monochromatic spin arm or a resolved contour passage from } B_r \text{ to } B_R^c) \leq C(r/R)^\rho. \quad (2.9)$$

*For the monochromatic event either color may be specified or both allowed. The annular construction may be conditioned on earlier reveals that remain outside its unpinned test collars. Fixed-factor changes of radii and bounded lattice drawing corrections change only the constants.*

*Proof.* Theorem 1 of [12] gives critical  $q = 2$  FK rectangle crossing bounds uniform over boundary wirings. Its Section 2.1 gives the domain Markov property, association, planar duality, and Edwards–Sokal coloring. A finite chain of overlapping rectangle crossings, including overlap-square crossings, constructs an FK connection through a prescribed corridor. The analogous finite construction gives an FK circuit in a band. All aspect ratios and the number of rectangles are fixed. A dual circuit in a disjoint outer collar isolates the constructed cluster from the boundary. Uniform conditional RSW permits this dual event and the inner open tests to be imposed in sequence. The isolated cluster then receives either prescribed spin color with probability  $1/2$ . For several opposite-color constructions keep their isolating collars disjoint; their clusters can be colored independently. Same-color connections may merge, and spin FKG combines the finitely many same-color tests. This proves positivity in the reference free or plane charts.

Mixed-sign spin pins are not arbitrary FK wirings: identifying the two notions would invalidate this argument. Instead, Proposition 2.1 transfers the reference lower bound through each unpinned collar. Reveal separated constructions successively; the remaining collars stay free and the lower bounds remain valid. For a single annular circuit one can first transfer the individual rectangular tests and then use pure-spin FKG. Consequently the argument does not require the entire annulus to be a square box avoiding all interior pins.

For the power bound choose  $m \geq c \log(R/r) - C$  annular bands of a fixed aspect ratio with separated test collars. Each has a conditional probability at least  $p > 0$  of a strong spin circuit of either specified color. A circuit of the opposite color blocks a monochromatic arm. A strong spin circuit of either color also blocks a contour passage, since disagreement edges cannot cross its equal-spin bonds. Conditional iteration therefore gives  $C(1-p)^m \leq C'(r/R)^\rho$ , with a union bound for two possible arm colors. This is a spin and contour deduction from FK RSW, coloring, and buffered comparison; no FK arm exponent is being identified with a spin arm exponent. Pins touching a corridor's terminals without a buffer are outside the assertion.  $\square$

Together, Proposition 2.12 and Proposition 2.13 provide, respectively, a bulk exponent strictly greater than two and a positive one-passage exponent. Section 5 will derive the additional estimate for two passages constrained to a half-plane, rather than importing a boundary-arm exponent for a different boundary law.

### 3 Uniform transfer of bounded spin functions

The scale construction requires a statement about every bounded local function, including functions that change with the lattice scale. We prove that, after integration across a large square annulus, only one even scalar direction can grow under the normalization appropriate to an interaction density. Convergence of any fixed collection of spin correlations would not by itself give this assertion. The first part of the proof supplies the uniform approximation that permits us to use those correlations.

For an integer  $s \geq 1$ , set

$$B_s = [-s, s]^2 \cap \mathbb{Z}^2, \quad B_s^\circ = \{x \in \mathbb{Z}^2 : \|x\|_\infty < s\}, \quad \partial B_s = B_s \setminus B_s^\circ.$$

Let  $\mathcal{X}_s$  be the real functions of  $\sigma_{B_s}$  invariant under spin reversal and the dihedral group of the square, modulo constant functions. Its norm is

$$\|f\|_q := \inf_{c \in \mathbb{R}} \|f - c\|_\infty = \frac{1}{2}(\max f - \min f). \quad (3.1)$$

A representative of norm  $\|f\|_q$  is obtained by subtracting the midpoint of its range. Subtracting  $\mathbb{E}_0 f$  instead gives a linear choice of representative with norm at most  $2\|f\|_q$ . For an integer  $L > 1$ , define

$$A_s f = L^2 \mathbb{E}_0[f \mid \sigma_{\partial B_{Ls}}]. \quad (3.2)$$

Here the conditional expectation integrates  $B_{Ls}^\circ$  at the pure critical coupling. Every boundary assignment has positive probability, so this definition specifies the function on every assignment, not merely almost surely. It also defines a map  $\mathcal{X}_s \rightarrow \mathcal{X}_{Ls}$ .

**Theorem 3.1** (Uniform transfer). *There is a constant  $C < \infty$  with the following property. For each  $a > 0$ , all sufficiently large fixed integers  $L$  admit an integer  $s_0$ , linear functionals  $\ell_s : \mathcal{X}_s \rightarrow \mathbb{R}$ , and vectors  $e_s \in \mathcal{X}_s$ , for every integer  $s \geq s_0$ , such that*

$$\|\ell_s\| \leq C, \quad \|e_s\|_q \leq C, \quad \ell_s(e_s) = 1, \quad (3.3)$$

$$\|A_s f\|_q \leq CL|\ell_s(f)| + a\|f\|_q, \quad (3.4)$$

$$|\ell_{Ls}(A_s f) - L\ell_s(f)| \leq a\|f\|_q. \quad (3.5)$$

The constant  $C$  is independent of  $a, L, s$ . The choices are made in the following order: first  $L$ , then a larger fixed measurement ratio  $H$ , then the approximation accuracies used below, and finally  $s_0$ . In particular, all estimates hold simultaneously for all  $s \geq s_0$  and all bounded functions in  $\mathcal{X}_s$ .

We first work in fixed rescaled square charts. A spin sum in two dimensions is normalized by  $s^{-15/8}$ ; the line sums appearing in Proposition 2.4 are normalized by  $s^{-7/8}$ . These two normalizations have different roles in the proof. Every spatial smearing used below is a fixed bounded, compactly supported Riemann-integrable function, chosen before the mesh limit, as in Proposition 2.7. Polygonal cell indicators belong to this class; their specified integer roundings will be controlled below. This convention concerns spatial averaging functions. The microscopic spin functions  $f$  and the exterior likelihoods remain arbitrary bounded functions.

### 3.1 A window common to all exterior likelihoods

We shall use boxes and square bands with a positive proportional buffer. An enlargement of a band moves both boundary components outward from the band, so its spin sets form an increasing family. The number of boundary components is fixed. All distances in this subsection, unless accompanied by a lattice scale, are in the rescaled chart.

Take such a chart with minus spins on its boundary and denote its law by  $\nu$ . Let  $D$  be a layer surrounding a smaller buffered region  $F_*$ ; for a band it is the union of the two surrounding layers. All altered boundary data are farther away than  $D$ . By Proposition 2.1, their marginal likelihoods on  $D$  relative to  $\nu$  are functions  $h$  with

$$0 \leq h \leq K, \quad \mathbb{E}_\nu h = 1, \quad (3.6)$$

where  $K$  depends only on the buffers. Conditional expectation from  $D$  to any intervening set  $F$  gives its likelihood there:  $a_F = P_F h$ , where  $P_F = \mathbb{E}_\nu[\cdot \mid \sigma_F]$ . These likelihoods are increasing functions of the spins of  $F$ . Indeed, relative to all-minus boundary, changing boundary spins multiplies the joint Ising density by an increasing exponential of boundary-adjacent spins; marginalization preserves increasing likelihood ratios for an attractive Ising law. This last assertion also follows directly by conditioning on two ordered configurations of  $F$  and using the monotone coupling in the remaining spins. Convex mixtures preserve the assertion.

**Lemma 3.2** (Simultaneous likelihood window). *Fix nested enlargements  $F_0 \subset F_*$  of the support of a test function, with all three relevant gaps positive: from the support to  $\partial F_0$ , from  $\partial F_0$  to  $\partial F_*$ , and from  $F_*$  to  $D$ . For each  $t > 0$  there is a finite menu of radial windows, each of positive rescaled width, such that one window  $[F_-, C]$  satisfies*

$$\sup_{F_- \subset F \subset C} \sup_h \|P_F h - P_C h\|_{L^2(\nu)} \leq t. \quad (3.7)$$

The second supremum is over all the likelihoods in (3.6). The menu and its widths are independent of the mesh and of the boundary assignments.

*Proof.* Let  $Q : L^2(\nu_D) \rightarrow L^2(\nu_{F_*})$  be conditional expectation from  $D$  to  $F_*$ . Its integral kernel is the density of the joint law of  $(\sigma_{F_*}, \sigma_D)$  relative to the product of its marginals. The buffered comparison bounds this density by a fixed constant, hence  $\|Q\|_{\text{HS}} \leq K_1$ . Choose  $m + 1$  equally spaced nested enlargements  $F_0 \subset \dots \subset F_m = F_*$ . Regarding all conditional expectations as orthogonal projections in  $L^2(\nu)$  gives

$$\sum_{j=1}^m \|(P_{F_j} - P_{F_{j-1}})Q\|_{\text{HS}}^2 \leq \|Q\|_{\text{HS}}^2 \leq K_1^2.$$

Choose  $m$  so that  $KK_1 m^{-1/2} \leq t$ . At least one summand is at most  $K_1^2/m$ . For its endpoints  $F_-, C$ , nesting implies, for every intermediate  $F$ ,

$$\|(P_C - P_F)h\|_2 \leq \|(P_C - P_{F_-})Q\|_{\text{op}} \|h\|_2 \leq t.$$

We used  $P_F h = P_F Q h$ , which holds since  $F \subset F_*$ . This proves the simultaneous, rather than likelihood-dependent, assertion.  $\square$

### 3.2 Gaussian observations and order-positive residuals

We now seek one observation that retains the influence of every exterior likelihood. Gaussian noise is useful because conditioning on it only adds external fields to the Ising law, preserving conditional attraction. The unobserved part of an increasing likelihood is therefore an order-positive signed source. If that residual stayed large, transmission would force a magnetic signal on every one of many separating cuts. Prediction from the observation and from nearby cuts will turn those signals into more orthogonal increments than its bounded norm permits.

Conditional on the spins, in the selected core  $C$  observe independent Gaussian variables

$$Y_x = M s^{-7/8} \sigma_x + Z_x, \quad x \in C, \quad (3.8)$$

where the  $Z_x$  are independent standard normal variables, independent of the spins. Write  $K_M v = \mathbb{E}_\nu[v \mid Y]$  and  $T_M v = \mathbb{E}[K_M v(Y) \mid \sigma]$ . Thus  $T_M = K_M^* K_M$  is a self-adjoint positive contraction on  $L^2(\nu_C)$ , preserves constants, and is a contraction on bounded functions.

**Lemma 3.3** (Uniform smoothing of likelihoods). *Given  $b > 0$ , the window tolerance in Lemma 3.2 can be chosen so that, for a sufficiently large fixed  $M$  and then all sufficiently large  $s$ ,*

$$\sup_h \|(I - T_M) P_C h\|_2 < b. \quad (3.9)$$

*All constants are uniform over the likelihood family and the finite menu of possible windows.*

*Proof.* Put  $a = P_C h$  and  $z = (I - T_M)a$ . Then  $\mathbb{E}_\nu z = 0$ ,  $|z| \leq 2K$ , and, for every increasing  $g$  of the core spins,

$$\mathbb{E}_\nu[zg] = \mathbb{E}[\text{Cov}_\nu(a, g \mid Y)] \geq 0. \quad (3.10)$$

To justify the last inequality, conditioning on  $Y = y$  changes the Ising weight by  $\exp(M s^{-7/8} \sum_x y_x \sigma_x)$  times a constant. The conditional law remains ferromagnetic, with external fields of arbitrary signs, so conditional FKG applies [13, Proposition 1 and pp. 98–99]. The function  $z$  itself need not be increasing. Positivity of  $I - T_M$  and its contraction bound also give

$$\langle z, a \rangle = \langle a, (I - T_M)a \rangle \geq \|z\|_2^2. \quad (3.11)$$

Suppose  $\|z\|_2 \geq b$ . Choose the window tolerance less than  $b^2/(4K)$ . For every  $F$  in that window,

$$\langle P_F z, h \rangle = \langle z, P_F h \rangle \geq b^2 - 2Kt \geq b^2/2.$$

Subdivide an inner fixed portion of the window into  $n$  equal radial steps. Denote the successive spin sets by  $F'_0, \dots, F'_n$  and choose a separating square contour, or the two contours of a band,  $J_i$  strictly between  $F'_i$  and  $F'_{i+1}$ . Its distance from  $F'_i$  is comparable to  $w = s/n$ , with a constant depending on the already fixed window. Define  $m_i = s^{-7/8} \sum_{x \in J_i} \sigma_x$ . The signed source  $P_{F'_i} z$  is order-positive: an increasing  $F'_i$ -measurable test has the same pairing with it as with  $z$ . Subdivide  $J_i$  into straight pieces of a sufficiently small fixed multiple of  $w$ . Their enlarged charts are pin-free, they have bounded overlap, and the target layer  $D$  is at distance comparable to  $s$ . These are exactly the admissible cuts of Proposition 2.4. For its fixed exponent  $\eta > 0$ , that proposition yields

$$\langle P_{F'_i} z, m_i \rangle \geq c_b n^{-7/8+\eta}. \quad (3.12)$$

The constant does not deteriorate further as  $n$  increases.

The observation predicts each  $m_i$  with arbitrarily small error when  $n$  is fixed. To see this quantitatively, average it over parallel nearby lines, including the corresponding short endpoint corrections, in a strip of width  $\theta s$  lying inside  $C$ . Call the average  $\bar{m}_i$ . All these lines have the same length and buffer bounds, with displacement at most  $C\theta s$ ; a square contour is treated side by side, including its short endpoint pieces. For fixed  $\theta$ , Proposition 2.5 is uniform over the averaging lines: otherwise choose a maximizing line at each mesh, obtaining a sequence of pairs that still satisfies that proposition with  $w/s = \theta$ . Minkowski's inequality therefore gives, even though the number of averaging lines grows with  $s$ ,

$$\limsup_{s \rightarrow \infty} \|m_i - \bar{m}_i\|_2 \leq C\theta^{3/8}.$$

The average is a linear combination of the signals  $s^{-7/8}\sigma_x$ , with coefficients of size  $O((\theta s)^{-1})$  on  $O(\theta s^2)$  sites. Replacing each signal by  $Y_x/M$  gives a  $Y$ -measurable estimator of  $\bar{m}_i$  with noise variance at most  $C/(M^2\theta)$ . Consequently

$$\limsup_{s \rightarrow \infty} \|m_i - \mathbb{E}[m_i | Y]\|_2 \leq C\theta^{3/8} + C/(M\sqrt{\theta}).$$

Since  $\langle z, m_i \rangle = \mathbb{E}[\text{Cov}(a, m_i | Y)]$ , first choosing  $\theta$  small and then  $M$  large makes this pairing smaller in absolute value than  $\frac{1}{2}c_b n^{-7/8+\eta}$ .

Set  $d_i = (P_{F'_{i+1}} - P_{F'_i})z$ . The variable  $m_i$  is measurable on  $F'_{i+1}$ , whereas  $d_i$  is orthogonal to all  $F'_i$ -measurable functions. Therefore

$$|\langle d_i, m_i - P_{F'_i} m_i \rangle| = |\langle z, m_i \rangle - \langle P_{F'_i} z, m_i \rangle| \geq \frac{1}{2}c_b n^{-7/8+\eta}.$$

Corollary 2.6, with a nearby contour in  $F'_i$  and with  $n$  fixed before  $s \rightarrow \infty$ , bounds its second factor by  $Cn^{-3/8}$ . Hence  $\|d_i\|_2 \geq c'_b n^{-1/2+\eta}$  for all sufficiently large  $s$ . Orthogonality of the increments gives the contradiction

$$4K^2 \geq \|z\|_2^2 \geq \sum_{i=0}^{n-1} \|d_i\|_2^2 \geq c''_b n^{2\eta}. \quad (3.13)$$

Choose  $n$  large enough before choosing  $\theta, M, s$ . Thus the complete order is  $b$ , the window tolerance and width,  $n, \theta, M$ , and finally the mesh threshold. Only finitely many contours and windows occur after  $n$  has been fixed, so their thresholds can be combined.  $\square$

This gives an approximation in exactly the norm needed for separated correlations. If  $v$  is a bounded function on the original support, then for every exterior likelihood,

$$|\mathbb{E}_\nu[(v - T_M v)h]| = |\langle v, (I - T_M)P_C h \rangle| \leq \|v\|_\infty b. \quad (3.14)$$

The same conclusion holds against a bounded exterior observable, by conditioning on its spin configuration and integrating the preceding bound. The remaining issue is that  $Y$  has a number of coordinates growing with  $s$ . We next reduce it to a fixed finite collection.

### 3.3 Overlap moments and finitely many observation modes

For two independent copies  $\sigma, \sigma'$  on a bounded rescaled region, put

$$Q_s = s^{-7/4} \sum_x \sigma_x \sigma'_x.$$

We shall use the following two estimates, also valid under any pure law whose density in the region is bounded relative to the plane law:

$$\sup_s \mathbb{E} \exp(t|Q_s|) < \infty \quad (t < \infty), \quad (3.15)$$

$$\lim_{\kappa \downarrow 0} \limsup_{s \rightarrow \infty} \mathbb{E}|Q_s - Q_{s,\kappa}|^2 = 0. \quad (3.16)$$

Here  $\Pi_\kappa$  is orthogonal projection of site vectors onto functions constant on the cells of a fixed rectangular partition of mesh  $\kappa$  in rescaled coordinates, intersected with the selected square or square-band core. These cells are polygonal and have boundaries of area zero. Retain the positive-area cells, assigning boundary sites to one adjacent cell by a fixed rule. The finite partition is fixed before the mesh limit. Define  $Q_{s,\kappa} = s^{-7/4} \langle \Pi_\kappa \sigma, \Pi_\kappa \sigma' \rangle$ .

We now control the growth in the moment order to prove the uniform integrability needed in these statements. Recall (2.6): for  $n \geq 2$  and sites that need not be distinct,

$$\left| \mathbb{E}_0 \prod_{i=1}^n \sigma_{x_i} \right| \leq C^n \prod_{i=1}^n (1 + \min_{j \neq i} |x_i - x_j|)^{-1/8}. \quad (3.17)$$

Expanding an even moment of  $Q_s$  squares the correlation in (3.17). After rescaling, it remains to bound the integral, with normalized lattice counting measure on a fixed bounded region, of  $\prod_i (\delta \vee r_i)^{-1/4}$ , where  $\delta = s^{-1}$  and  $r_i = \min_{j \neq i} |x_i/s - x_j/s|$ . Take  $h = n^{-1/2}$ . If  $\delta > h$ , the deterministic bound  $|Q_s| \leq C s^{1/4}$  suffices. Otherwise let  $\rho_i = \min\{h, \max\{\delta, r_i\}\}$ , and randomly order the  $n$  labels. Write  $D_i$  for the distance to the nearest predecessor, with the same cutoffs and with  $D_i = h$  if there is no predecessor. A selected nearest neighbor precedes label  $i$  with probability  $1/2$ . Thus Jensen's inequality gives

$$\prod_i \rho_i^{-1/4} \leq h^{-n/4} \mathbb{E}_{\text{order}} \prod_i (h/D_i)^{1/2}. \quad (3.18)$$

For any fixed ordering, integrate positions in reverse order. For each last remaining position the union of disks of radius  $r \geq \delta$  around its predecessors has normalized counting measure at most  $Cnr^2$ . The layer-cake integral for  $(h/D_i)^{1/2}$  is bounded by a constant, since  $nh^2 = 1$ ; the atom at the cutoff  $\delta$  obeys the same bound. Repeating the integration proves

$$\mathbb{E}_0 Q_s^n \leq C^n n^{n/8} \quad (n \text{ even}). \quad (3.19)$$

Cauchy–Schwarz gives the corresponding absolute-moment bound for all integer orders. Its exponential series converges for every fixed coefficient, proving (3.15).

For (3.16), the identity for the second moment of an overlap expresses the error through the covariance operator. More explicitly, with  $C_s$  the covariance matrix of  $s^{-7/8}(\sigma_x)_x$  in ordinary Euclidean coordinates,

$$\mathbb{E}_0 |Q_s - Q_{s,\kappa}|^2 = \text{tr} \left( ((I - \Pi_\kappa) C_s (I - \Pi_\kappa))^2 \right) \leq \|C_s - \Pi_\kappa C_s \Pi_\kappa\|_{\text{HS}}^2.$$

The two-point convergence and its bound  $C(1 + |x - y|)^{-1/4}$  identify the limiting kernel with a constant times  $|u - v|^{-1/4}$ . Its square is locally integrable in two dimensions. Removing a small neighborhood of the diagonal, taking Riemann sums, and then restoring that neighborhood proves Hilbert–Schmidt convergence and the claim. Buffered density bounds transfer the nonnegative error and the exponential estimates to the reference laws.

The coarse overlaps also have the exponential bounds uniformly in  $\kappa$ . Under the plane law, if  $m_n = \mathbb{E}[(s^{-7/8}\sigma)^{\otimes n}]$ , then

$$\mathbb{E}Q_{s,\kappa}^n = \|\Pi_\kappa^{\otimes n} m_n\|^2 \leq \|m_n\|^2 = \mathbb{E}Q_s^n \quad (n \text{ even}).$$

Again density comparison gives the assertion in a buffered reference chart. It follows from (3.16), truncation, and these exponential bounds that, for fixed  $M$ ,

$$\mathbb{E} \left| e^{M^2 Q_s} - e^{M^2 Q_{s,\kappa}} \right| \longrightarrow 0 \quad \text{as } s \rightarrow \infty \text{ and then } \kappa \downarrow 0. \quad (3.20)$$

We now apply this calculation to observations. Relative to standard white noise  $\gamma_s$  on the sites of  $C$ , the observation density weighted by a function  $v$ ,  $|v| \leq 1$ , is

$$R_v(y) = \mathbb{E}_\nu \left[ v(\sigma) \exp \{ M \langle y, s^{-7/8} \sigma \rangle - \frac{1}{2} M^2 \| s^{-7/8} \sigma \|^2 \} \right].$$

Let  $R_v^\kappa = \mathbb{E}_{\gamma_s} [R_v \mid \Pi_\kappa y]$ . Gaussian integration and the orthogonal-projection identity give exactly

$$\|R_v - R_v^\kappa\|_{L^2(\gamma_s)}^2 = \mathbb{E}_{\nu \otimes \nu} \left[ v(\sigma) v(\sigma') (e^{M^2 Q_s} - e^{M^2 Q_{s,\kappa}}) \right]. \quad (3.21)$$

The right side tends to zero uniformly over  $|v| \leq 1$ . The full and coarse posterior means are  $R_v/R_1$  and  $R_v^\kappa/R_1^\kappa$ . They are bounded by one, and

$$\int R_1 \left| \frac{R_v}{R_1} - \frac{R_v^\kappa}{R_1^\kappa} \right| d\gamma_s \leq \|R_v - R_v^\kappa\|_1 + \|R_1 - R_1^\kappa\|_1. \quad (3.22)$$

This uses  $|R_v^\kappa| \leq R_1^\kappa$  and requires no positive uniform lower bound on  $R_1$ .

Write  $X_{s,\kappa} = s^{-7/8} \Pi_\kappa \sigma$  in an orthonormal cell basis, and let  $q_{s,v}$  be the coarse posterior as a function of the projected observation. Its absolute value is at most one. Conditioning back on the spins gives the replacement

$$F_{s,v}(X_{s,\kappa}), \quad F_{s,v}(x) = \int q_{s,v}(Mx + z) d\gamma_{d_\kappa}(z),$$

where  $d_\kappa$  is the fixed number of retained cells and  $\gamma_{d_\kappa}$  is standard Gaussian measure in that dimension. Thus it is the Gaussian convolution of the bounded posterior that is uniformly Lipschitz, with a constant depending on  $M$  and the partition but not on  $v$  or  $s$ : bound the difference by the total variation distance between the two translated Gaussians. For a fixed positive-area polygonal cell  $E$ , put  $n_{s,E} = |sE \cap C|$ , with the prescribed boundary assignment. In its normalized cell basis the corresponding coordinate is

$$s^{-7/8} n_{s,E}^{-1/2} \sum_{x \in sE \cap C} \sigma_x = \left( \frac{s^2}{n_{s,E}} \right)^{1/2} s^{-15/8} \sum_{x \in sE \cap C} \sigma_x.$$

Here  $n_{s,E}/s^2 \rightarrow \text{area}(E) > 0$ , so the cell is nonempty for all sufficiently large  $s$  and the scalar normalization converges. Its spatial smearing is the fixed Riemann-integrable indicator of  $E$ . Equation (3.22) gives an  $L^1(\nu)$  error, which is sufficient against every uniformly bounded likelihood.

**Lemma 3.4** (Bounded replacement across a gap). *Fix finitely many pairs of positively separated supports, each a square or a square band, with buffered enlargements that remain disjoint. For every  $t > 0$ , each lattice function  $v$  of norm at most one on any one support can be replaced, for its full-plane correlations against all functions of norm at most one on the paired supports, with error*

at most  $t$ , by a function of finitely many smeared spin sums in its enlargement. The replacement has norm at most one. The number of sums, their shapes, and a modulus of continuity for the replacing functions belong to a fixed finite menu, independent of  $v$  and of all sufficiently fine meshes. Along any mesh sequence, further subsequences of these replacements converge to bounded continuous tests of finitely many continuum smeared spins. Replacements on both sides of each gap can be made together, and preserve spin-flip and square symmetries by averaging.

*Proof.* Use Lemma 3.3, then (3.14), and finally the finite-mode construction. To pass from the reference law  $\nu$  to a full-plane pairing, choose the buffered reference chart inside the specified enlargement, disjoint from the paired supports, and condition the plane law on its exterior spins. The conditional marginal on  $D$  has a likelihood  $h$  in (3.6), and its core likelihood is  $P_C h$ . The paired observable is measurable outside that chart, so integrating the bound against its signed values gives the same correlation estimate under  $\mu_0$ . For the finite-mode error, every such core likelihood is bounded by  $K$ ; its  $L^1(\nu)$  error therefore costs at most that fixed factor. The replacing function is the same for every exterior configuration. For finitely many required supports the menu is their finite product, and errors are divided by their number. The bounded functions of the finite-dimensional sums are equicontinuous; tightness of those sums and compactness on bounded coordinate sets give the subsequential assertion by a diagonal argument. These replacements and their smeared coordinates are now evaluated under  $\mu_0$ . Their limiting field law is therefore the full-plane spin-field law: Proposition 2.7 gives joint moment convergence for the fixed polygonal cell indicators. The prescribed rounding errors lie in vanishing neighborhoods of their finitely many edges. The uniform  $L^2$  bounds on fixed-order rescaled correlation kernels proved there, and the vanishing lattice counting measure of these neighborhoods, make the errors tend to zero by Cauchy–Schwarz. The finite menu of spatial smearings is fixed before the mesh limit. The unsquared version of the ordering argument (3.18) supplies exponential integrability of each fixed smeared sum and hence moment determinacy. Averaging over the finite symmetry group preserves all bounds. Applying the result to one side and then the other costs at most the sum of the two errors.  $\square$

The lemma asserts approximation for correlations across a gap. No pointwise recovery of an arbitrary spin function from the magnetization field is asserted or needed.

### 3.4 The radial Hilbert space

We have reduced lattice correlations of bounded functions to continuum field tests with controlled ordinary  $L^2$  norm. We now identify the part of their radial spectrum below degree three. The explicit spin formula will first give expansions for polynomial tests; reflection positivity will then turn those expansions into an operator bound.

Use the normalization of Proposition 2.7, so the continuum two-point function is  $|z - w|^{-1/4}$ , and write  $\Phi(\psi)$  for the spin field smeared against a bounded compactly supported Riemann-integrable spatial function  $\psi$ . All spatial smearings in this section belong to this class. Let  $\mathcal{A}$  be the algebra of polynomials in such smeared fields supported strictly inside the unit disk. Initially their supports may be kept away from the origin. This causes no loss upon completion: multiply each spatial function by a smooth cutoff vanishing on a disk of radius  $\varepsilon$  and equal to one outside the disk of radius  $2\varepsilon$ . As  $\varepsilon \downarrow 0$  these functions converge almost everywhere, remain uniformly bounded, and have common compact support. The integrable fixed-order correlation bounds give convergence of each smeared variable in every fixed finite  $L^p$ . Telescoping a polynomial difference and applying Hölder’s inequality therefore gives ordinary  $L^2$  convergence of every fixed smeared polynomial.

The spatial class is preserved by the weighted coordinate changes used below, when the support stays away from their poles. If  $g$  is a Möbius or anti-Möbius map, denote its local length scale

by  $|Dg(z)|$ , equal to  $|g'(z)|$  in the conformal case, and set  $T_g\sigma(z) = |Dg(z)|^{1/8}\sigma(g(z))$  on point insertions. Changing variables in a smearing gives

$$T_g\Phi(\psi) = \Phi(\psi_g), \quad \psi_g(w) = \psi(g^{-1}(w))|Dg^{-1}(w)|^{15/8}. \quad (3.23)$$

The area Jacobian has exponent 2 and the primary spin weight subtracts  $1/8$ , giving the exponent  $15/8$  in this formula. On the relevant compact neighborhoods the coordinate map and its inverse are smooth with bounded derivatives, preserve null sets, and have a bounded continuous Jacobian weight. Thus  $\psi_g$  is bounded, compactly supported, and Riemann integrable. In particular inversion gives the spatial weight  $|w|^{-15/4}\psi(1/\bar{w})$ .

Cutoffs are made before applying a map with a pole. For a polynomial  $F$  in the cutoff smears, correlation covariance gives  $\|T_g F\|_2 = \|F\|_2$ . After removing the cutoffs, this isometry defines the transformed polynomial by an ordinary  $L^2$  limit. For example, inverted supports need not remain in one compact set as an origin cutoff is removed. The extension uses the isometry and asserts no lattice sampling theorem for the resulting unbounded-support expression.

Polynomials are dense in the  $L^2$  space of any finite collection of smeared fields. Here is the determinacy argument in the precise form needed. Apply the random predecessor ordering argument of (3.18) to the unsquared nearest-distance bound, with exponent  $1/8$  in place of  $1/4$ . For even  $n$  it gives

$$\mathbb{E}|\Phi(\psi)|^n \leq C_\psi^n n^{n/16}.$$

Cauchy–Schwarz gives the same bound, after enlarging  $C_\psi$ , for every integer  $n \geq 1$ . Thus  $\mathbb{E}e^{t|\Phi(\psi)|} < \infty$  for every fixed  $t$ . If an  $L^2$  function  $g$  of a finite vector of smears is orthogonal to all its polynomials, its signed distribution has a Laplace transform analytic in a neighborhood of the origin, by Cauchy–Schwarz and exponential integrability. Every derivative there is zero. Analytic continuation to the imaginary axes and uniqueness of Fourier transforms give  $g = 0$ . Thus bounded continuous tests in Lemma 3.4 can be approximated in ordinary  $L^2$  by elements of  $\mathcal{A}$ .

We use the reflection-positive Hilbert-space construction of Osterwalder and Schrader [20, Section 4.1] in its radial form, verifying contraction and continuity below. Let  $\iota(z) = 1/\bar{z}$ . On point insertions, interpreted through smearing, reflection in the unit circle acts as

$$\Theta\sigma(z) = |z|^{-1/4}\sigma(\iota(z)).$$

It acts multiplicatively on polynomials. Define

$$(F, G)_{\text{rad}} = \mathbb{E}[(\Theta\bar{F})G], \quad \|F\|_{\text{rad}}^2 = (F, F)_{\text{rad}}. \quad (3.24)$$

The field is invariant under this transformation by its Möbius covariance. Line reflection positivity from Lemma 2.8, transported to the circle by a Möbius map, makes this form positive semidefinite. Cauchy–Schwarz in the ordinary probability space gives

$$\|F\|_{\text{rad}} \leq \|F\|_2. \quad (3.25)$$

Let  $\mathcal{H}$  be its completion after quotienting out the null space.

For  $t \geq 0$ , let  $D_t$  dilate insertions inward by  $r = e^{-t}$ , including their primary weights:  $D_t\sigma(z) = r^{1/8}\sigma(rz)$ . The covariance formula shows that  $D_t D_u = D_{t+u}$ , that  $D_t$  is symmetric for (3.24), and that  $\|D_t F\|_2 = \|F\|_2$ . To verify the needed contraction without assuming a spectral representation, use symmetry and Cauchy–Schwarz:

$$\|D_t F\|_{\text{rad}}^2 = (F, D_{2t} F)_{\text{rad}} \leq \|F\|_{\text{rad}} \|D_{2t} F\|_{\text{rad}}.$$

Iteration, followed by (3.25), gives

$$\|D_t F\|_{\text{rad}} \leq \|F\|_{\text{rad}}^{1-2^{-k}} \|F\|_2^{2^{-k}}.$$

Letting  $k \rightarrow \infty$  proves contraction, and proves as well that  $D_t$  descends through the null space. For  $r \rightarrow 1$ , the spatial smearing associated with  $D_{-\log r}$  is  $r^{-15/8}\psi(\cdot/r)$ . It converges to  $\psi$  at every continuity point of  $\psi$ , hence almost everywhere, and the functions are uniformly bounded on a common compact set. Dominated convergence against the integrable fixed-order correlation bounds gives convergence of each smeared variable in every fixed finite  $L^p$ . Telescoping polynomial differences and using Hölder's inequality gives ordinary  $L^2$  convergence of every fixed smeared polynomial. By (3.25) this proves strong continuity on  $\mathcal{A}$ , then on  $\mathcal{H}$  by contraction. The self-adjoint contraction-semigroup theorem [5, Theorem 7.3.10, pp. 337–338] therefore gives

$$D_t = e^{-t\mathcal{D}}, \quad \mathcal{D} \geq 0, \quad (3.26)$$

with  $\mathcal{D}$  self-adjoint. The cited real-Hilbert-space theorem applies to the underlying real space; complex linearity of the semigroup makes its generator complex linear as well. Square symmetries and spin reversal commute with this semigroup.

### 3.5 The complex determinant identity and the scalar gap

**Lemma 3.5** (Charged sums at complex points). *Let  $z_1, \dots, z_{2m}$  be distinct complex points, with  $m \geq 1$ , and put*

$$W_z(u) = \prod_{i < j} |z_i - z_j|^{u_i u_j / 2}, \quad A_q(z) = \sum_{\substack{u_i \in \{-1, 1\} \\ \sum_i u_i = q}} W_z(u),$$

where an empty charge class has sum zero. Define

$$A_0^d(z) = \sum_{\sum_i u_i = 0} W_z(u) \left| \sum_i u_i z_i \right|^2.$$

Then

$$A_0^d A_0 = 4A_2^2. \quad (3.27)$$

If all points lie in the unit disk, then  $A_2/A_0 \leq 1$ , and all ratios  $A_q/A_0$  are bounded by one.

*Proof.* Set  $w_i = (\prod_{j \neq i} |z_i - z_j|)^{-1}$  and  $C_z = \prod_{i < j} |z_i - z_j|^{1/2}$ . For the positive subset  $I = \{i : u_i = 1\}$ , direct multiplication gives

$$W_z(u) = C_z |\Delta(z_I)|^2 \prod_{i \in I} w_i, \quad (3.28)$$

where  $\Delta$  is the Vandermonde determinant. Let  $Z_k$  be the sum of the determinant weights on the right, without  $C_z$ , over  $|I| = k$ ; set  $Z_0 = 1$ . For the positive measure  $\sum_i w_i \delta_{z_i}$  on this finite set of complex points, let  $p_j$  be the monic orthogonal polynomial of degree  $j$ , and let  $h_j = \sum_i w_i |p_j(z_i)|^2$ . Cauchy–Binet applied to the evaluation matrix gives

$$Z_k = h_0 \cdots h_{k-1}, \quad 1 \leq k \leq 2m. \quad (3.29)$$

The weight in (3.28) without  $C_z$  is also  $\prod_{i \in I, j \notin I} |z_i - z_j|^{-1}$ . It is unchanged under complementation. Thus  $Z_{2m-k} = Z_k$  and

$$h_m h_{m-1} = Z_{m+1}/Z_{m-1} = 1. \quad (3.30)$$

Consider the probability distribution on  $m$ -subsets proportional to these weights. Its inclusion probabilities are the minors of the orthogonal projection  $P$  onto polynomials of degree at most  $m-1$ , viewed on the finite weighted point set. This is another direct Cauchy–Binet identity: summing squared minors containing a prescribed set leaves the corresponding principal minor of  $P$ . Let  $M_z$  denote multiplication by  $z$  on that space. Expanding the absolute-square variance with the one- and two-point inclusion probabilities gives

$$\begin{aligned}\text{Var}\left(\sum_{i \in I} z_i\right) &= \text{tr}(PM_z^* M_z) - \text{tr}(PM_z^* PM_z) \\ &= \|(I - P)M_z P\|_{\text{HS}}^2 = h_m/h_{m-1}.\end{aligned}$$

The last equality uses the orthonormal polynomial basis: multiplication by  $z$  sends its first  $m-1$  vectors into  $\text{ran } P$ , and the last vector has complement equal to  $p_m/\sqrt{h_{m-1}}$ . This argument uses complex absolute-square variance; no assumption that the points lie on a real line is present.

Complementation shows that  $\mathbb{E} \sum_{i \in I} z_i = \frac{1}{2} \sum_i z_i$ . Since  $\sum_i u_i z_i = 2 \sum_{i \in I} z_i - \sum_i z_i$ , we obtain

$$\frac{A_0^d}{A_0} = 4 \frac{h_m}{h_{m-1}} = 4h_m^2, \quad \frac{A_2}{A_0} = \frac{Z_{m+1}}{Z_m} = h_m,$$

which proves (3.27). If  $|z_i| \leq 1$ , the minimizing property of monic orthogonal polynomials, applied to  $zp_{j-1}$ , gives  $h_j \leq h_{j-1}$ . Together with (3.30) this implies  $h_m \leq 1$ . Ratios  $Z_{m+j}/Z_m = h_m \cdots h_{m+j-1}$  are at most one, and complementation treats negative charges.  $\square$

**Proposition 3.6** (Radial spectral gap). *On the subspace of  $\mathcal{H}$  generated by even, quarter-turn invariant spin tests, the spectrum of  $\mathcal{D}$  below 3 consists of the constant state at 0 and one nonzero one-dimensional state at 1. There is a real linear functional  $\ell$  with  $|\ell(F)| \leq \|F\|_2$  such that, for real even quarter-turn invariant tests  $F, G$  of finite ordinary  $L^2$  norm and  $0 < r \leq 1$ ,*

$$|\mathbb{E}[(\Theta F)D_{-\log r} G] - \mathbb{E} F \mathbb{E} G - r \ell(F) \ell(G)| \leq r^3 \|F\|_2 \|G\|_2. \quad (3.31)$$

The assertion extends from smeared polynomials to bounded field tests by ordinary  $L^2$  approximation.

*Proof.* We compute first with an even number  $n = 2m$  of inner insertions  $z_i$  and  $n' = 2m'$  reflected outer insertions  $y_j$ . Both insertion sets are inside the unit disk and have distinct points. After including the dilation and reflection weights, the squared spin formula in Proposition 2.7 becomes

$$2^{-(n+n')/2} \sum_q r^{q^2/4} \sum_{\substack{u_i=q \\ \sum v_j=-q}} W_z(u) W_y(v) \prod_{i,j} |1 - rz_i \bar{y}_j|^{u_i v_j/2}. \quad (3.32)$$

For example, the power of  $r$  is the primary factor  $r^{n/4}$  in the squared expression times the internal-distance factor  $r^{(q^2-n)/4}$ . All powers of  $|y_j|$  from inversion cancel. This verifies the normalization of the radial degree.

Angular frequency refers here to the terms  $e^{ij\theta}$  produced by rotating one whole insertion set,  $z_i \mapsto e^{i\theta} z_i$ . Divide (3.32) by its constant term, and abbreviate the charged sums of the  $y$  variables by  $B_q, B_0^d$ . The  $q = \pm 2$  terms have first-order coefficient

$$p = 2(A_2/A_0)(B_2/B_0).$$

In the charge-zero term the first-order cross factor is proportional to  $\text{Re}[(\sum_i u_i z_i)(\overline{\sum_j v_j y_j})]$  and vanishes after summing either sign family. To find the angularly constant second-order coefficient, use

$$\log \prod_{i,j} |1 - rz_i \bar{y}_j|^{u_i v_j / 2} = -\frac{r}{2} \text{Re}(X \bar{Y}) - \frac{r^2}{4} \text{Re}(X_2 \bar{Y}_2) + O(r^3),$$

where  $X = \sum u_i z_i$ ,  $X_2 = \sum u_i z_i^2$ , and similarly for  $Y, Y_2$ . Squaring the linear term in the exponential contributes  $r^2 |X|^2 |Y|^2 / 16$  in angular frequency zero. Its normalized sum is

$$q_0 = \frac{A_0^d B_0^d}{16 A_0 B_0}.$$

Every other second-order term has angular frequency one or two: frequency one comes from the first cross correction in the  $q = \pm 2$  sector, and frequency two from the remaining charge-zero terms. Charges of absolute value at least four begin at  $r^4$ .

The correlation itself is the positive square root. Its scalar second-order coefficient is  $q_0/2 - p^2/8$ . By Lemma 3.5,  $q_0 = p^2/4$ , so this coefficient is zero. Averaging one insertion set over quarter turns kills the nonzero angular frequencies just listed. The first-order coefficient in the unsquared correlation is

$$(S(z_1, \dots, z_n) A_2 / A_0) (S(y_1, \dots, y_{n'}) B_2 / B_0). \quad (3.33)$$

It factors across the gap. An empty insertion set contributes only the constant state.

These computations survive smearing. For a fixed number of insertions and  $|r|$  sufficiently small, expand the real-analytic factors into their absolutely convergent power series in  $r$  and the angular variables. The ratios of charged sums are bounded by Lemma 3.5; the cross factors and their first three derivatives are bounded uniformly when  $|z_i|, |y_j| \leq 1$  and  $|r| \leq r_0 < 1$ . The remaining integrable dominating function is a constant times the product of the within-set spin correlations. Thus every even quarter-turn invariant smeared polynomial has the bilinear expansion

$$(F, D_{-\log r} G)_{\text{rad}} = \mathbb{E} \bar{F} \mathbb{E} G + r \overline{c(F)} c(G) + O_{F,G}(r^3), \quad (3.34)$$

with the linear functional  $c$  given by (3.33) and linear extension.

We explain why the polynomial-dependent remainder in this calculation implies the uniform bound in the proposition. In a self-pairing the spectral functional calculus [5, Theorem 6.4.1 and Section 6.5, especially Theorem 6.5.4] represents the left side of (3.34) as  $\int_{[0,\infty)} r^d d\mu_F(d)$  with  $\mu_F$  positive. Letting  $r \downarrow 0$  identifies its mass at zero as  $|\mathbb{E} F|^2$ . After subtracting it, divide by  $r$ . Positivity excludes mass on  $(0, 1)$  and identifies the mass at 1 as  $|c(F)|^2$ . After subtracting this atom as well, the remainder is positive and  $O_F(r^3)$ , which excludes mass on  $(1, 3)$ : any positive mass in a compact interval with upper endpoint below 3 would violate that bound. Density of the polynomial tests excludes these spectral intervals on the entire stated subspace.

Polarization and the two factorizations in (3.34) show that the projections at 0 and 1 have rank one, provided the latter is nonzero. It is nonzero already for a two-spin test smeared in two small disjoint regions: choose nonzero nonnegative smooth compactly supported bumps  $\psi_1, \psi_2$  in those regions and use  $\Phi(\psi_1)\Phi(\psi_2)$ . The coefficient is strictly positive since  $S(z_1, z_2)A_2/A_0 > 0$  for distinct points; square-group averaging preserves this positive scalar coefficient. Choose a unit real vector in that eigenspace and denote the corresponding coefficient by  $\ell$ . Then  $|\ell(F)| \leq \|F\|_{\text{rad}} \leq \|F\|_2$ . On the remaining spectral subspace, the norm of  $D_{-\log r}$  is at most  $r^3$ . Cauchy–Schwarz proves (3.31). Finally (3.25) and polynomial density give the asserted extension.  $\square$

In particular there exist bounded continuous, even, square-invariant tests with a nonzero degree-one coefficient. Start with the smeared two-spin test built from the smooth bumps in the proof and

average it under the square group. Continuous bounded truncations converge to this test in ordinary  $L^2$  and are functions of the same finite vector of smooth smears. Since the coefficient is continuous in this norm, a sufficiently large fixed truncation has nonzero coefficient. Choose the bumps in a fixed annular subset of the disk. Their square images remain in that annulus, so reflection keeps all transformed spatial smears smooth and compactly supported in a bounded annulus.

### 3.6 Radial coefficients for square supports

We first record how to use Proposition 3.6 for square supports. If a continuum test  $F$  is supported in a fixed disk of radius  $a_0$ , define its degree-one coefficient by

$$\Lambda(F) = a_0 \ell(D_{\log a_0} F),$$

taking  $a_0 \geq 1$  large enough that the dilated support is strictly inside the unit disk. This definition does not depend on  $a_0$ : the unit degree eigenstate scales by  $e^{-t}$  under  $D_t$ . In particular  $|\Lambda(F)| \leq C\|F\|_2$  on any fixed bounded family of supports. For a test  $G$  supported in a fixed annulus with inner radius  $b_0 > 0$ , choose a slightly smaller positive  $b < b_0$  and put

$$\Gamma(G) = b^{-1} \ell(\Theta D_{\log b} G).$$

Here a dilation with a negative parameter is simply the primary coordinate change on compactly supported tests; it is used before placing the test in the radial Hilbert space. The transformed test  $\Theta D_{\log b} G$  is supported strictly inside the unit disk. The same scaling calculation makes  $\Gamma$  independent of  $b$ . Write  $G_R = D_{-\log R} G$  for the outward rescaling of  $G$ . For fixed support shapes and sufficiently large  $R$ , Proposition 3.6 gives

$$\left| \text{Cov}(F, G_R) - R^{-1} \Lambda(F) \Gamma(G) \right| \leq C R^{-3} \|F\|_2 \|G\|_2. \quad (3.35)$$

Indeed the radius ratio after the preceding two normalizations is  $a_0/(bR)$ ; its first-order factor is exactly the one displayed. The constants depend only on the fixed support enlargements, and not on  $R$ . All these assertions hold after square-group averaging.

### 3.7 One lattice measurement for every bounded test

Fix once and for all a bounded continuous test  $W$  of finitely many smeared fields in an annulus outside the unit disk, even and square-invariant, with

$$g := \Gamma(W) \neq 0. \quad (3.36)$$

Such a test is obtained by reflecting the bounded test following Proposition 3.6, whose support may be kept away from zero. Also fix a bounded continuous, even, square-invariant test  $V$  from the same smooth-bump construction, supported strictly inside  $[-1, 1]^2$ , with  $\Lambda(V) \neq 0$ . The spatial smears defining  $V$  and  $W$  are smooth, compactly supported, and fixed once and for all. For an integer  $t$ , denote by  $W_t$  and  $V_t$  their lattice versions at scale  $t$ . Each smeared coordinate is formed by the normalized sum  $t^{-15/8} \sum_x \psi(x/t) \sigma_x$ , including the fixed lattice normalization from Proposition 2.7, using those same fixed spatial functions at every scale and averaging over the square group if necessary. Their norms are uniformly bounded and their joint laws, together with other fixed smears, converge to their continuum counterparts.

Choose a fixed large integer  $H$ , to be specified after  $L$ , and define for every  $s$

$$\ell_s(f) = \frac{H}{g} \text{Cov}_0(f, W_{Hs}). \quad (3.37)$$

This is a linear functional annihilating constants. It depends on a full-plane law and fixed bulk tests only.

We make the uniform meaning of this measurement explicit. For any sequence of functions  $f_s$  with  $\|f_s\|_q \leq 1$ , choose their midrange representatives, so  $\|f_s\|_\infty \leq 1$ . For any fixed finite list of large radius ratios, including  $H$ , Lemma 3.4 gives a common continuum proxy  $F$  along a subsequence, with  $\|F\|_\infty \leq 1$ , for all the required pairings. Applying (3.35) at ratio  $H$  gives

$$\ell_s(f_s) = \Lambda(F) + O(H^{-2}) + o(1) + O(Ht). \quad (3.38)$$

The  $t$  is the chosen replacement error, the  $o(1)$  is the subsequent mesh error, and the constants are uniform in the sequence. This is an estimate for the pairings of a proxy, not a claim that  $f_s$  itself converges as a random field test. The same proxy must be used for the finite list, which is why the simultaneous assertion in Lemma 3.4 was retained.

Taking  $t$  arbitrarily small after fixing  $H$  proves a uniform bound on  $\ell_s$  for all sufficiently large  $s$ , independent of large  $H$ . Likewise, with the fixed test  $V_s$ ,

$$\ell_s(V_s) = \Lambda(V) + O(H^{-2}) + o(1).$$

For all sufficiently large fixed  $H$  and then large  $s$  this is bounded away from zero by  $|\Lambda(V)|/2$ . Therefore

$$e_s = V_s / \ell_s(V_s) \quad (3.39)$$

is a uniformly bounded right inverse. Both constants are independent of the subsequently prescribed error tolerance and of  $L$ .

### 3.8 Boundary uniformity and compatibility across scales

The measured coefficients and their bounded right inverses are now fixed. It remains to establish the two operator estimates. We first turn an arbitrary boundary assignment into a bounded test across the annulus. Let  $\tau$  be any fixed assignment on  $\partial B_{Ls}$  and take the intermediate contour  $I_s = \partial B_{\lfloor Ls/2 \rfloor}$ . Its marginal under that boundary condition has a density  $b_{s,\tau}$  relative to its full-plane marginal satisfying

$$K^{-1} \leq b_{s,\tau} \leq K, \quad \mathbb{E}_0 b_{s,\tau} = 1, \quad (3.40)$$

by Proposition 2.1. The constant  $K$  is independent of  $L, s, \tau$ : the contour-to-boundary ratio is fixed. The spatial Markov property gives the exact identity

$$\mathbb{E}_0[f \mid \sigma_{\partial B_{Ls}} = \tau] - \mathbb{E}_0 f = \text{Cov}_0(f, b_{s,\tau}). \quad (3.41)$$

When  $f$  is even and square-invariant, we may average  $b_{s,\tau}$  under those symmetries without changing this pairing or its bounds.

Consider any sequence of functions and boundary assignments in (3.41). Apply bounded replacement to the inner support, to a fixed proportional band around  $I_s$ , and to the measuring test at ratio  $H$ . The enlargements around the inner square have fixed size in units of  $s$ ; the band enlargement has fixed size in units of  $Ls$  and stays a positive distance from zero in those coordinates. For  $L$  large enough they are disjoint. The resulting proxies  $F, G$  satisfy  $\|F\|_\infty \leq 1$  and  $\|G\|_\infty \leq K$ . Equation (3.35) therefore implies

$$L^2 |\text{Cov}(F, G_L)| \leq CL |\Lambda(F)| + C/L.$$

Substituting (3.38), and letting the replacement accuracy tend to zero after fixing  $L, H$ , proves

$$\limsup_{s \rightarrow \infty} \sup_{\|f\|_q \leq 1} \left( \|A_s f\|_q - CL|\ell_s(f)| \right) \leq C/L + CL/H^2. \quad (3.42)$$

For clarity, the centered supremum in (3.41) bounds  $\|A_s f\|_q$  from above. Uniformity in (3.42) follows by contradiction: any failure supplies a sequence of bounded functions and a maximizing boundary assignment, and the preceding common-proxy argument applies to that sequence. Thus no compactness of the space of microscopic boundary assignments is being assumed.

The compatibility estimate uses an exact Markov identity before any approximation. Since  $W_{HLs}$  is supported outside  $B_{Ls}$ ,

$$\begin{aligned} \ell_{Ls}(A_s f) &= \frac{HL^2}{g} \text{Cov}_0(\mathbb{E}_0[f \mid \sigma_{\partial B_{Ls}}], W_{HLs}) \\ &= \frac{HL^2}{g} \text{Cov}_0(f, W_{HLs}). \end{aligned} \quad (3.43)$$

The second equality is conditional independence of the inside and outside given the boundary; it is not a formal composition of unrelated conditional expectations. Use a common proxy for  $f$  in its pairings at ratios  $H$  and  $HL$ . Equation (3.35) then gives respectively

$$\begin{aligned} \ell_s(f) &= \Lambda(F) + O(H^{-2}) + o(1) + O(Ht), \\ \ell_{Ls}(A_s f) &= L\Lambda(F) + O((LH^2)^{-1}) + o(1) + O(HL^2t). \end{aligned}$$

After fixing  $L, H$  and taking the accuracies in their stated order, we obtain

$$\limsup_{s \rightarrow \infty} \sup_{\|f\|_q \leq 1} |\ell_{Ls}(A_s f) - L\ell_s(f)| \leq CL/H^2. \quad (3.44)$$

The constants in these two displays depend on the fixed chart shapes and measuring test, and are independent of  $L, H$  once both are large.

*Proof of Theorem 3.1.* Choose  $L$  large enough that the term  $C/L$  in (3.42) is less than  $a/4$  and the fixed support enlargements are disjoint. Next choose an integer  $H > L$  large enough that all terms  $CL/H^2$  are less than  $a/4$ , and that the lower bound needed in (3.39) holds. Choose every replacement accuracy small compared with the now fixed amplification factors  $H, L^2, HL^2$ . In each replacement construction, choose the window, its finite subdivision, the observation strength, and the coarse partition before taking the mesh small. Finally choose  $s_0$  so that the uniform upper limits (3.42) and (3.44) are within their remaining error margins, and so that the norm and right-inverse bounds hold for every  $s \geq s_0$ . Homogeneity gives the estimates for arbitrary  $f$ .

All square contours and cell boundaries used here are specified by integer rounding. They remain in the same fixed buffered charts; their rescaled locations converge to the prescribed contours and cells. Changes of cell indicators are confined to shrinking neighborhoods of their finitely many polygonal edges; their normalized lattice counting measure tends to zero. The uniform  $L^2$  bounds on fixed-order rescaled correlation kernels and Cauchy–Schwarz therefore make the resulting smearing errors tend to zero. Spatial test functions and their finite menus are fixed before the mesh limit. Only these specified geometric roundings and the convergent cell-normalization scalars depend on the mesh, not arbitrary varying Riemann-integrable spatial functions. The separate line-continuity estimate controls the auxiliary square contours. Consequently changing an auxiliary contour by a bounded number of lattice layers affects none of the limiting estimates or their uniform thresholds. The conditioning convention in (3.2), however, is fixed throughout, so the Markov identity (3.43) is exact at every scale.  $\square$

For the analytic construction in the next section one may complexify these spaces. The complex quotient norm is  $\|f\|_q = \inf_{c \in \mathbb{C}} \|f - c\|_\infty$ ; its linear mean-zero representative has norm at most  $2\|f\|_q$ . Extend  $A_s$  and  $\ell_s$  complex linearly. Applying the real estimates to real and imaginary parts changes universal constants by at most a factor two, and the arbitrarily small error  $a$  can be chosen smaller beforehand. Thus the same conclusions, with harmless fixed constants, hold for the complex spaces used to justify analyticity.

## 4 An exact scale flow and the temperature curve

Theorem 3.1 controls the linear effect of moving a bounded potential to a larger square. We now implement that operation by an exact change of the bulk interaction. Its nonlinear error will be quadratic, and its only expanding direction will be removed by choosing the temperature. After estimating the residual caused by periodic evaluation, reversing the flow will couple the resulting torus law to the pure law by edits in independently marked squares.

The bulk scale parameters will use only the pure full-plane law  $\mu_0$ , fixed scale ratios, and the given interaction  $U$ , independently of the target domain and its boundary data. All square coordinates are lattice coordinates. We write

$$B_r(x) = \{z \in \mathbb{Z}^2 : \|z - x\|_\infty \leq r\}, \quad B_r^\circ(x) = \{z \in \mathbb{Z}^2 : \|z - x\|_\infty < r\}, \\ \partial B_r(x) = B_r(x) \setminus B_r^\circ(x).$$

Conditional integration over a square means integration over its open square  $B_r^\circ(x)$ . In the pure model its exterior enters through  $\partial B_r(x)$ . Enlarging an edit square by a fixed number of lattice steps covers any alternate convention for this outer layer.

### 4.1 The finite-state mechanism

The conditional expectation in the scale flow has an exact probabilistic meaning. On a finite spin space, write  $d\mu_t = Z_t^{-1} e^{H_t} d\mu_{\text{ref}}$ , with a fixed strictly positive reference law. Fix update sets  $I_x$  and bounded real functions  $f_x$ , and let  $P_{t,x}$  condition under  $\mu_t$  on the complement of  $I_x$ . If the log-density path satisfies

$$\dot{H}_t = c \sum_x (P_{t,x} f_x - f_x), \quad c \geq 0,$$

then each summand has zero  $\mu_t$ -mean. Differentiating  $Z_t$  therefore gives the probability derivative in the following lemma. Adding a constant to  $\dot{H}_t$  gives the same normalized derivative. The lemma shows why moving a local function toward its conditional expectation can be reversed by refreshes at rates independent of the current spins.

**Lemma 4.1** (Reverse generator identity). *Let  $\mu_t$  be a differentiable family of strictly positive probability laws on a finite spin space, and let  $I_x$  be specified update sets. Let  $f_x$  be fixed bounded real functions, with  $\|f_x\|_\infty \leq a$ ,  $a \geq 0$ , and let  $P_{t,x}$  denote conditional expectation under  $\mu_t$  given the complement of  $I_x$ . Suppose*

$$\dot{\mu}_t(\sigma) = \mu_t(\sigma) c \sum_x (P_{t,x} f_x(\sigma) - f_x(\sigma)), \quad c \geq 0. \quad (4.1)$$

If  $a > 0$ , define the refresh law

$$Q_{t,x}(d\eta_{I_x} \mid \sigma_{I_x^c}) = \mu_t(d\eta_{I_x} \mid \sigma_{I_x^c}) \left[ 1 + \frac{f_x(\eta_{I_x} \sigma_{I_x^c}) - P_{t,x} f_x(\sigma)}{2a} \right]. \quad (4.2)$$

It is a probability law. Updating set  $I_x$  from this law at rate  $2ac$ , independently for each  $x$ , has generator  $\mathcal{L}_t$  with  $\mathcal{L}_t^* \mu_t = -\dot{\mu}_t$ . Thus using  $\mathcal{L}_{1-t}$  from time zero to one carries the law at time one exactly to the law at time zero. If  $a = 0$ , the derivative in (4.1) vanishes and no refresh is needed.

*Proof.* Since both  $f_x$  and its conditional expectation lie in  $[-a, a]$ , the bracket in (4.2) lies in  $[0, 2]$ . Its conditional integral equals one. For a configuration  $\sigma$ , the incoming mass of an  $x$ -refresh applied to  $\mu_t$  is

$$\mu_t(\sigma) \left[ 1 + \frac{f_x(\sigma) - P_{t,x} f_x(\sigma)}{2a} \right].$$

Indeed, summing the old interior configuration first leaves its exterior marginal, which is then multiplied by the refresh conditional law. Subtract the outgoing mass  $\mu_t(\sigma)$  and multiply by  $2ac$ . The result is  $c\mu_t(\sigma)(f_x - P_{t,x} f_x)$ . Summing over  $x$  proves the adjoint identity, including its sign and normalization. The finite-state forward equation has a unique solution, so the reversed family is the law of the stated time-inhomogeneous chain. The zero case is immediate.  $\square$

We will realize this operation on translation-invariant bulk interaction densities at lattice scale  $s$ , with  $c = s^{-2}$  and update squares of radius  $LS$  for a fixed scale ratio  $L$ . The next construction controls the new interactions generated by their conditional expectations. Its periodic evaluation has a small residual relative to the finite-state identity; that residual will be estimated before the lemma is used on tori.

## 4.2 Spaces of labeled potentials

An interaction created by conditional integration need not remain supported in one square. We retain a covering list as part of each coefficient, rather than estimating an interaction only by the diameter of its support.

For an integer  $s \geq 1$ , a rooted list of length  $m \geq 1$  is an ordered tuple

$$\ell = (z_1, \dots, z_m), \quad z_1 = 0, \quad z_i \in \mathbb{Z}^2,$$

such that the graph joining  $i$  and  $j$  when  $B_s(z_i) \cap B_s(z_j) \neq \emptyset$  is connected. Repetitions of centers are allowed. Put  $S_s(\ell) = \bigcup_i B_s(z_i)$ . A coefficient on  $\ell$  is a function of the spins in  $S_s(\ell)$ , modulo constants. Whenever a representative is needed, we use the bounded linear choice

$$[v] \mapsto v - \mathbb{E}_0 v.$$

If  $\|v\|_q = \inf_{c \in \mathbb{C}} \|v - c\|_\infty$ , this representative has norm at most  $2\|v\|_q$ . Thus working with these representatives gives an equivalent norm to the quotient norm used in Theorem 3.1. The same choice works for complex coefficients.

For  $b > 0$ , let  $\mathcal{A}_{s,b}$  be the space of families  $V = (v_\ell)_\ell$  of these centered coefficients with

$$\|V\|_{s,b} := \sum_\ell e^{b(|\ell|-1)} \|v_\ell\|_\infty < \infty. \quad (4.3)$$

It is a Banach space: it is a weighted  $\ell^1$  sum of finite-dimensional spaces. For each fixed list length there are only finitely many lists, since every center is within  $2s(m-1)$  of the root. We use its closed subspace invariant under spin flip and the square group; the group acts on the lists and their functions together. In particular its length-one coefficient, denoted  $f = \pi_s V$ , is an even square-invariant function on  $B_s$ . Different lists may describe the same function. No independence of these descriptions is asserted or required.

The translation convention is fixed by the root: if  $(\tau_x \sigma)_z = \sigma_{x+z}$ , the log-density addition represented by  $V$  is

$$\mathcal{H}_s(V) = s^{-2} \sum_{x \in \mathbb{Z}^2} \sum_{\ell} v_{\ell}(\tau_x \sigma). \quad (4.4)$$

Only conditional differences, or the corresponding finite torus sum, are used; the full-plane sum itself is not assigned a numerical value. Recentring a local coefficient by a lattice translation leaves this translation sum unchanged. We will not need such recentring: our changes of covering lists always keep the root at zero. Additive constants in the coefficients likewise have no effect on any normalized measure.

Fix an integer  $L \geq 9$ . For an exterior configuration  $\eta$  outside  $I = B_{Ls}^\circ$ , let

$$H_{I,s}(V)(\sigma_I, \eta) = s^{-2} \sum_{\ell} \sum_{x: (x+S_s(\ell)) \cap I \neq \emptyset} v_{\ell}(\tau_x(\sigma_I \eta)). \quad (4.5)$$

Including terms that happen to be constant in  $\sigma_I$  is harmless. For a fixed list of length  $m$ , the number of translations in this sum is at most  $C_L m s^2$ : for at least one of its boxes to meet  $I$ , its center must lie in  $B_{(L+1)s}$ , which has at most  $C_L s^2$  sites. Consequently

$$\|H_{I,s}(V)\|_{\infty} \leq C_L \sum_{\ell} |\ell| \|v_{\ell}\|_{\infty}. \quad (4.6)$$

The constants in this and subsequent one-step estimates do not depend on  $s$ .

Let  $P_{Ls}^0$  be the pure conditional expectation over  $I$ , and define

$$P_{Ls}^V f(\eta) = \frac{P_{Ls}^0(f \exp H_{I,s}(V))(\eta)}{P_{Ls}^0(\exp H_{I,s}(V))(\eta)}. \quad (4.7)$$

For real  $V$ , this is the conditional expectation in the perturbed specification. For complex  $V$  with a sufficiently small right-hand side in (4.6), the denominator is uniformly separated from zero. Indeed its distance from 1 is at most  $\exp(\|H_{I,s}(V)\|_{\infty}) - 1$ . Equation (4.7) is then defined for every exterior spin configuration, including configurations of zero full-plane probability.

### 4.3 The conditional differential equation

During one step we hold the initial single-square function  $f$  fixed and solve, modulo constants,

$$\frac{d}{dt} V(t) = -f + P_{Ls}^{V(t)} f, \quad V(0) = V, \quad 0 \leq t \leq 1. \quad (4.8)$$

The right-hand side is interpreted as a labeled family by the expansion below. This is the translation-density version of the finite-state logarithmic derivative above: applying  $\mathcal{H}_s$  sums its translates with coefficient  $s^{-2}$ . Holding  $f = \pi_s V$  fixed is essential: it cancels the original single-square coefficient exactly at time one.

**Lemma 4.2** (Analytic conditional flow). *Fix  $L \geq 9$  and  $d > 0$ . There are  $r > 0$  and  $C < \infty$ , independent of the integer  $s$ , such that for  $\|V\|_{s,d} < r$ , Equation (4.8) has a specified analytic solution taking values in  $\mathcal{A}_{s,d/2}$ . Uniformly in  $0 \leq t \leq 1$ ,*

$$\sum_{\ell} |\ell| e^{(d/2)(|\ell|-1)} \|v_{\ell}(t)\|_{\infty} \leq C \|V\|_{s,d}. \quad (4.9)$$

Its endpoint has the form

$$V(1) = (V - f) + P_{Ls}^0 f + \mathcal{R}_s(V), \quad \|\mathcal{R}_s(V)\|_{s,d/2} \leq C\|V\|_{s,d}^2. \quad (4.10)$$

On a smaller ball the remainder is analytic and satisfies  $\|D\mathcal{R}_s(V)\| \leq C\|V\|_{s,d}$  and  $\|D^2\mathcal{R}_s(V)\| \leq C$ , with these operator norms from  $\mathcal{A}_{s,d}$  to  $\mathcal{A}_{s,d/2}$ . The construction preserves real coefficients and the stated symmetries.

*Proof.* Choose a connected covering list  $\mathcal{C}$  of  $B_{Ls}$  by  $c_L \geq 2$  squares of radius  $s$ , all contained in  $B_{Ls}$ , including the root square. Such a list can be obtained from centers  $(as, bs)$ ,  $-L+1 \leq a, b \leq L-1$ . It has size depending only on  $L$ , and its union is exactly  $B_{Ls}$ .

Expand the numerator in (4.7) in its exponential series. Expand the reciprocal denominator as

$$\frac{1}{P^0 e^H} = \sum_{j \geq 0} (-P^0 (e^H - 1))^j.$$

The absolute scalar series is bounded by

$$\|f\|_\infty \frac{e^h}{2 - e^h}, \quad h = \|H\|_\infty < \log 2.$$

Its coefficient of degree  $p$  is bounded by  $K_0 K_1^p \|f\|_\infty$  for fixed numerical  $K_0, K_1$ . Each degree- $p$  coefficient is a sum of products of pure conditional expectations with precisely  $p$  inserted translates of coefficients of  $V$ . It depends only on the spins in the covering list  $\mathcal{C}$  and the inserted lists. Give it the concatenated list consisting of  $\mathcal{C}$ , followed by all inserted lists, retaining repeated boxes. An inserted list meets the integrated square and hence is connected to  $\mathcal{C}$ . If its input lengths are  $m_1, \dots, m_p$ , its output length is exactly

$$n = c_L + \sum_{i=1}^p m_i. \quad (4.11)$$

Center each output coefficient with the fixed linear projection. This costs a factor of at most two in its sup norm. The placement bound (4.6) supplies a factor  $C_L m_i$  per insertion; the factors  $s^2$  have canceled with the translation normalization. Taking the square-group average of this rule makes it equivariant, without increasing a sum of absolute norms. Spin flip commutes with every pure conditional integration.

Here is a convergent majorant, including the dependence on list size. Let  $a_m(t)$  be the sum of nonnegative coefficient majorants on lists of length  $m$ , and put

$$X_b(t) = \sum_{m \geq 1} m e^{b(m-1)} a_m(t), \quad Y_b(t) = \sum_{m \geq 1} m(m-1) e^{b(m-1)} a_m(t).$$

The majorant includes the absolute values of the initial coefficients and of both fixed forcing terms  $-f$  and  $P^0 f$ . By (4.11), for  $d/2 \leq b \leq d$  the contribution of terms with  $p \geq 1$  insertions to its  $X_b$  derivative is at most

$$K \|f\|_\infty K^p \left[ c_L X_b^p + p(X_b + Y_b) X_b^{p-1} \right], \quad (4.12)$$

where  $K = K_{L,d}$ . To verify this bound, the exponential weight of an output is the product of the input weights times  $e^{b(c_L - 1 + p)}$ , absorbed in  $K^{p+1}$ . Its additional size factor  $n$  is the sum of  $c_L$  and the  $m_i$ . Choosing the latter term at the  $i$ -th insertion replaces one factor  $\sum m e^{b(m-1)} a_m = X_b$  by  $\sum m^2 e^{b(m-1)} a_m = X_b + Y_b$ . This proves (4.12), including the unbounded size factor.

Set  $b(t) = 3d/4 - dt/8$ , reserving a final exponent gap  $b(1) - d/2 = d/8$ . As long as  $KX_{b(t)} \leq 1/2$ , summing the geometric series and its derivative in (4.12) gives

$$D^+ X_{b(t)}(t) \leq -\frac{d}{8} Y_{b(t)}(t) + C_{L,d} \|f\|_\infty (1 + Y_{b(t)}(t)). \quad (4.13)$$

Furthermore

$$X_{3d/4}(0) \leq \left( \sup_{m \geq 1} m e^{-d(m-1)/4} \right) \|V\|_{s,d}.$$

Choose the input radius so that  $C_{L,d} \|f\|_\infty \leq d/16$  and the resulting bound  $X_{b(t)}(t) \leq X_{3d/4}(0) + C_{L,d} \|f\|_\infty t$  is strictly below  $1/(2K)$ . A first-exit argument then gives this bound through time one. In particular it gives (4.9).

For completeness, the same estimates construct the solution rather than only bounding a hypothetical one. First retain only output lists of length at most  $M$ . By (4.11), every nonlinear coefficient equation at length  $n$  uses only coefficients of lengths strictly smaller than  $n$ ; only the fixed forcing occurs at length one or at the fixed cover length. The truncated equations therefore have successive explicit integral solutions. Their coefficients are polynomials in the finitely many input coefficients of lengths at most  $M$ . Truncations agree on all smaller lengths. The corresponding nonnegative triangular majorants obey (4.13) and increase to a majorant for all coefficients. Their tails satisfy, uniformly on the complex input ball,

$$\sup_{0 \leq t \leq 1} \sum_{m > M} e^{(d/2)(m-1)} a_m(t) \leq \frac{C \|V\|_{s,d}}{M+1}.$$

Thus the truncations converge uniformly into  $C([0, 1], \mathcal{A}_{s,d/2})$ . They are holomorphic polynomials in finite-dimensional projections of the input, so their locally uniform limit is analytic. The reserved exponent gap also gives uniform convergence of the derivatives in  $\mathcal{A}_{s,d/2}$ : the vector field is bounded in  $\mathcal{A}_{s,5d/8}$  by  $C \|f\|_\infty [1 + \sum_{p \geq 1} (KX_{5d/8})^p]$ , so its tail above length  $M$ , measured at exponent  $d/2$ , is at most  $C \|f\|_\infty e^{-dM/8}$ . Absolute convergence permits termwise conditional integration and time differentiation. This proves (4.8); uniqueness for this labeled expansion follows successively in each length. This construction is also valid for real data and commutes with symmetry averaging.

Integrating the two fixed forcing terms gives  $(V - f) + P^0 f$  at time one. Every remaining term contains one factor  $f$  and at least one coefficient of  $V(t)$ . Its  $\mathcal{A}_{s,d/2}$  norm is bounded by  $C \|f\|_\infty \sum_{p \geq 1} (KX_{d/2}(t))^p$ , which is  $O(\|V\|_{s,d}^2)$ . Integration proves (4.10). The same analytic bound on a complex ball, followed by the Cauchy estimates on a smaller ball, gives the two derivative estimates. All constants used above are independent of  $s$ .  $\square$

The loss of exponential weight has a specific purpose: it pays for the number of placements of a long interaction. The next geometric step restores that weight when the scale is increased.

**Lemma 4.3** (Compression of covering lists). *For every integer  $L \geq 9$ , there is a linear relabeling map  $C_s : \mathcal{A}_{s,d/2} \rightarrow \mathcal{A}_{Ls,d}$  that leaves the local functions and their roots unchanged and satisfies*

$$\|C_s W\|_{Ls,d} \leq \|W\|_{s,d/2}.$$

*If a declared union is contained in  $B_{Ls}$ , it is relabeled as a single square. The map commutes with the specified symmetries. The rewriting map  $\mathcal{S}_s = L^2 C_s$  preserves the translation density:*

$$\mathcal{H}_{Ls}(\mathcal{S}_s W) = \mathcal{H}_s(W). \quad (4.14)$$

*On input lists of length at least two it also satisfies*

$$\|\mathcal{S}_s W\|_{Ls,d} \leq L^2 e^{-d/2} \|W\|_{s,d}. \quad (4.15)$$

*Proof.* Choose a spanning tree of the connected box graph using its vertex labels, and traverse its doubled edges starting at the root. The resulting walk has  $2(m-1)$  steps, each of sup-norm length at most  $2s$ . Put  $q = \lfloor (L-1)/2 \rfloor$ , and retain the centers at times  $0, q, 2q, \dots$ , through the last such time in the walk. Every old center occurs in the walk and is at distance at most  $2s(q-1)$  from the preceding retained center. Its radius- $s$  square is therefore contained in the radius- $Ls$  square at that retained center. Consecutive retained squares overlap. Removing repetitions leaves a connected covering list, containing the root, of length  $M$  with

$$M-1 \leq \left\lfloor \frac{2(m-1)}{q} \right\rfloor \leq \frac{m-1}{2}.$$

In the final inequality we used  $q \geq 4$ . If the original union fits in  $B_{Ls}$ , choose that single square instead. The same inequalities hold. The procedure uses only the labeled graph, so it can be chosen square-equivariantly; alternatively one can average its conjugates under the square group. Such averaging preserves all norm bounds.

Relabeling a function does not change it. The inequality  $e^{d(M-1)} \leq e^{(d/2)(m-1)}$ , summed over lists, proves the first bound. For  $m \geq 2$ , comparison with the original weight gives the extra factor  $e^{-d(m-1)/2} \leq e^{-d/2}$ , proving (4.15). Finally, the factor  $L^2$  cancels the change from  $s^{-2}$  to  $(Ls)^{-2}$  in the translation normalization, which proves (4.14) term by term.  $\square$

Define the one-step map  $\Phi_s(V) = \mathcal{S}_s V(1)$ . Its linearization is

$$T_s V = A_s f + \mathcal{S}_s(V - f), \quad A_s f = L^2 P_{Ls}^0 f, \quad f = \pi_s V, \quad (4.16)$$

where the first term is a single-square coefficient at scale  $Ls$ . Indeed the pure conditional term is supported in the fixed cover of  $B_{Ls}$ , so the special single-square rule of Lemma 4.3 applies. Lemmas 4.2 and 4.3 give

$$\Phi_s(V) = T_s V + R_s(V), \quad \|R_s(V)\|_{Ls,d} \leq C \|V\|_{s,d}^2, \quad \|DR_s(V)\| \leq C \|V\|_{s,d}, \quad (4.17)$$

uniformly in  $s$ , on a common ball. The large, but fixed, factor  $L^2$  is included in  $C$ .

#### 4.4 A decaying graph for the nonautonomous maps

We next solve the recurrence  $V_{k+1} = \Phi_{s_k}(V_k)$ . The spaces and maps vary with  $k$ , so we give the sequence-space argument rather than appeal to an autonomous stable-manifold theorem.

Write  $s_k = L^k s_0$  and  $\mathcal{X}_k = \mathcal{A}_{s_k,d}$ . Extend the functional  $\ell_{s_k}$  of Theorem 3.1 to  $\mathcal{X}_k$  by  $\ell_k V = \ell_{s_k}(\pi_{s_k} V)$ . The same theorem, and the equivalence of our norm with the quotient norm, allow the following choices. First choose its target error  $a > 0$  sufficiently small, then  $L$  sufficiently large, then the farther measurement ratio  $H$ , approximation accuracies, and  $s_0$  in the order prescribed there. All sufficiently large  $s$ , including every  $s_k$ , then satisfy

$$\|A_s f\|_\infty \leq C_1 L |\ell_s f| + a \|f\|_\infty, \quad |\ell_{Ls} A_s f - L \ell_s f| \leq a \|f\|_\infty. \quad (4.18)$$

Constants have been enlarged, and the input error decreased, to account for centering representatives. The norms of  $\ell_s$  and the norms of some right inverses are bounded independently of  $s$  and  $L$ . After fixing  $L$ , take  $d$  large enough that  $\kappa = L^2 e^{-d/2}$  is as small as needed. Increasing  $s_0$  further does not affect any of these bounds. This freedom will permit the fixed pure annulus estimates in Section 5 to be imposed before we construct the trajectory.

**Lemma 4.4** (Decaying sequence graph). *With the choices just described, there are bounded single-square vectors  $w_k \in \mathcal{X}_k$ ,  $\ell_k w_k = 1$ , and a real analytic function  $h$  on a neighborhood of zero in  $\ker \ell_0$ , with  $h(0) = 0$ , such that each sufficiently small vector*

$$V_0 = h(u_0)w_0 + u_0, \quad u_0 \in \ker \ell_0, \quad (4.19)$$

*has a trajectory  $V_{k+1} = \Phi_{s_k}(V_k)$  satisfying*

$$\sup_{k \geq 0} 2^k \|V_k\|_{s_k, d} \leq C \|u_0\|_{s_0, d}.$$

*In a sufficiently small ball these are exactly the initial conditions with a trajectory of finite sufficiently small norm  $\sup_k 2^k \|V_k\|_{s_k, d}$ . The tangent space of the graph at zero is exactly the set of initial conditions whose linear trajectories  $Z_{k+1} = T_{s_k} Z_k$  have finite norm  $\sup_k 2^k \|Z_k\|_{s_k, d}$ .*

*Proof.* Take a bounded single-square right inverse  $w_0$ . Recursively define

$$a_k = \ell_{k+1} T_{s_k} w_k, \quad w_{k+1} = a_k^{-1} T_{s_k} w_k.$$

The vectors remain single-square by (4.16). If  $\|w_k\| \leq M$ , Equation (4.18) gives

$$|a_k - L| \leq aM, \quad \|w_{k+1}\| \leq \frac{C_1 L + aM}{L - aM}.$$

Choose a fixed  $M$  larger than the initial right-inverse bound and  $2C_1 + 1$ , and choose the parameters to make the displayed ratio at most  $M$ . Induction gives  $\sup_k \|w_k\| \leq M$  and  $a_k \geq L/2 \geq 4$ . We have used real vectors, so the scalar multipliers are positive; their complexifications use the same fixed multipliers.

Every  $V_k$  has the unique decomposition  $V_k = \alpha_k w_k + u_k$ , with  $\alpha_k = \ell_k V_k$  and  $u_k \in \ker \ell_k$ . These coordinates have norms uniformly equivalent to  $\|V_k\|$ . For  $u \in \ker \ell_k$ , its single-square part has  $\ell_{s_k} \pi_{s_k} u = 0$ . Consequently (4.18) and (4.15) imply

$$\|T_{s_k} u\| \leq (a + \kappa) \|u\|.$$

Set

$$b_k u = \ell_{k+1} T_{s_k} u, \quad D_k u = T_{s_k} u - w_{k+1} b_k u.$$

Thus  $D_k : \ker \ell_k \rightarrow \ker \ell_{k+1}$ . The uniform functional and vector bounds give  $\|b_k\| + \|D_k\| \leq C(a + \kappa)$ . Choose  $a$  and then  $\kappa$  so that  $\|D_k\| \leq 1/8$ . The coordinate form of the linear recurrence is exactly triangular:

$$\alpha_{k+1} = a_k \alpha_k + b_k u_k, \quad u_{k+1} = D_k u_k. \quad (4.20)$$

There is no term from  $\alpha_k$  into the second equation, by the definition of  $w_{k+1}$ .

We record the inverse of the inhomogeneous recurrence. Given  $u_0$  and forcing terms  $g_k \in \mathbb{C}$ ,  $v_k \in \ker \ell_{k+1}$  with finite norm  $\sup_k 2^k \max\{|g_k|, \|v_k\|\}$ , its only decaying solution is

$$u_k = D_{k-1} \cdots D_0 u_0 + \sum_{j=0}^{k-1} D_{k-1} \cdots D_{j+1} v_j, \quad (4.21)$$

$$\alpha_k = - \sum_{j=k}^{\infty} \frac{b_j u_j + g_j}{a_k a_{k+1} \cdots a_j}. \quad (4.22)$$

An empty product in these expressions is the identity. The first formula is ordinary forward substitution. Iterating the scalar equation to time  $n$ , dividing by  $a_k \cdots a_n$ , and letting  $n \rightarrow \infty$  gives

the second; the terminal term vanishes for a decaying sequence. Conversely, absolutely convergent sums in (4.21)–(4.22) satisfy both equations. In the norm

$$\|(\alpha, u)\|_* = \sup_{k \geq 0} 2^k \max\{|\alpha_k|, \|u_k\|\},$$

they are a bounded linear function of  $u_0$  and the forcing norm  $\sup_k 2^k \max\{|g_k|, \|v_k\|\}$ . For example the forward convolution is bounded by the geometric sum  $2 \sum_{j \geq 0} (2/8)^j$ , and the backward convolution by  $\sum_{j \geq 0} 4^{-j-1} 2^{-j}$ . These constants are uniform in  $k$  and in the lattice scales.

For the nonlinear recurrence use the forcing

$$g_k = \ell_{k+1} R_{s_k}(\alpha_k w_k + u_k), \quad v_k = (I - w_{k+1} \ell_{k+1}) R_{s_k}(\alpha_k w_k + u_k).$$

Equation (4.17) implies that on  $\|(\alpha, u)\|_* \leq \rho$  this forcing has norm at most  $C\rho^2$  and Lipschitz constant at most  $C\rho$ . To see the latter explicitly, at time  $k$  both arguments have norm at most  $C\rho 2^{-k}$ , and their difference has norm at most  $C2^{-k} \|(\alpha, u) - (\tilde{\alpha}, \tilde{u})\|_*$ . Multiplication by  $2^k$  leaves a factor at most  $C\rho$ .

Insert this forcing into the two sums. Their bounded linear inverse is a contraction on a ball of radius  $C\|u_0\|$  when  $u_0$  is small enough. It produces the asserted unique sequence. Iteration of the contraction also proves analytic dependence on complex  $u_0$ : the iterates are analytic and converge uniformly on smaller balls. Set  $h(u_0) = \alpha_0$ . The construction respects conjugation, so  $h$  is real analytic on real data.

Finally the derivative at zero is given by the same sums with the quadratic forcing deleted. It parametrizes exactly all decaying solutions of (4.20). Uniform equivalence of coordinates proves the assertion about the tangent space and the trajectory estimate.  $\square$

## 4.5 Energy transversality and the choice of temperature

We apply the graph to the physical interaction. To fix its normalization, let

$$E(\sigma) = \frac{1}{2} \sum_{z: \|z\|_1=1} \sigma_0 \sigma_z.$$

Its translation sum counts each nearest-neighbor bond once. By Lemma 1.2, the interaction from  $U$  has a bounded finite-support even square-invariant density  $u_U$  such that its translation sum is the full sum of disagreement monomials, up to exterior-only constants when spins are pinned. One direct choice is to sum  $U(S) \prod_{e \in S} d_e(\sigma)$ , with  $d_e$  as in Section 1, over one representative of each translation class of nonempty edge sets, then average the density over the square group. There are finitely many such classes because  $U$  has finite range. Translation invariance shows that this averaging preserves the translation sum exactly. Enlarge  $s_0$ , if needed, so that all these representatives are supported in  $B_{s_0}$ .

Relative to the pure interaction at  $\beta_0 = \log(1 + \sqrt{2})/(2J)$ , the initial potential is the single-square vector

$$V_0(\beta, \lambda) = s_0^2 [J(\beta - \beta_0)E - \beta \lambda u_U], \quad (4.23)$$

with its additive constant removed. The minus sign agrees with the contour penalty. In particular this formula retains every interaction crossing a domain boundary: it is a bulk density whose conditional law is the law in Lemma 1.2, not a density obtained by discarding boundary terms.

**Theorem 4.5** (Domain-independent decaying trajectory). *For every  $J > 0$  and finite-range real potential  $U$  satisfying the symmetries of Theorem 1.1, the preceding scale parameters can be fixed*

independently of all domains and boundary data. The initial scale  $s_0$  may be enlarged by any prescribed finite amount before the following construction. For each sufficiently small  $\epsilon > 0$  there is  $\lambda_* > 0$  and a real analytic function  $\beta_c : (-\lambda_*, \lambda_*) \rightarrow (0, \infty)$ , with  $\beta_c(0) = \beta_0$ , for which the initial potential (4.23) has a trajectory  $V_{k+1} = \Phi_{s_k}(V_k)$  satisfying

$$\|V_k\|_{s_k, d} \leq \epsilon 2^{-k}, \quad k \geq 0. \quad (4.24)$$

Its intermediate one-step trajectories obey

$$\sup_{0 \leq t \leq 1} \sum_{\ell} |\ell| e^{(d/2)(|\ell|-1)} \|v_{k, \ell}(t)\|_{\infty} \leq C \epsilon 2^{-k}. \quad (4.25)$$

The same construction applies to both signs of  $\lambda$ .

*Proof.* Let  $G(V) = \ell_0 V - h((I - w_0 \ell_0)V)$ . The graph in Lemma 4.4 is exactly  $G(V) = 0$ . We prove that the derivative of  $G(V_0(\beta, \lambda))$  in  $\beta$  at  $(\beta_0, 0)$  is nonzero.

The temperature derivative of (4.23) is the single-square vector  $Z_0 = J s_0^2 E$ , modulo constants. Its successive linear images remain single-square and are

$$Z_k = J s_k^2 \mathbb{E}_0[E \mid \sigma_{\partial B_{s_k}}], \quad k \geq 1, \quad (4.26)$$

modulo constants. For the induction step, the pure Markov property gives

$$\mathbb{E}_0[\mathbb{E}_0[E \mid \sigma_{\partial B_{s_k}}] \mid \sigma_{\partial B_{s_{k+1}}}] = \mathbb{E}_0[E \mid \sigma_{\partial B_{s_{k+1}}}].$$

Although the two boundary sigma-fields are not nested, this identity follows by first conditioning on the full exterior of  $B_{s_k}^\circ$ . The larger boundary is measurable there, and the conditional expectation of  $E$  given that exterior is the inner-boundary conditional expectation. Multiplication by  $L^2$  at each step proves (4.26).

Put  $y_k = (2s_k, 0)$  and  $E_{y_k} = E \circ \tau_{y_k}$ . This bounded observable is exterior to  $B_{s_k}^\circ$ , so another use of the pure Markov property yields

$$\text{Cov}_0(Z_k, E_{y_k}) = J s_k^2 \text{Cov}_0(E, E_{y_k}). \quad (4.27)$$

Proposition 2.9 says that the right-hand side converges to a strictly positive number. Its square symmetrization is precisely what rules out cancellation between bond orientations. On the other hand

$$|\text{Cov}_0(Z_k, E_{y_k})| \leq 2 \|E\|_{\infty} \|Z_k\|_q.$$

Thus the quotient norms of  $Z_k$  do not tend to zero. If  $Z_0$  were tangent to  $G = 0$ , Lemma 4.4 would imply  $\|Z_k\| \leq C 2^{-k}$ , a contradiction. This proves the required nonzero derivative.

The scalar analytic implicit-function theorem applied to  $G(V_0(\beta, \lambda)) = 0$  now gives a unique local real analytic branch  $\beta = \beta_c(\lambda)$  through  $(\beta_0, 0)$ . All data are real, so the branch is real on a two-sided real interval. Shrink the interval so that  $\beta_c(\lambda) > \beta_0/2 > 0$  and the initial point lies in the small graph neighborhood. The analytic sequence constructed in Lemma 4.4 vanishes at  $\lambda = 0$ , hence its sequence norm is at most  $C_U |\lambda|$  on a smaller interval. Shrink it once more to obtain (4.24). Equation (4.25) is then Lemma 4.2. Every ingredient in this construction is a bulk object fixed before a domain or exterior configuration is chosen.  $\square$

The notation  $\beta_c$  refers to this constructed temperature. The remaining sections establish its claimed curve-scaling property. No sign condition on  $u_U$ , or on  $\lambda$ , was used to construct it.

## 4.6 Reverse refreshes and independent marked squares

For an integer  $N$ , let  $\mathbb{T}_N = (\mathbb{Z}/N\mathbb{Z})^2$ , and write  $\mu_{0,N}$  for its pure critical spin law. Evaluating local functions on periodic spin configurations defines the finite Hamiltonian

$$\mathcal{H}_{s,N}(V)(\sigma) = s^{-2} \sum_{x \in \mathbb{T}_N} \sum_{\ell} v_{\ell}(\tau_x \sigma), \quad d\mu_{s,N}^V = \frac{\exp \mathcal{H}_{s,N}(V)}{\mathbb{E}_{0,N} \exp \mathcal{H}_{s,N}(V)} d\mu_{0,N}.$$

These sums converge absolutely in sup norm for the potentials above. For  $V_0$  from Theorem 4.5, the law  $\mu_{s_0,N}^{V_0}$  is the physical perturbed torus law, denoted  $\mu_{\lambda,N}$ , for all sufficiently large  $N$ .

The scale index  $k$  labels successive interaction densities; within each step,  $t \in [0, 1]$  is the time variable. We will use the refresh laws of Lemma 4.1 on each path, with the following estimate controlling the error in its probability equation.

Our flow was constructed in the plane, so its periodic evaluation need not solve that finite-volume equation exactly. We quantify the discrepancy before applying the identity.

**Lemma 4.6** (Periodization error). *Suppose  $Ls \leq N/10$ , and suppose a real one-step trajectory obeys*

$$\sup_{0 \leq t \leq 1} \sum_{\ell} |\ell| e^{(d/2)(|\ell|-1)} \|v_{\ell}(t)\|_{\infty} \leq r_s, \quad \|f\|_{\infty} \leq r_s,$$

*with  $r_s$  in the fixed small ball. Let  $\mu_t = \mu_{s,N}^{V(t)}$ , and let  $P_{t,x}^N$  be its exact torus conditional expectation over  $B_{Ls}^{\circ}(x)$ . There is a signed measure  $e_t$ , of total mass zero, such that*

$$\dot{\mu}_t = \mu_t s^{-2} \sum_{x \in \mathbb{T}_N} (P_{t,x}^N f_x - f_x) + e_t, \quad \|e_t\|_{\text{TV}} \leq Cr_s (N/s)^2 e^{-cN/s}. \quad (4.28)$$

*The constants  $C, c > 0$  depend on the fixed flow parameters, but not on  $s, N, t$ .*

*Proof.* A connected list of  $m$  radius- $s$  squares has sup-norm diameter at most  $2sm$ . Set  $M = \lfloor N/(8s) \rfloor$ , changing fixed constants if needed. Every list of length less than  $M$  then has diameter less than  $N/2$ . Such a translated support cannot intersect two distinct periodic copies of  $B_{Ls}^{\circ}$ : the distance between these copies is at least  $N - 2Ls \geq 4N/5$ , greater than that diameter. Accordingly, for the potential truncated to these lists, the plane conditional Hamiltonian with periodic exterior data is the same as the torus conditional Hamiltonian, up to terms constant in the updated spins. This is an identity of local Hamiltonians, not an assertion that plane and torus Gibbs measures are equal.

For the discarded lists, the weighted estimate gives

$$\sum_{|\ell| \geq M} |\ell| \|v_{\ell}(t)\|_{\infty} \leq Cr_s e^{-cN/s}. \quad (4.29)$$

The placement count of (4.6) therefore bounds the sup norm of their plane conditional Hamiltonian by  $Cr_s e^{-cN/s}$ . The same bound holds for the torus conditional Hamiltonian. To see the latter, enumerate every torus translation that can depend on the update spins by choosing one of its declared boxes meeting a periodic copy of the update square. The number of distinct translations is at most  $C_L m s^2$ ; counting a translation more than once only increases the upper bound. This proves the torus version of the same placement estimate, including supports with repeated periodic images.

If two real conditional Hamiltonians differ by at most  $h$  after subtracting constants, their normalized conditional densities have ratio between  $e^{-2h}$  and  $e^{2h}$ . Thus their expectations of a bounded function differ by at most  $C\|f\|_{\infty} h$  for  $h \leq 1$ . Apply this observation to truncation of

each Hamiltonian. The exact identity for short lists and (4.29) yield, uniformly in the entire spin configuration,

$$\left| (P_{Ls}^{V(t)} f)_x \Big|_{\text{periodic}} - P_{t,x}^N f_x \right| \leq C r_s e^{-cN/s}. \quad (4.30)$$

We used  $\|f\|_\infty \leq r_s \leq C$  to weaken the bound and simplify notation. For a long list, periodic evaluation of the plane conditional function can involve an image of an updated spin in what was the plane exterior. This causes no problem: the comparison is pointwise, and the entire dependence in question is included in the discarded tail.

Differentiate the finite Gibbs density and use Equation (4.8) with periodic evaluation. Replacing each plane conditional term by its exact torus counterpart changes the logarithmic derivative by at most  $C r_s (N/s)^2 e^{-cN/s}$ , before centering. The torus counterpart has zero  $\mu_t$ -mean. Subtracting the mean of the error costs at most another factor two. Multiplying by  $\mu_t$  proves (4.28).  $\square$

**Proposition 4.7** (Coupling by independent marked squares). *Let the physical torus law  $\mu_{\lambda,N}$  be at the temperature of Theorem 4.5, with trajectory bound (4.24). Let*

$$K_N = \max\{k \geq 0 : Ls_k \leq N/10\}$$

*for all sufficiently large  $N$ . There is a coupling of  $\sigma^{(0)} \sim \mu_{0,N}$ ,  $\sigma^{(\lambda)} \sim \mu_{\lambda,N}$ , and Bernoulli marks  $M_{k,x}$ ,  $0 \leq k \leq K_N$ ,  $x \in \mathbb{T}_N$ , with the following properties. The marks are mutually independent and independent of  $\sigma^{(0)}$ , and*

$$\mathbb{P}(M_{k,x} = 1) \leq C \epsilon 2^{-k} s_k^{-2}. \quad (4.31)$$

*Except on an event of probability  $o(1)$  as  $N \rightarrow \infty$ ,  $\sigma^{(\lambda)}$  can be obtained from  $\sigma^{(0)}$  by changing spins only inside the union of the marked squares  $B_{Ls_k}(x)$ . The values of the changed spins may depend on all previous refreshes. The constant and the exceptional probability are independent of test domains and exterior configurations. Enlarging these squares by any fixed number of lattice steps preserves the conclusion.*

*Proof.* For the duration of this proof put  $K = K_N$ . At each scale retain the deterministic intermediate potential  $V_k(t)$  supplied by Lemma 4.2, with  $f_k = \pi_{s_k} V_k$  fixed. The endpoint Gibbs law is exactly the starting law for the next scale:

$$\mu_{s_k,N}^{V_k(1)} = \mu_{s_{k+1},N}^{V_{k+1}},$$

by the density identity (4.14), also after periodic evaluation.

First bound the final perturbation. Maximality of  $K$  gives  $N/s_{K+1} < 10L$ , whereas  $K \rightarrow \infty$  with  $N$ . Hence

$$\|\mathcal{H}_{s_{K+1},N}(V_{K+1})\|_\infty \leq (N/s_{K+1})^2 \|V_{K+1}\|_{s_{K+1},d} \leq C_L \epsilon 2^{-K-1} = o(1). \quad (4.32)$$

The normalized density relative to  $\mu_{0,N}$  lies between the exponentials of plus and minus twice this bound. Its total variation distance from  $\mu_{0,N}$  therefore tends to zero. Couple a sample from this endpoint law and a pure sample so that they agree outside an event of that probability.

For each site and each scale, now independently generate a Poisson process on  $[0, 1]$  of rate

$$r_{k,x} = 2s_k^{-2} \|f_k\|_\infty.$$

These rates are deterministic; they do not involve the current spins, the exterior data of a refresh, or its time within the step. If  $f_k = 0$  we take no clocks at that scale. Generate all these processes

independently of the coupled endpoint and pure samples, and take  $M_{k,x} = 1$  exactly when its process has at least one point. Then

$$\mathbb{P}(M_{k,x} = 1) = 1 - e^{-r_{k,x}} \leq 2s_k^{-2} \|f_k\|_\infty \leq 2\epsilon 2^{-k} s_k^{-2},$$

and all asserted independence properties already hold. Additional randomness used to choose refreshed spins can likewise be generated in advance; no independence assertion about those spin values is needed.

Run scales in the order  $K, K-1, \dots, 0$ . At scale  $k$ , traverse the deterministic measure path backwards in time. At a clock of site  $x$ , refresh  $B_{L_{s_k}}^\circ(x)$  with the law in (4.2), using the exact torus conditional of  $\mu_{s_k,N}^{V_k(t)}$  and the fixed function  $(f_k)_x$ . The normalization and nonnegativity proved in Lemma 4.1 apply at every time, irrespective of whether the evolving sample has the instantaneous Gibbs law. There are finitely many Poisson clocks on a finite torus, so this constructs a finite-state time-inhomogeneous Markov chain. It changes spins only in marked squares.

Lemma 4.6, with  $r_s = C\epsilon 2^{-k}$  from (4.25), bounds the error of its desired reverse probability equation by

$$C\epsilon 2^{-k} (N/s_k)^2 e^{-cN/s_k}.$$

These errors add in total variation. Explicitly, for a prescribed probability path  $p_t$  and a Markov evolution with generator  $\mathcal{L}_t$ , the difference satisfies the inhomogeneous forward equation with forcing  $\dot{p}_t - \mathcal{L}_t^* p_t$ . Variation of constants expresses the difference as the initial difference propagated by a Markov kernel, plus the integral of the propagated forcing. Markov kernels contract total variation of signed measures, so its norm is bounded by the initial norm plus the integrated forcing norm. Apply this identity on each unit-time reverse step and use the exact equality of consecutive endpoint laws. The output law differs from  $\mu_{\lambda,N}$  by at most

$$C \sum_{k=0}^{K_N} \epsilon 2^{-k} (N/s_k)^2 e^{-cN/s_k}. \quad (4.33)$$

This quantity tends to zero. For each fixed  $k$ ,  $N/s_k \rightarrow \infty$ . Moreover  $r^2 e^{-cr}$  is bounded for  $r \geq 0$ , so the summands, extended by zero for  $k > K_N$ , are dominated by a constant times the summable sequence  $\epsilon 2^{-k}$ . Dominated convergence proves the assertion without requiring the largest-scale error to vanish at a fixed radius ratio.

Finally maximally couple the output with a sample from the exact desired law. Its probability of disagreement is bounded by (4.33). This final correction can be sampled conditionally on the already constructed output and leaves the joint law of the pure input and all marks unchanged. On the complement of the endpoint-coupling failure and this final disagreement, the target sample is obtained from the pure sample by precisely the described marked-square refreshes. Both exceptional probabilities tend to zero, which proves the proposition.  $\square$

The coupling has the particular independence needed in Section 5: the possible edit locations can be sampled before the pure spin field is examined. Conditional refreshed values remain unrestricted. Thus a later estimate that permits arbitrary spin assignments in the marked squares applies to this exact perturbed law, up to the vanishing exceptional probability already quantified above.

## 5 Stability of bulk crossings under the scale edits

Proposition 4.7 represents a perturbed configuration, apart from an event of vanishing probability, by changes inside independently marked squares of a pure critical configuration. The values assigned in

those squares depend on the preceding changes. We therefore prove a stronger statement: with high probability, every assignment in their union gives the same answer to each prescribed macroscopic test. Four contour passages control changes in the interior of a test, two passages confined to a half-plane control changes near an edge, and one passage controls changes near a vertex. The different numbers of possible edit locations explain the thresholds two, one, and zero in the estimates below.

## 5.1 Tests, passages, and independent marks

Draw in every unit square of the spin lattice the diagonal joining its northwest and southeast vertices. Interpolate the spin values affinely on the resulting triangles. The zero set consists of disjoint polygonal contours; its connections at a checkerboard square give precisely the north–east and south–west pairing in Section 1. Here the diagonals specify connectivity and introduce no interaction in the spin Hamiltonian. A path of a specified color means a path in this triangulation all of whose vertices have that spin. Inside a bulk chart, the corresponding pieces of these zero contours and the locally rounded contours of Section 1 have matched oriented parametrizations at distance at most a constant times the mesh.

A *polygonal corridor* is a closed topological rectangle with piecewise linear boundary and two disjoint opposite sides designated as its gates. Its crossing test asks for a path of a specified color in the corridor joining those gates. A *polygonal band* is the closed region between two disjoint simple polygons, and its circuit test asks for a path of a specified color separating the two boundary components. In both definitions we use ordinary approximations by subcomplexes of the triangulation, with the gates approximated as marked boundary arcs. The approximations have Hausdorff error  $O(\delta)$  and preserve the prescribed cyclic order. We can and do choose polygons with the elementary nondegeneracies required for these approximations; arbitrarily close polygonal choices with those properties suffice in our applications.

Here is the convention for counting contour passages. A passage between two disjoint terminal sets is a contour subarc whose endpoints lie on the two sets and whose interior lies in the permitted region between them. Several passages must have disjoint open parameter intervals on every contour. Their endpoints may coincide at terminal hits, but a subarc of positive length cannot be used twice. Thus passages of a single contour are counted with their multiplicities. Cropping an arc at its last hit of the initial terminal before its first hit of the final terminal gives this convention. A reversal of a contour can supply two passages; it is not suppressed by counting only contour components.

All planar distances in this section are physical distances at mesh  $\delta$ ; a superscript  $\text{lat}$  explicitly denotes lattice units. For a point  $z$ , a direction  $\theta$ , and  $0 < r < R$ , write

$$S_r(z, \theta) = z + e^{i\theta}[-r, r]^2, \quad A(z, \theta; r, R) = S_R(z, \theta) \setminus \text{int } S_r(z, \theta).$$

The three local events used in the proof are

$$\begin{aligned} F_4(z, \theta; r, R) &= \{\text{at least four contour passages across } A(z, \theta; r, R)\}, \\ H_2(z, \theta; r, R) &= \{\text{at least two such passages contained in } z + e^{i\theta}\{\text{Im } w \geq 0\}\}, \\ F_1(z, \theta; r, R) &= \{\text{at least one contour passage across } A(z, \theta; r, R)\}. \end{aligned} \tag{5.1}$$

The straight part of the boundary in  $H_2$  imposes confinement; the inner and outer square boundaries are its terminals. We use the same subcomplex convention, or intersect the interpolated contours with the indicated polygons. Enlarging the first radius by  $O(\delta)$  and reducing the second by  $O(\delta)$  compares these conventions. Later all rings have fixed proportional gaps, so this change is absorbed before a probability estimate is applied.

Let  $\mu_{0,N}$  denote the pure critical law on the lattice torus of side  $N$ , and sample  $\sigma^0$  from this law. Independently sample Bernoulli variables  $M_{k,x}$ , independent also of one another, with

$$\mathbb{P}(M_{k,x} = 1) \leq C\epsilon 2^{-k} s_k^{-2}, \quad s_k = L^k s_0. \quad (5.2)$$

A marked index permits an arbitrary change in a square of radius  $b_k^{\text{lat}} = Ls_k + O(1)$  centered at  $x$ . Only the levels in Proposition 4.7 are present; in particular these squares have radius less than a fixed small fraction of the torus period. For a set of marked indices  $M$ , let  $\mathcal{C}(\sigma^0, M)$  be all spin configurations equal to  $\sigma^0$  outside the union of their squares. For a test  $T$  with values in  $\{0, 1\}$ , set

$$\begin{aligned} T^+(\sigma^0, M) &= \mathbf{1}\{\text{some } \sigma \in \mathcal{C}(\sigma^0, M) \text{ has } T(\sigma) = 1\}, \\ D_T(\sigma^0, M) &= \mathbf{1}\{T \text{ is not constant on } \mathcal{C}(\sigma^0, M)\}. \end{aligned} \quad (5.3)$$

Both are increasing functions of the set of permitted marks, regardless of whether  $T$  is increasing in the spins. In particular,  $T^+$  for a contour-passage event is not being treated as a monotone spin event.

**Theorem 5.1** (Bulk stability). *After increasing  $s_0$  if necessary, there is  $\epsilon_* > 0$ , depending only on the fixed constants in Proposition 4.7, with the following property. Fix any finite family  $\mathcal{T}$  of polygonal corridor crossings, polygonal-band circuits, and four-passage square annulus tests. Let  $N = N(\delta)$  be integers with  $\delta \downarrow 0$  and  $\inf_\delta \delta N(\delta) > T_0(\mathcal{T})$ , where  $T_0(\mathcal{T})$  is sufficiently large that all tests and their fixed collars lie in one planar chart of the physical torus. If  $\epsilon \leq \epsilon_*$  and the marks satisfy (5.2), then*

$$\mathbb{P}(D_T(\sigma^0, M) = 1 \text{ for some } T \in \mathcal{T}) \longrightarrow 0 \quad \text{as } \delta \downarrow 0. \quad (5.4)$$

The threshold  $\epsilon_*$  is independent of the family, its polygonal complexity, and all its fixed positive widths. The mesh threshold and the constants in the convergence bound may depend on that geometry.

Consequently the pure and perturbed torus laws at the trajectory of Theorem 4.5 have asymptotically identical joint laws for every such finite family. In particular, pure positive lower bounds for any fixed finite compatible system of signed buffered corridor and circuit events, and pure upper limits for four-passage events, hold for the perturbed law as well.

Compatibility in the last assertion includes the separated signed corridors and bands of Proposition 2.13. It is not a claim that mutually obstructing prescribed crossings can all occur. The proof of the theorem occupies the rest of this section.

## 5.2 Local changes and the required passages

**Lemma 5.2** (Passages forced by a pivotal square). *Let  $T$  be a fixed polygonal crossing, circuit, or one of the three annular tests in (5.1). Suppose that two spin configurations agree off a square of radius  $b$  but give different values to  $T$ . Enlarge this square by a fixed factor to contain all contour segments affected by the change. There is a positive geometry scale  $a_T$  with the following properties. Set*

$$u = \min\{a_T, \max\{b, \text{dist}(z, \mathcal{E}_T)\}\}, \quad v = \min\{a_T, \max\{b, \text{dist}(z, \mathcal{V}_T)\}\},$$

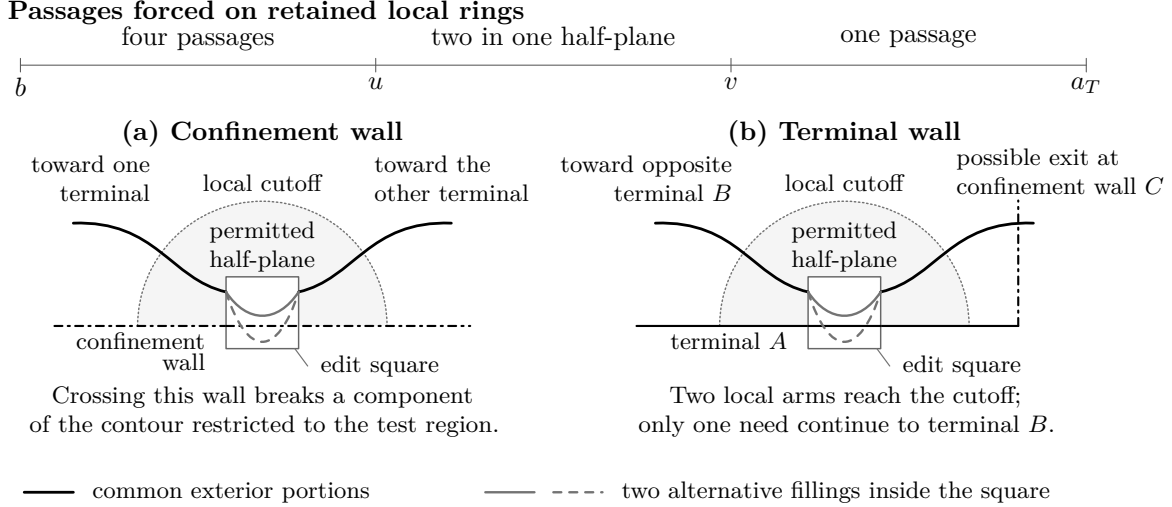
where  $\mathcal{E}_T$  and  $\mathcal{V}_T$  are the finite collections of sides and vertices of the test polygons, and  $z$  is the edit center. For  $H_2$ , these collections also include the confinement line segments and their intersections with the two square boundaries. For  $b < a_T$ , the common contours outside the enlarged edit square must supply the following checks, after fixed crops:

- (i) four passages on rings between scales  $b$  and  $u$ ;

- (ii) two passages confined to an appropriate half-plane on rings between scales  $u$  and  $v$ ;
- (iii) one passage on rings between scales  $v$  and  $a_T$ .

Rings within a fixed multiplicative distance of transition scales may be omitted. The half-plane checks may be centered on the nearby side and the last checks near the nearby vertex. For a fixed test only a bounded number of further rings need be omitted. The constants are uniform over translations, rotations, and dilations of each of the three fixed annular shapes.

Figure 1 distinguishes the two roles of a wall in the intermediate regime.



**Figure 1:** Local geometry in Lemma 5.2. The scale strip records the four-, two-, and one-passage regimes; intervals may be empty, and rings near their endpoints are omitted as in the lemma. Passages are counted by disjoint parameter intervals and may lie on a single contour. The panels show the intermediate regime, with the local cutoff below the next vertex and the other terminal. Black contour portions agree outside the edit square; the two gray fillings are alternatives. Crossing a confinement wall ends a component restricted to the permitted region, whereas crossing a terminal wall creates a terminal hit. In (b), one local arm may later exit at the confinement wall  $C$ , so the two local arms need not form two full  $A$ – $B$  passages. The semicircular cutoffs are schematic; the ring checks use the fixed crops and recenterings specified in the lemma.

*Proof.* Choose  $a_T$  smaller than the separation of distinct nonincident sides, the lengths of all sides, and the distances between the disjoint terminal sets, with additional fixed reductions depending on the finitely many vertex angles. Then a ball below this scale meets no test wall, just one straight wall, or walls incident to one nearby vertex. This choice is essential for a thin polygon: distance to a vertex alone need not exclude a second, nonincident side. Capping at  $a_T$  removes that difficulty. Enlargement and recentering cost only fixed factors for this test.

First work in a neighborhood meeting no test wall. Cut the common exterior contours at their intersections with the enlarged edit square, making an arbitrarily small displacement of the cut to avoid vertices. Follow every exterior end to its first return to the square or to the local outer cutoff. An exterior connection which reaches that cutoff supplies two long ends; short returns can be contracted for this argument. The number of long ends is even. If it is zero, the edit is screened within the cutoff. If it is two, the interior filling, together with the short exterior returns, must connect these same two ends in either configuration. It can add small closed curves but cannot change their through connection or the two side colors seen outside the cutoff. Since there is no terminal in this neighborhood, these small curves cannot create a terminal visit. A changed passage

count therefore requires at least four long contour portions. This argument is about cut parameter intervals, and allows all four portions to come from one loop.

More explicitly, the filling together with short exterior returns is a degree-two network whose only free ends are the long ends. With two such ends it has one connecting path and terminal-free cycles. Restrict each resulting contour to the permitted test region, breaking its parameter interval wherever it exits that region. On each remaining component record the successive inner and outer terminal visits in parameter order, cyclically on a closed component, collapsing consecutive visits of the same type. The maximal number of disjoint passages is the number of changes of terminal type within these components. A confinement-wall exit ends a component; visits on opposite sides of such an exit are not paired. Replacing the terminal-free connecting path in the present wall-free neighborhood preserves these sequences and their component endpoints. This also covers repeated visits by one contour and small terminal-free loops.

For a color-crossing test the same conclusion also follows by juxtaposing a successful path and the blocking path of the opposite color in the configuration where the test fails. The triangulation gives the exact crossing–blocking alternative. Both paths must meet the edit square. Outside it they supply four alternating color continuations, whose intervening contours give four long portions. Cutting a band along a radial path gives the identical argument for a circuit and its radial blocker.

Next suppose that only one straight wall is present. There are two different wall roles for passage tests. At a confinement wall, a passage disrupted by the edit has a part on either side of the disrupted portion leading back to the local cutoff in the permitted half-plane; its terminal walls lie beyond that cutoff. These are the two required portions. At a terminal wall, stop every passage at its first terminal hit. A portion which hits this terminal before entering the edit neighborhood is unchanged. Among the remaining portions arriving from the opposite terminal, a terminal hit can disappear only if the new filling pairs its long end with another long end on the permitted side. Pairing with an end on the other side still crosses the terminal wall and preserves a hit. If there is only one long end on the permitted side, neither pairing can remove the hit. Consequently a changed terminal visit supplies two long portions in the permitted half-plane. A hairpin tip moved across a terminal is covered by this case: its two local arms supply the required half-plane portions. In a confined test only one arm may continue all the way to the opposite terminal, so the passage count need not change by two.

For a crossing and its blocker, at most the color arm directed towards this single wall can terminate before the local cutoff. The other three occur in alternating order on the permitted side. The two transitions between them give two disjoint contour portions in that half-plane. The circuit argument is the same after the radial cut. In each case start the half-plane rings at a fixed multiple of the distance to the wall and stop before the next vertex scale; the contours in question stay on one consistent permitted side. If the test permits either side, taking the union of the two checks costs at most a factor two per retained ring.

Above the vertex scale we retain only one contour portion. For a crossing and blocker, two adjacent incident walls can terminate two of the color continuations, but a transition between the remaining colors persists towards the separated terminal. For a passage test, an affected passage still has a portion towards its other terminal, at distance at least  $a_T$ . Equivalently, if no contour exits the local cutoff, every change is confined to components which cannot connect the separated terminal sets or alter a separating circuit. Thus a pivotal change forces one passage up to  $a_T$ .

At every step the portions are cropped at the appropriate first or last hits. No positive-length interval is used twice, even when a contour returns repeatedly to a wall. The fixed-factor omissions accommodate the chosen cuts, recentering, and the transition between the three geometric regimes. An  $O(\delta)$  displacement of a wall is absorbed by increasing the starting square and, for the half-plane checks, translating its line outwards by  $O(\delta)$ .  $\square$

### 5.3 A half-plane estimate under bulk conditions

The pivotal-square lemma requires a bound  $o(r/R)$  for two passages constrained to one side of a bulk line. We obtain it by counting macroscopic subarcs, bounding the area of their near-support contacts, and then separating off a four-passage alternative. Propositions 2.12 and 2.13 give, for some  $c > 0$ ,

$$\limsup_{\delta \downarrow 0} \mathbb{P}_0(F_4(z, \theta; r, R)) \leq C(r/R)^{2+c}, \quad \limsup_{\delta \downarrow 0} \mathbb{P}_0(F_1(z, \theta; r, R)) \leq C(r/R)^c. \quad (5.5)$$

The ratios are fixed before the mesh limit. Buffered comparisons transfer these statements between the plane, a larger disk with plus boundary, and bulk torus charts, with fixed multiplicative constants. We next supply the estimate between them. It concerns a geometric half-plane inside a bulk chart, not a model with boundary conditions on the straight line.

**Lemma 5.3** (Integrable numbers of macroscopic subarcs). *Fix a compact region  $K$  and a length  $a > 0$ , in a bulk chart containing a fixed  $20a$ -neighborhood of  $K$ . There is a random finite list  $\mathcal{L}_\delta$  of compact contour parameter-subarcs, chosen without reference to a test center or direction, with  $\sup_\delta \mathbb{E}|\mathcal{L}_\delta| < \infty$  for sufficiently fine meshes. The list covers the following visits, with their contour multiplicities. If a contour visits  $x \in K$  and reaches distance  $10a$  from  $x$  in both parameter directions, a member of the list contains that visit and lies between the first such distant visit in each direction. Subarcs are retained only in a fixed enlargement of  $K$ . The same assertion holds for the limiting complete loop collection, with a list of finite expected cardinality. Fixed enlargements of the chart and fixed crops of the excursion lengths only change the bound.*

*Proof.* First consider a rectangle of fixed shape, with an unexamined buffer around it. Contour crosscuts between its horizontal sides, confined between two vertical sides, are linearly ordered from left to right. After cropping their ends, each crosscut has two colored side routes. One obtains these routes by following the open spin regions next to the interpolated contour and then following vertices in the triangulation. The two routes associated with one crosscut have opposite signs. Reading the routes in their planar order and retaining successive sign changes produces an alternating list whose length is at least a fixed fraction of the number of crosscuts. Opposite signs cannot meet. Routes of the same sign separated by an opposite route are disjoint as crossings. The fixed end crop and side margins absorb the interpolation distance. Thus it suffices to bound the number of ordered alternating spin crossings of a fixed rectangle.

Explore the leftmost crossing of a prescribed sign, if one exists. The exploration examines only the sites on its left and the crossing itself. For example, reveal the vertices reachable from the examining side before a crossing of that sign blocks the exploration; their outer envelope gives the same leftmost path. An alternative path lying farther left somewhere is detectable on this examined side by splicing at its first and last common vertices. Consequently the transcript leaves the spins to the right with the ordinary Ising conditional law, and the only newly exposed spins facing that region on the separating path have the prescribed sign. Repeat for the opposite sign in the remaining right region. Previously examined sites lie behind the most recent spanning path.

There is a uniform probability  $c_0 > 0$ , conditional on each successful transcript, that no further opposite crossing can occur. We construct the blocking event in a deterministic rectangle, rather than asking RSW for a crossing ending on the random exposed path. For one fixed choice of proportions, take the counting slab to be  $[-1, 1] \times [-2, 2]$ , and let  $\Gamma$  be the exposed simple color- $s$  crossing from its bottom to its top. Use the central rectangle  $[-2, 2] \times [-1, 1]$  for a horizontal test, with end gates on  $x = -2$  and  $x = 2$ , and surround it by the larger rectangle  $[-4, 4] \times [-3/2, 3/2]$ . Fixed changes of these proportions cover the original rectangle shapes.

In the larger rectangle, fill the side to the left of  $\Gamma$  by color  $s$ , leave its right side free, and put color  $-s$  on the outer boundary. The side is determined by the bottom-to-top crosscut in the counting slab; its restriction to the central levels suffices, even when it has bulges or crosses a horizontal level several times. Nearest-neighbor conditional independence makes the filled left side invisible from the right except through the color- $s$  path. All earlier revealed sites are screened by that path. No sites to the right of the counting slab have been examined, so the lateral extension retains a free region past the right gate. Thus, for increasing tests when  $s = +$ , the actual remaining law dominates this right-side law; putting the outer pins at minus only lowers it. For  $s = -$  the order is reversed.

The filled comparison law in turn dominates, in the color- $s$  order, the ordinary law in the same larger rectangle with all boundary spins  $-s$  and no interior pins. In this last, deterministic law, a color- $s$  horizontal crossing of the central test has probability at least  $c_0$  by Proposition 2.13; its gates and test collars are a positive distance from the outer pins. Such a crossing must meet  $\Gamma$ , since its gates lie on opposite sides of the counting slab. Its portion after the last meeting with  $\Gamma$  connects the exposed path to the right gate. It therefore blocks another color- $-s$  vertical crossing confined to the slab. This suffix uses only the unexplored side, so the two monotone comparisons prove the asserted conditional stopping probability. The central vertical crop leaves clearance from the endpoints of  $\Gamma$ ; lattice rounding is absorbed by the fixed margins.

The chance of continuing the alternating exploration after  $m$  successful crossings is therefore at most  $(1 - c_0)^{m-1}$ , with a factor two for the initial color. This gives uniformly bounded moments of all orders for the number of contour crosscuts in a fixed cropped rectangle.

Cover a fixed enlargement of  $K$  by finitely many squares of diameter much smaller than  $a$ . Every departure by distance  $a$  from one such square contains a passage through one of finitely many buffered rectangles; choose the first side through which an appropriate enlargement is left. Disjoint parameter intervals yield distinct passages in this finite collection. Their total number thus has bounded expectation, indeed bounded moments.

Cut the contours at the boundary of an enlarged chart containing the  $20a$ -neighborhood of  $K$ , and retain only components that meet  $K$  and have diameter at least  $10a$ . Subdivide each such parameter interval successively when its distance from the last subdivision point first reaches  $a$ . For a closed component choose a root without reference to the later center or direction and subdivide cyclically. Retain every union of five consecutive subdivision intervals whose middle interval meets  $K$ . The interval containing a qualifying visit is therefore the middle of one listed union. That union lies within distance less than  $10a$  of the visit, after an absolute reduction of the subdivision distance if needed. It therefore lies between the two distant visits in the statement. The buffer from  $K$  to the cutting boundary ensures that these intervals exist on both sides of every qualifying visit. Endpoint remainders and cyclic intervals through a root are included. The rectangle count bounds the number of retained intervals, including separate traversals of the same geometric region. This constructs the required list without subdividing any part of a contour far from the fixed chart.

Complete-resolution convergence from Theorem 2.11, with slightly relaxed rectangles, preserves every fixed finite collection of these macroscopic passages. Fatou's lemma transfers the expectation bound to the limit. Alternatively perform the same successive subdivision on the limiting loops: uniform continuity makes the list finite on each loop, and the relaxed passage counts bound its expected length. The construction lists no microscopic loops unrelated to the excursions under consideration.  $\square$

**Lemma 5.4** (Area of near-support contacts). *Fix a bulk region and a positive excursion length. For the pure contours at mesh  $\delta$ , let  $G_K(z, \theta, u)$  be the event that a contour subarc comes within  $Cu$  of  $z$ , travels the fixed excursion length on both sides of this visit, and throughout those two portions*

stays above the line of direction  $\theta$  through  $z$  lowered by  $CKu$ . For each fixed  $K$ ,

$$\limsup_{\delta \downarrow 0} \mathbb{P}_0(G_K(z, \theta, u)) = o_K(u), \quad (5.6)$$

as  $u \downarrow 0$  with  $K$  fixed, uniformly over centers and directions in a smaller bulk chart. Fixed crops or fixed changes of the constants have the same conclusion.

*Proof.* Use a larger round disk with plus boundary as a reference and average the center over the lattice translates of a small disk in its interior. Translation invariance of the plane law and pure buffered density comparison bound the plane probability at a specified center by a constant times this averaged reference probability. Off-lattice centers are covered by an  $O(\delta)$  relaxation of the distances. This step uses only lattice translations. Complete loop convergence with relaxed distances then bounds its upper limit by the analogous continuum event averaged over area. Indeed every witnessing subarc has a fixed positive diameter, so it belongs to the macroscopic part of the loop ensemble; subsequential uniform parametrized convergence preserves its closed near-visit and half-plane conditions after the relaxation. The reference disk and the averaging disk are rotationally invariant. We may therefore average also over the direction in this continuum bound. For directions or centers varying with the mesh, take convergent subsequences and relax the geometric inequalities before applying loop convergence. This gives the same uniform bound and does not assert rotational invariance of the lattice.

Use Lemma 5.3 with a subdivision length smaller than one tenth of the fixed excursion length. If the event occurs, one member  $C$  of its list contains the near visit and lies entirely between the distant visits. This list is independent of  $z$  and  $\theta$ . Write  $n_\theta$  and  $t_\theta$  for the upward normal and the unit tangent to the line, and put

$$m_C(\theta) = \min_{w \in C} n_\theta \cdot w, \quad C_{\theta,h} = \{w \in C : n_\theta \cdot w \leq m_C(\theta) + h\}.$$

For almost every  $\theta$  the support line of the compact convex hull of  $C$  has a unique contact point. One elementary justification is to use the support function of that convex hull: it is Lipschitz on the circle and differentiable almost everywhere, and at a differentiability direction the two extreme tangent coordinates of the supporting face coincide. Compactness then gives

$$\text{diam}\{t_\theta \cdot w : w \in C_{\theta,h}\} \longrightarrow 0 \quad (h \downarrow 0) \quad (5.7)$$

at each such direction.

If  $C$  supplies a near visit, its height above its minimum at that visit is at most  $C_K u$ . The normal coordinate of  $z$  lies in an interval of length  $C_K u$  about  $m_C(\theta)$ , and its tangent coordinate lies within  $Cu$  of the projection of  $C_{\theta,C_K u}$ . The area of the possible centers is therefore bounded by

$$C_K u (\text{diam}\{t_\theta \cdot w : w \in C_{\theta,C_K u}\} + Cu).$$

By (5.7) this is  $o_K(u)$  for almost every direction and is at most  $C_K u$  times a fixed chart diameter. The expected number of arcs in the list is finite. Dominated convergence first over directions and then over that list proves the averaged  $o_K(u)$  bound. The preceding buffered comparison and rotational averaging yield (5.6).  $\square$

**Proposition 5.5** (Two bulk passages confined to a half-plane). *For the events in (5.1), in any buffered pure bulk chart,*

$$\limsup_{\delta \downarrow 0} \mathbb{P}_0(H_2(z, \theta; r, R)) = o(r/R) \quad \text{as } r/R \downarrow 0. \quad (5.8)$$

*The ratio is fixed before taking the mesh limit. The assertion is uniform in translations and rotations of the fixed geometry, and continues to hold with fixed proportional crops.*

*Proof.* Normalize  $R = 1$  and write  $u = r/R$ . Select two confined passages. If there are fewer than four unconstrained passages from radius  $u$  to radius  $Ku$ , they belong to the same loop and their inner ends are joined inside radius  $O(Ku)$ . To verify the statement including multiplicities, first suppose that the selected passages belong to different loops. The complementary arc on each loop supplies a second passage to  $Ku$ , giving four in total. On a single loop the two complementary arcs either both join an inner end to an outer end, again giving four, or join inner to inner and outer to outer. In the latter case an excursion of the inner connecting arc to  $Ku$  supplies two further passages. These may share their outer hit, as allowed by our convention, and have disjoint interiors. Thus that excursion is excluded.

Concatenate the two confined passages and their inner connecting arc. This subarc has a near visit to  $z$  and goes a fixed positive distance on both sides, all above the original line lowered by  $CKu$ . Its probability is  $o_K(u)$  by Lemma 5.4. In the complementary case there are four passages from  $u$  to  $Ku$ , and the original confined passages still give two confined passages from  $C_0Ku$  to 1. Here  $C_0$  is a fixed buffer constant. Condition on the spins outside radius  $C_0Ku$ . Pure buffered comparison bounds the conditional four-passage probability by  $CK^{-2-c}$ , uniformly in those outside spins, by (5.5).

Let  $p(u)$  be the mesh upper limit of the plane confined-passage probability, with the uniform translation and direction convention just described. We have proved, for every sufficiently large fixed  $K$  and all sufficiently small  $u$ ,

$$p(u) \leq o_K(u) + CK^{-2-c}p(C_0Ku). \quad (5.9)$$

All constants associated with buffers and relaxed squares are fixed before  $K$  is enlarged. Square and Euclidean radii differ by fixed factors and give the same recursion after changing  $C_0$ .

For completeness, put  $q(u) = p(u)/u$ ,  $d = C_0K$ , and  $\vartheta = CC_0K^{-1-c}$ . Fix  $K$  so large that  $\vartheta < 1$ . Then  $q(u) \leq e_K(u) + \vartheta q(du)$ , where  $e_K(u) \rightarrow 0$ . Iterate until  $d^m u$  first reaches a fixed small interval bounded away from zero. The terminal value is bounded because  $p \leq 1$ , and

$$q(u) \leq \sum_{j=0}^{m-1} \vartheta^j e_K(d^j u) + \vartheta^m q(d^m u).$$

This first gives boundedness of  $q$  near zero. Splitting the sum at a fixed index, letting  $u \downarrow 0$ , and then sending that index to infinity gives  $q(u) \rightarrow 0$ . This proves the plane assertion. A final fixed buffered density comparison proves the same assertion in the other pure bulk charts.  $\square$

The three pure probabilities now decay more strongly than their respective location counts. We need their consequence at one large fixed annulus ratio, together with a product rule. Neither step will refer to the law of a configuration after it has been edited.

## 5.4 Conditional ring products with independent marks

At a given stage allow only marks of radius at most  $b$ . A check on an annulus is determined by the original spins in that annulus with an  $O(\delta)$  margin, and by the mark variables whose squares meet this support. It does not require the assigned spins outside this support. Denote its support by  $A_j$  and this finite set of mark indices by  $\mathcal{J}_j$ .

**Lemma 5.6** (Ring product bound). *There is a fixed constant  $D$  with the following property. Take annular checks of the types (5.1), whose inner radii exceed  $Cb$ , and whose enlarged neighborhoods*

and proportional inner and outer collars are disjoint. More precisely, after absorbing rotations and lattice margins, each check has a collar  $W_j$  of the form

$$S_{2R_j}(z_j, \theta_j) \setminus S_{r_j/2}(z_j, \theta_j),$$

the other check supports lie outside  $W_j$ , and the sets  $\mathcal{J}_j$  are disjoint. All collars lie in bulk torus charts. For arbitrary independent mark probabilities at or below this cutoff,

$$\mathbb{P} \left( \bigcap_{j=1}^m T_j^+ \right) \leq \prod_{j=1}^m D \mathbb{P}(T_j^+). \quad (5.10)$$

The constant  $D$  depends on the fixed collar proportions and is independent of the annular aspect ratios and the mark probabilities.

*Proof.* Condition on the original pure spins outside  $W_j$  and on all marks outside  $\mathcal{J}_j$ . For each fixed value of the remaining local marks,  $T_j^+$  is simply an event of the pure spins on  $A_j$ . Corollary 2.2 gives the pointwise density bound

$$\frac{d\mu_{0,N}(\sigma_{A_j} \mid \sigma_{W_j^c})}{d\mu_{0,N}(\sigma_{A_j})} \leq D. \quad (5.11)$$

For clarity, the two parts of this comparison have fixed aspect: the outer data are separated from  $A_j$  by the collar outside  $S_{R_j}$ , and the inner data by the collar inside  $S_{r_j}$ . For the second comparison interchange the terminal sets in the symmetric density-odds bound and regard the already fixed outer data as common pins. The two losses are independent of  $R_j/r_j$ . No pins occur in the intervening free collars.

For rotated squares this comparison uses a fixed patch cover, as in [19, Corollary 2.6], without rotating the lattice. Write  $S_t = z + e^{i\theta}[-t, t]^2$ . Partition  $A = S_R \setminus S_r$  into a bounded number of sets, each contained in an axis-aligned square of radius  $cR$ , with  $c > 0$  fixed and small. Each patch is separated from the changed data outside  $S_{2R}$  by a distance comparable to  $R$ , uniformly in  $\theta$ . Factor the probability of the configuration on  $A$  into successive patch conditionals. Earlier patches and all inner data are common pins, allowed by Proposition 2.1, so the bounded product costs a fixed factor. To change the inner data, partition  $S_{r/2}$  into a bounded number of patches of diameter  $cr$  and change them one at a time. Each is separated from the whole observed annulus by distance comparable to  $r$ . Terminal symmetry gives the same bound, with the remaining inner patches and the outer data common. These two finite products are uniform in  $R/r$  and  $\theta$ . Finally mix the reference exterior data to obtain (5.11). Fixed positive patch clearances absorb lattice rounding.

The marks in  $\mathcal{J}_j$  are independent of the conditioning and of the original spins. Integrating the pointwise bound over their unchanged distribution gives

$$\mathbb{P}(T_j^+ \mid \sigma_{W_j^c}, M_{\mathcal{J}_j^c}) \leq D \mathbb{P}(T_j^+).$$

All other  $T_i^+$  are measurable with respect to this conditioning. Remove the factors one at a time to obtain (5.10). This proof uses conditional laws of  $\sigma^0$  only. In particular, it neither defines nor assumes a conditional mixing estimate for an edited spin law.  $\square$

Choose a large fixed gap factor  $G$ . In a range of available scales retain rings of ratio  $A$  whose successive inner radii have ratio  $GA$ . For  $G$  large enough their collars and mark-index neighborhoods satisfy Lemma 5.6, including the harmless recenterings of Lemma 5.2. Fix also a monitoring constant  $C_{\text{mon}}$  larger than the minimum-radius constant in Lemma 5.6, enlarged to allow the fixed crops

and recenterings. Around an inserted square of radius  $b$ , begin each retained ring only when its inner radius is at least  $C_{\text{mon}}b$ . Removing the preceding rings costs a bounded number of rings, absorbed in the pivotal bound below. Thus every retained check is among the checks monitored at that activation stage; this is the scale closure used in the bootstrap. First choose these constants and the density constant  $D$ . Then choose  $A$  large. By (5.5) and Proposition 5.5, the three pure probabilities at ratio  $A$  can be made small enough that there are numbers

$$p_4 > 2, \quad p_h > 1, \quad p_1 > 0, \quad (5.12)$$

pure probability bounds  $q_4, q_h, q_1$ , and positive tolerances  $\tau_4, \tau_h, \tau_1$  such that

$$D(q_i + \tau_i) < (GA)^{-p_i}, \quad i \in \{4, h, 1\}. \quad (5.13)$$

Increase  $D$  first to include the possible two choices of half-plane and any other fixed union of cropped checks. The assertion for  $h$  uses  $q_h = o(A^{-1})$ : after  $G, D$  are fixed, one can choose  $A$  with  $Dq_h < (GA)^{-1}$  with strict slack, and then choose  $p_h > 1$  sufficiently close to one. It does not require a previously known half-plane power law. The assertion for 4 uses the strict exponent above two. The assertion for 1 uses any positive one-passage exponent.

All these choices precede  $s_0$ . At this fixed ratio, increase  $s_0$  so that the pure finite-mesh bounds  $q_i$  hold uniformly for every translated and rotated check with inner lattice radius at least a fixed multiple of  $LS_0$ , in the stated torus charts. Such a threshold follows from the mesh-first pure estimates and buffered comparison. If uniformity in the frame failed, take convergent directions and normalized offsets and use slightly relaxed radii; their fixed margins give the same estimate. This is a choice for three fixed shapes, rather than a uniform claim over an increasing family of final polygonal tests.

Suppose for the moment that, with marks of radii at most  $b$ , the probabilities of all needed possibility checks are at most  $q_i + \tau_i$ . Pack the retained rings in the three ranges of Lemma 5.2. Omitting only the bounded number of transition rings, (5.10) and (5.13) give the following bound for the possibility of a pivotal insertion of a square of radius  $b$ :

$$C_T \left(\frac{b}{u}\right)^{p_4} \left(\frac{u}{v}\right)^{p_h} \left(\frac{v}{a_T}\right)^{p_1}, \quad b \leq u \leq v \leq a_T. \quad (5.14)$$

This probability does not include the mark of the inserted square. If its addition first permits a different test value, choose a witnessing assignment afterwards and revert the new square to the old values, retaining all other assignments. The two configurations agree outside that square. The required checks outside it are therefore possible using the previous marks. They are exactly the events to which the product bound applies.

## 5.5 The area sum and the finite-size bootstrap

**Lemma 5.7** (Counting potential insertion centers). *Assume the possibility bounds just used. For a fixed polygonal test  $T$ , the total contribution to its change probability from all marks of level  $k$  is at most*

$$C_T \epsilon 2^{-k} \min\{1, (\delta b_k^{\text{lat}}/a_T)^{c_*}\}, \quad (5.15)$$

*for a fixed  $c_* > 0$ . The assertion also holds during any spatially nonuniform partial activation of that level. For the three fixed annular shapes the constant is uniform in their scale, position, and rotation.*

*Proof.* Use physical lengths and rescale  $a_T$  to one. Write  $b = \delta b_k^{\text{lat}}/a_T$ , and similarly rescale  $u, v$ . Discard centers whose edit square does not meet the finite spin support of  $T$ : their insertion cannot affect the test. For  $b \leq 1$ , divide the insertion centers into dyadic classes with truncated distances comparable to  $u, v$ , where  $b \leq u \leq v \leq 1$ . The area of one such class is at most  $C_T uv$ . Indeed, when  $v < 1$  the relevant portion of an edge has length  $O_T(v)$  near a vertex, and its  $u$ -neighborhood has area  $O_T(uv)$ . When  $v$  is comparable to one, the total finite edge length is  $O_T(1)$ . Centers far from the edges belong to the capped class  $u = v = 1$ . Lower truncation at  $b$  allows the same estimate after enlarging by one lattice cell, so there are at most  $C_T uv/\delta_T^2$  centers, where  $\delta_T = \delta/a_T$  is the rescaled mesh.

The marking probability is bounded by  $C_L \epsilon 2^{-k} \delta_T^2/b^2$ . Multiplication by the number of centers and (5.14) gives exactly

$$\begin{aligned} C_T \epsilon 2^{-k} \frac{uv}{b^2} (b/u)^{p_4} (u/v)^{p_h} v^{p_1} \\ = C_T \epsilon 2^{-k} (b/u)^{p_4-2} (u/v)^{p_h-1} v^{p_1}. \end{aligned} \quad (5.16)$$

This equality is where both location-count thresholds enter. Let  $m = \min\{p_4 - 2, p_h - 1, p_1\} > 0$ . The three base ratios in the last line belong to  $(0, 1]$  and their product is  $b$ . Every term is therefore at most  $C_T \epsilon 2^{-k} b^m$ . There are  $O_T((1 + \log(1/b))^2)$  dyadic pairs. Absorbing this factor into  $b^{m/2}$  proves (5.15), with  $c_* = m/2$ .

If  $b \geq 1$ , the centers whose squares can affect the test occupy area at most  $C_T b^2$ . The crude bound one on pivotality, combined with the marking probability, gives  $C_T \epsilon 2^{-k}$ . The same crude count covers  $b$  within a fixed factor of one. Omitted transition rings at smaller scales have already been absorbed into  $C_T$  in (5.14). The computation uses only upper bounds on individual mark probabilities; partial activation can therefore be arbitrary in space. For a fixed annular shape the geometric constants scale with its inner radius and are independent of its location or orientation.  $\square$

**Proposition 5.8** (Uniform bootstrap of the three checks). *There is  $\epsilon_* > 0$ , chosen from the three fixed annular shapes and the fixed constants of the mark law, such that the following holds on every sufficiently fine finite torus. Activate the independent marks in order of increasing radius. At a stage with maximum permitted radius  $b$ , every check of ratio  $A$  with inner radius at least  $C_{\text{mon}} b$ , and with its prescribed bulk collars, satisfies*

$$\mathbb{P}(T_i^+) \leq q_i + \tau_i, \quad i \in \{4, h, 1\}. \quad (5.17)$$

*This remains true throughout continuous, possibly spatially nonuniform activation of marks of the current radius.*

*Proof.* There are finitely many mark variables on the torus. Activate one level at a time, increasing its Bernoulli parameters continuously from zero to their prescribed values. Conditional on the pure spins, the expectation of an increasing event of these marks is a polynomial in their parameters. Differentiating in a parameter gives the probability that this index is pivotal when its own mark is omitted. Integration is the usual finite-product interpolation identity and requires no monotonicity in the spins. For any collection of checks, even a continuum of geometric frames, these expectations are uniformly Lipschitz in the finitely many parameters: the sum of their absolute parameter changes bounds the change of every event probability. Thus their supremum is continuous as well.

Before any marks are activated, the estimates hold with strict slack, by the choice of  $s_0$ . At each stage monitor only checks with inner radius at least  $C_{\text{mon}}$  times the current radius. When the cutoff increases, some checks leave this collection. They are not needed again: the rings used for a

later insertion start at a larger fixed multiple of that insertion's radius. Conversely, any check still monitored at a given stage has been monitored throughout all earlier stages.

Suppose there were a first failure of (5.17). Up to that stage all ring possibility estimates used to control an insertion satisfy (5.17); by continuity the same weak bounds hold at the candidate first failure. Each pivotal insertion for any monitored check therefore satisfies (5.14). Apply Lemma 5.7 to that check, summing all previous levels and the activated fraction of the current level. Since these checks are translates, rotations, and dilates of only three fixed shapes, their total probability increase is at most

$$C_{\text{boot}}\epsilon \sum_{k \geq 0} 2^{-k} \leq 2C_{\text{boot}}\epsilon.$$

Choose  $\epsilon_*$  so that this is less than  $\frac{1}{2} \min_i \tau_i$ . The first failure cannot occur. The same estimate, being uniform over the frames, rules out a failure occurring only as a supremum. This proves the bootstrap on each finite torus, before taking any limit.

The order of the choices is now explicit: fix the collars and gap factor, choose  $A$  and the tolerances, enlarge  $s_0$ , and then choose  $\epsilon_*$ . The finitely many universal shapes determine  $C_{\text{boot}}$ . No later polygonal test enters this choice. The enlargement of  $s_0$  is made before the critical trajectory of Theorem 4.5 is constructed.  $\square$

## 5.6 Vanishing for every fixed family

*Proof of Theorem 5.1.* Fix one of the tests  $T$  in the statement. During increasing activation of the marks, apply the finite-product interpolation identity to  $D_T$ . Its initial value is zero. If a new mark first makes  $D_T = 1$ , choose two permitted assignments with different test values and revert the newly permitted square in a witnessing assignment. Before its addition all assignments gave one value. We obtain two configurations which differ only inside that square and change the test. Lemma 5.2 applies, and its required outside checks are possibility events using the older marks. Proposition 5.8 controls these checks whenever the insertion is smaller than the test. For larger insertions use the crude overlapping-square count in Lemma 5.7. We conclude that

$$\mathbb{P}(D_T = 1) \leq C_T \epsilon \sum_{k \geq 0} 2^{-k} \min \left\{ 1, \left( \frac{\delta(Ls_k + O(1))}{a_T} \right)^{c_*} \right\}. \quad (5.18)$$

Absent torus levels can be set to zero; the displayed infinite sum is a convenient upper bound. For every fixed  $k$ , its factor in braces tends to zero with  $\delta$ . Every summand is dominated by  $2^{-k}$ , whose sum is finite. Dominated convergence therefore proves  $\mathbb{P}(D_T = 1) \rightarrow 0$ . Equivalently first make the level tail small, and then refine the mesh for the finitely many remaining levels. This argument keeps  $\epsilon$  fixed; it does not require a new smallness condition depending on  $C_T$ .

A union bound proves (5.4) for the fixed finite family. Under Proposition 4.7, the perturbed configuration belongs to  $\mathcal{C}(\sigma^0, M)$  except with probability  $o(1)$ . On the complement of that exceptional event and all  $D_T$ , the entire test vector agrees with the pure vector. Its total-variation distance from the pure test-vector law tends to zero. The joint positivity and four-passage conclusions follow respectively from Proposition 2.13 and Proposition 2.12.  $\square$

We record the precise quantifiers in the upper-bound consequence. For every fixed buffered square annulus of radii  $r, R$ ,

$$\limsup_{\delta \downarrow 0} \mathbb{P}_{\text{pert}}(F_4(z, \theta; r, R)) \leq \limsup_{\delta \downarrow 0} \mathbb{P}_{0,N}(F_4(z, \theta; r, R)) \leq C(r/R)^{2+c}. \quad (5.19)$$

Only after this mesh limit do we send  $r/R$  to zero. In particular, for a fixed compact chart and fixed  $R$ , cover the possible centers by an  $r$ -grid, enlarge the inner squares by a fixed factor, and reduce the outer radius from  $R$  to  $R/2$ . The finite union for each fixed  $r$  has at most  $Cr^{-2}$  members. Applying (5.19) to that finite family and then sending  $r \downarrow 0$  gives

$$\lim_{r \downarrow 0} \limsup_{\delta \downarrow 0} \mathbb{P}_{\text{pert}}(\text{some center in the compact chart has four passages from } r \text{ to } R) = 0. \quad (5.20)$$

Thus the family grows only after its own mesh limit has been taken. Later fixed-buffer density comparisons will transfer this statement from torus charts to the interior of the specified finite domains.

## 6 An exact representation and comparison at boundary incidences

The bulk comparison of Section 5 concerns unconditioned torus laws. We now obtain the two tools needed to use it in a domain with prescribed exterior spins. The first orders spins across certain *strict* changes of boundary incidences. The second compares all interior spin events, pointwise in their densities, across a fixed free buffer. Neither tool requires positive association of the perturbed spin law with arbitrary pins.

Throughout this section the inverse temperature is the value supplied by Theorem 4.5. We abbreviate the conditional spin law of Lemma 1.2 by  $\nu_{\Omega, \lambda}^\tau = \nu_{\Omega, \beta_c(J, U, \lambda), \lambda}^\tau$ , and write  $\mu_{\lambda, N}$  for the unconditioned law on a square torus of side  $N$ . We take tori larger than the fixed interaction range and than all the charts used below. All thresholds in the local construction depend only on the fixed interaction, not on a domain or its pins.

### 6.1 Private auxiliary variables for a local factor

Write  $K_0 = \beta_0 J$ . The spin weight can be expressed as

$$\exp \left( K_0 \sum_{uv} \sigma_u \sigma_v \right) \prod_{\alpha, x} F_\alpha(\sigma_{x+A_\alpha}), \quad \max_{\alpha} \|F_\alpha - 1\|_\infty \longrightarrow 0 \quad (\lambda \longrightarrow 0). \quad (6.1)$$

Here the sum is over nearest-neighbor bonds, the index  $\alpha$  ranges over a fixed finite list of finite supports, and  $x$  ranges over their translates. Every  $F_\alpha$  is positive. Lemma 1.2 gives the factors coming from  $U$ ; the change  $\beta_c J - K_0$  is included as another bond factor. Enlarging a support does not change its factor. Thus we may fix, for each  $\alpha$ , a finite nearest-neighbor tree  $T_\alpha$  whose vertex set contains  $A_\alpha$ . Singleton supports are allowed.

We use binary variables with values in  $\{-1, +1\}$ . A variable placed on a bond will be called a *bit*, to distinguish it from a physical spin at a lattice vertex. For a fixed integer  $M$ , put  $M$  bits  $b_{e,1}, \dots, b_{e,M}$  on every nearest-neighbor bond  $e = uv$ . Choose  $a > 0$  so that

$$\frac{M}{2} \log \cosh(2a) = K_0. \quad (6.2)$$

Then summing the bits in  $\prod_{i=1}^M \exp(a(\sigma_u + \sigma_v)b_{e,i})$  gives a spin-independent constant times  $\exp(K_0 \sigma_u \sigma_v)$ . In particular, every bit has a strictly positive coupling to both ends of its bond.

The reason for using these variables is a strict distinction between free spins and pins. At the unperturbed vertex, summing the physical spin gives  $2 \cosh(a \sum_i b_i)$ , whereas fixing it to either sign gives  $\exp(\pm a \sum_i b_i)$ . In the first factor the conditional odds of one bit increase with every other incident bit. Its logarithmic increments lie strictly between those of the minus and plus factors. We

will retain these finite-table inequalities under perturbation. The construction below must therefore realize each local factor exactly while changing every vertex table by a quantity tending to zero.

**Proposition 6.1** (Exact vertex representation). *There is a finite choice of  $M$  such that, for all sufficiently small  $|\lambda|$ , positive functions*

$$W_v(\sigma_v, b_{\mathcal{B}(v)})$$

*represent the spin weight in (6.1) exactly, up to a constant, as the spin marginal of  $\prod_v W_v$ . Here  $\mathcal{B}(v)$  is the set of the  $4M$  bits on bonds incident to  $v$ . The same functions are used whether a physical spin is free or pinned. There are only finitely many local tables, and uniformly in their entries*

$$W_v(\sigma, b) \longrightarrow W_v^0(\sigma, b) := \exp \left( a\sigma \sum_{i \in \mathcal{B}(v)} b_i \right) \quad \text{as } \lambda \longrightarrow 0. \quad (6.3)$$

*The representation applies to either sign of every sufficiently small local perturbation.*

*Proof.* We construct one factor on a tree  $T$ , and then give different constructions disjoint sets of bits. For each complete spin pattern  $t \in \{-1, +1\}^{V(T)}$ , reserve two private bits on every edge of  $T$ , with the same coupling  $a$ . Conditional on all the physical spins, these bits are independent in the unperturbed representation. For the two bits on  $e = uv$ , set  $D_e = b_{e,1} - b_{e,2}$ . Their conditional moments are

$$\mathbb{E}[D_e \mid \sigma] = 0, \quad q_e(\sigma_u, \sigma_v) := \mathbb{E}[D_e^2 \mid \sigma] = 2 \operatorname{sech}^2(a(\sigma_u + \sigma_v)) > 0. \quad (6.4)$$

At a vertex  $v$  of this private copy of  $T$ , multiply its vertex factor by

$$1 + c_{v,t} \mathbf{1}_{\{\sigma_v = t_v\}} \prod_{\substack{e \in E(T) \\ e \ni v}} D_e. \quad (6.5)$$

An empty product is one. In the expansion of the product over vertices, a chosen subset  $S \subseteq V(T)$  contributes one copy of  $D_e$  for each end of  $e$  in  $S$ . If  $S$  is nonempty and proper, the connectedness of  $T$  gives an edge with exactly one end in  $S$ . Its centered difference makes this term vanish upon summing the private bits. The only surviving terms are therefore the empty selection and the full selection. Their sum is

$$1 + \mathbf{1}_{\{\sigma_{V(T)} = t\}} \left( \prod_{v \in V(T)} c_{v,t} \right) \left( \prod_{uv \in E(T)} q_{uv}(t_u, t_v) \right). \quad (6.6)$$

The same formula holds for a singleton tree, with an empty edge product.

For a target factor  $F$  on a subset of  $V(T)$ , put

$$\varepsilon_t = F(t) - 1, \quad Q_t = \prod_{uv \in E(T)} q_{uv}(t_u, t_v) > 0.$$

Choose coefficients whose product is  $\varepsilon_t/Q_t$ . For example, if  $m = |V(T)|$ , take absolute values  $|\varepsilon_t/Q_t|^{1/m}$  at every vertex, place the sign of  $\varepsilon_t$  at one prescribed vertex, and take all coefficients zero when  $\varepsilon_t = 0$ . Every coefficient tends to zero as  $F \rightarrow 1$ , for either sign of  $\varepsilon_t$ . Equation (6.6) then realizes  $1 + \varepsilon_t \mathbf{1}_{\{\sigma_{V(T)} = t\}}$ . Use distinct private bits for different patterns. Conditional independence

allows their contributions to be multiplied, and exactly one pattern is active at a given physical spin configuration. Hence

$$\prod_t \left(1 + \varepsilon_t \mathbf{1}_{\{\sigma_{V(T)}=t\}}\right) = F(\sigma).$$

There are finitely many trees, finitely many patterns on each, and only finitely many translated trees using a given bond. First count these uses, and choose  $M$  large enough to assign all their pairs of private bits disjointly. Unused bits retain their original factor. This allocation can be made translation covariant; its number is independent of the size of the torus. Choose  $a$  only after this finite allocation, using (6.2). The product of all the factors (6.5) at  $v$ , times  $W_v^0$ , defines  $W_v$ . Each  $D_e$  has absolute value at most two. Since a vertex belongs to only finitely many assigned trees, all correction factors are positive when their coefficients are small enough. Their tables then converge uniformly to one, proving (6.3). Summing all private channels, with the physical spins fixed, proves the claimed exact identity.  $\square$

The construction need not be analytic in  $\lambda$ : taking a root of  $|\varepsilon_t|$  provides the required continuity. The analytic construction of the critical temperature was completed in Section 4 before this representation was introduced.

## 6.2 Strict local margins and their comparison consequence

For a positive function  $H$  of bits, let

$$d_i \log H(b_{-i}) = \log H(b_i = +1, b_{-i}) - \log H(b_i = -1, b_{-i}).$$

The function is *log-supermodular* if every such increment is nondecreasing in every other coordinate. On a Boolean cube this is equivalent to  $H(b \wedge b')H(b \vee b') \geq H(b)H(b')$ ; it suffices to sum the two-coordinate differences along coordinate paths. A positive probability mass with this property is also called  $\text{MTP}_2$ .

Summing or fixing the spin at  $v$  gives respectively the factors

$$F_v(b) = W_v(+1, b) + W_v(-1, b), \quad P_v^+(b) = W_v(+1, b), \quad P_v^-(b) = W_v(-1, b).$$

The conditional probability of a free physical spin being plus is

$$k_v(b) = \frac{W_v(+1, b)}{F_v(b)}.$$

We distinguish these vertex factors from the conditional odds of a bit in the entire measure, which also involve its other endpoint.

**Lemma 6.2** (Finite-table margins). *After decreasing the allowed interval of  $\lambda$ , every free factor  $F_v$  is strictly log-supermodular in each pair of distinct incident bits, and every kernel  $k_v$  is strictly increasing in each incident bit. For every incident coordinate  $i$ , moreover,*

$$\sup_b d_i \log P_v^-(b) < \inf_b d_i \log F_v(b) \leq \sup_b d_i \log F_v(b) < \inf_b d_i \log P_v^+(b). \quad (6.7)$$

*The gaps are bounded below by a positive constant independent of the volume and the pins.*

*Proof.* At  $\lambda = 0$ , put  $H = a \sum_{j \neq i} b_j$ . Then

$$F_v^0(b) = 2 \cosh \left( a \sum_j b_j \right), \quad d_i \log F_v^0 = \log \cosh(H + a) - \log \cosh(H - a).$$

Because  $-1 < \tanh u < 1$  for finite  $u$ ,

$$-2a < d_i \log F_v^0 < 2a. \quad (6.8)$$

The increments of  $\log P_v^{-,0}$  and  $\log P_v^{+,0}$  are exactly  $-2a$  and  $2a$ . For two distinct bits the mixed difference of  $\log F_v^0$ , writing  $H$  now for the contribution from all the remaining bits, is

$$\log \cosh(H + 2a) + \log \cosh(H - 2a) - 2 \log \cosh H > 0,$$

by strict convexity. Finally, the log odds of the free spin are  $2a \sum_j b_j$ , with increment  $4a > 0$  in each bit. There are only finitely many entries in these tables. Their strictly positive minimum gaps persist under the uniform positive perturbation in (6.3). This proves all the assertions.  $\square$

The free factors therefore give an attractive law on bits when all physical spins are free. The factors of a pinned spin have not been claimed attractive. The strict inequalities (6.7) instead provide exactly the cross-law comparisons needed below. The comparison uses the order-preserving Markov-chain argument underlying Holley's theorem [17, Theorem (6)]; here the strict incidence conditions will order the single-bit conditional probabilities.

**Proposition 6.3** (Comparison at strict incidences). *Consider two laws formed from the same vertex factors  $W_v$ , with possibly different sets of pinned physical spins. Label the laws lower and upper. Let  $V$  be a union of nearest-neighbor connected components of their common free vertex set. At each vertex outside  $V$  adjacent to  $V$ , require one of the following possibilities:*

| <i>lower law</i>    | <i>upper law</i>   |
|---------------------|--------------------|
| <i>free</i>         | <i>pinned plus</i> |
| <i>pinned minus</i> | <i>free</i>        |
| <i>pinned minus</i> | <i>pinned plus</i> |

*Then there is a coupling in which all bits on bonds touching  $V$ , and all physical spins in  $V$ , are lower than or equal to their upper-law counterparts. The assertion remains valid if arbitrary values of bits on bonds not touching  $V$  have been conditioned on separately in the two laws. It consequently remains valid for arbitrary mixtures of these exterior bit conditions.*

*Proof.* Let  $\mathcal{B}(V)$  consist of all bits on bonds with at least one endpoint in  $V$ . Condition in each law on all other bits. After summing every free physical spin, the conditional weight on  $\mathcal{B}(V)$  is a product of vertex factors, restricted to the conditioned values where necessary. A vertex in  $V$  has every incident bit in  $\mathcal{B}(V)$ . Its factor is the identical attractive function  $F_v$  in the two laws. Its contribution to the conditional log odds of a bit is therefore ordered whenever the two configurations of varying bits are ordered.

At an endpoint outside  $V$ , the table in the statement and (6.7) order the two contributions to those log odds, regardless of the values of all other incident bits. This last qualification allows both the separately conditioned bits and the other varying bits at that endpoint to differ. Adding the contributions of the two endpoints orders the full single-bit conditional log odds.

Run single-bit heat-bath chains in the two finite state spaces, updating the same coordinate and using the same uniform random number at each step. The ordered conditional probabilities preserve the coordinatewise order. Start from any ordered pair, and take a subsequential stationary limit of the coupled chains. Strict positivity makes each marginal chain irreducible and gives its prescribed Gibbs measure as its unique stationary law. We have thus obtained an ordered coupling of the bits.

Conditional on all bits, the free physical spins are independent, with the kernels  $k_v$ . At the vertices of  $V$  these kernels are the same in both laws and are increasing by Lemma 6.2. Another

family of common uniform random numbers gives the ordered physical spins. Finally, the argument worked for every pair of exterior bit configurations. Integrating these couplings against the two exterior distributions proves the mixture assertion.  $\square$

Two equal-sign pins at a boundary vertex are deliberately absent from the table. Their perturbed factors need not have ordered increments when their other incident bits differ. In particular, Proposition 6.3 does not imply unrestricted boundary monotonicity or the FKG inequality for pinned physical spins.

### 6.3 Stopped reveals and geometric shielding

The spin paths used here lie in the fixed triangulation obtained by adding the northwest–southeast diagonal to every elementary square. On bulk faces this realizes the contour pairing in Section 1. Its edges do not cross, and it contains every nearest-neighbor bond. We record the precise role of a reveal, so that conditioning on a crossing does not introduce an unexamined assumption about a spatial Markov property of the spin interaction.

**Lemma 6.4** (Reveals with a monochromatic facing boundary). *Suppose spins are queried one at a time, each next query and the stopping decision being determined by the preceding queries and their answers. Conditional on a complete transcript, the remaining law is exactly the original law with the revealed spins pinned. This remains true after conditioning on any event determined by the transcript.*

*In a finite triangulated disk with a prescribed inner disk, one can explore from its outer boundary for a shield of either color  $s$ . On success, no inner-disk spin has been queried, and every queried neighbor of the remaining component containing that disk has color  $s$ . A color- $s$  circuit in a narrower annular band separating the two disks guarantees success. All queried sites are on the exterior side of that circuit or on the circuit itself.*

*The corresponding exploration in a triangulated topological rectangle, from one side toward the opposite side, determines whether there is a color- $s$  crossing between the other two sides. On success its facing boundary contains such a crossing, and the exploration queries no spins strictly beyond it toward the opposite side. If that crossing extends past the ends of a comparison region, every nearest-neighbor incidence from the region across the explored side meets a pin of color  $s$ .*

*Proof.* Fix a deterministic query order to resolve all choices. A transcript lists queried vertices  $v_1, \dots, v_m$  and their values  $s_1, \dots, s_m$ . Any configuration agreeing with these values causes the same successive choices and the same stopping decision, by induction on the number of queries. Thus the transcript event is the cylinder  $\{\sigma_{v_i} = s_i : 1 \leq i \leq m\}$ . Conditioning on a transcript-measurable event adds nothing once a compatible transcript has been specified. Independent exploration randomness, if used, can first be conditioned on and then averaged out.

For the annular construction, query the outer graph boundary. Keep a queue of all revealed vertices of color  $-s$ , and query every previously unqueried triangulation-neighbor of a queued vertex before removing that vertex from the queue. Newly discovered vertices of color  $-s$  join the queue; vertices of color  $s$  do not. Declare failure just before a query would enter the protected inner disk. Declare success when the queue has been exhausted. This is a finite adaptive procedure of the kind just considered.

On success let  $V$  be the unqueried nearest-neighbor component containing the inner disk, or the union of such components if the chosen inner set has more than one component. An unqueried nearest neighbor of  $V$  belongs to  $V$ . A queried neighbor cannot have color  $-s$ , since every neighbor of such a vertex was queried before its queue entry was removed. Hence every boundary incidence

of  $V$  meets a revealed  $s$  vertex. If a color- $s$  circuit separates the protected disk from the outer boundary, every queued vertex has a color- $-s$  path to the outer boundary and so lies outside the circuit. A further queried neighbor cannot lie strictly inside: the edge to it would cross the circuit without meeting a circuit vertex. Planarity forbids this, and expansion stops at vertices of the circuit because they have color  $s$ . The positive separation of the narrower band from the protected disk proves the asserted implication from a circuit to success.

For a topological rectangle, seed the exploration on the inspecting side and expand color  $-s$  within the rectangle. A color- $-s$  path reaching the opposite side certifies failure. Otherwise the boundary of the reached color- $-s$  set facing that side gives a color- $s$  crossing between the two end sides. To check this finite planar assertion directly, join the reached components by a thin deterministic strip immediately outside the inspecting side. Trace the boundary of a small polygonal neighborhood of this connected set on the side facing the opposite boundary. Since the reached set does not meet that boundary, this trace has a component running between the two end sides. Every graph vertex encountered just beyond the reached set was queried and has color  $s$ , because otherwise it would have been added to the queue. As the trace passes through a triangular face, consecutive such vertices coincide or are joined by the remaining edge of that face. They consequently give a walk of color  $s$  between the end sides. Erasing closed subwalks gives a simple crossing. Conversely a crossing between the end sides blocks any opposite-color path between the inspecting and opposite sides, since the two planar paths would have to share a vertex. Boundary arcs may be separated at their corners by one lattice step; the prescribed buffered gates allow this harmless choice of finite rectangle convention. Thus the existence of the desired crossing is determined by the transcript, and on success no queries pass strictly beyond its facing boundary. In applying this construction, the two end gates lie past the comparison region, so a nearest-neighbor path cannot go around a gate within that region.

Finally, a nearest-neighbor bond cannot cross a simple path or circuit of the triangulation in the interior of an edge. If its ends lie on opposite sides, it must meet a path vertex. This proves the incidence assertion even when the separating path uses diagonal steps.  $\square$

Only the facing pins enter Proposition 6.3. Examined spins on the other side of a shield may have arbitrary signs, and all of them remain part of the conditional law. Likewise, when a crossing test uses vertices outside one of its two comparison domains, those vertices may be filled virtually by the extremal color for the purpose of testing the increasing or decreasing event. Such a filling does not change any physical pin or any interaction. The comparison applies to the common free vertices after its incidence hypotheses have been checked.

## 6.4 From an extreme odds ratio to all interior densities

Our final comparison will cut the auxiliary-bit graph along a square inside the free buffer. Conditional on all bit channels crossing that cut, the interior law is the same in the domain and on the torus; the exterior data change only the mixing law on the cut. We must therefore bound how much any two cut assignments can change the interior density.

For the bits of the *unconditioned* torus,  $\text{MTP}_2$  reduces this task to a single extreme odds ratio. Two nested monochromatic shields will bound that ratio using the incidence comparison and the circuit probabilities from Section 5. We begin with the finite algebraic reduction. The four-difference identity below is the one used in Lemma 2.1, “Extreme rectangles in a ferromagnetic marginal”, in Section 2 of [19]; its proof applies to any positive  $\text{MTP}_2$  mass.

**Lemma 6.5** (Rectangles of an  $\text{MTP}_2$  mass). *Let  $p(i, j) > 0$  be an  $\text{MTP}_2$  probability mass on two disjoint finite sets of binary coordinates  $I, J$ . Let  $i_-, i_+, j_-, j_+$  denote their constant minus and plus*

configurations, set  $f = \log p$ , and put

$$\Lambda = f(i_+, j_+) + f(i_-, j_-) - f(i_+, j_-) - f(i_-, j_+) \geq 0. \quad (6.9)$$

Then, for arbitrary configurations  $a, a'$  on  $I$  and  $b, b'$  on  $J$ ,

$$|f(a, b) + f(a', b') - f(a, b') - f(a', b)| \leq \Lambda. \quad (6.10)$$

Consequently every two conditional laws of  $I$  given  $J$ , and every two mixtures of those conditional laws, have pointwise density ratio between  $e^{-\Lambda}$  and  $e^{\Lambda}$ .

*Proof.* For  $u \leq v$  and  $s \leq t$ , define

$$\Delta(u, v; s, t) = f(u, s) + f(v, t) - f(u, t) - f(v, s) \geq 0.$$

Writing  $D = f(a, b) + f(a', b') - f(a, b') - f(a', b)$ , expansion gives the exact identity

$$\begin{aligned} \Lambda - D &= \Delta(i_-, a'; j_-, b) + \Delta(a, i_+; b', j_+) \\ &\quad + \Delta(a', i_+; j_-, b') + \Delta(i_-, a; b, j_+). \end{aligned}$$

Every term is nonnegative; no order between  $a, a'$ , or between  $b, b'$ , is required. Interchanging  $b, b'$  proves the other bound.

If  $q_b, q_{b'}$  are the two conditional masses on  $I$ , (6.10) says that  $\log(q_b/q_{b'})$  has oscillation at most  $\Lambda$ . Its density ratio has mean one under  $q_{b'}$ , so its minimum is at most one and its maximum is at least one. Both bounds in the statement follow. They persist after taking arbitrary mixtures in either law.  $\square$

We will also use preservation of  $\text{MTP}_2$  under marginalization. Here is a short proof in the finite setting at hand. Sum one binary coordinate  $z$  out of a positive  $\text{MTP}_2$  mass  $h$ , and check the increment ratio in another coordinate  $i$ . That ratio for the summed mass is the conditional average, under  $h$  with  $i = -1$ , of

$$r(z, y) = \frac{h(i = +1, z, y)}{h(i = -1, z, y)},$$

where  $y$  denotes the other coordinates.  $\text{MTP}_2$  makes  $r$  nondecreasing in  $z$  and in every coordinate of  $y$ . The conditional probability of  $z = +1$ , with  $i = -1$ , also increases when one of those coordinates of  $y$  increases, by the corresponding two-coordinate inequality for  $h$ . Thus the averaged ratio increases. This verifies each two-coordinate square for the marginal. Iterating proves the assertion. In particular, summing the physical spins from the unconditioned representation gives the bit mass  $\prod_v F_v$ , which is  $\text{MTP}_2$ , and any marginal of this bit mass is  $\text{MTP}_2$  as well.

**Proposition 6.6** (Interior comparison across a fixed buffer). *Fix a square  $Q$  and a larger square  $\tilde{Q}$ , with  $\bar{Q} \subset \text{int } \tilde{Q}$ , in a fixed physical torus chart. Take lattice mesh  $\delta \downarrow 0$ , with torus side  $N$  chosen so that this chart has a fixed macroscopic buffer from its translates. Suppose every lattice vertex in  $\tilde{Q}$  is free in a domain spin law  $\nu_{\Omega, \lambda}^\tau$ . For all sufficiently small mesh there is a constant  $C < \infty$ , depending on the two fixed squares and the chart but not on  $\Omega, \tau$ , or the mesh, such that*

$$C^{-1} \leq \frac{\nu_{\Omega, \lambda}^\tau(\sigma_{Q \cap \delta \mathbb{Z}^2} = \eta)}{\mu_{\lambda, N}(\sigma_{Q \cap \delta \mathbb{Z}^2} = \eta)} \leq C \quad \text{for every spin configuration } \eta. \quad (6.11)$$

*The same bounds hold for every event determined by these spins. The exterior pins may be arbitrary, including those induced by any admissible exterior contour configuration.*

*Proof.* All sets in this proof are chosen in physical coordinates before the mesh tends to zero. We identify a lattice vertex with its rescaled position when using  $Q$  and  $\tilde{Q}$ . Let  $I$  contain all bits incident to the vertices of  $Q$ . Choose two surrounding square annular bands  $A_1, A_2$ , with  $A_1$  inside  $A_2$ , and then a square cut outside  $A_2$ , all within  $\tilde{Q}$ . Leave positive distances between successive objects. The inner protected disk for an exploration in  $A_1$  contains the endpoints of every bit in  $I$ ; the inner protected disk for an exploration in  $A_2$  contains all of  $A_1$ . Choose slightly wider exploration bands around  $A_1, A_2$ , still disjoint and separated from the cut. A circuit in the original, narrower band then implies exploration success by Lemma 6.4.

More precisely, take a lattice square vertex set  $S$  containing both exploration bands, with its boundary separated from them, and let  $J$  be *all* bit channels on the nearest-neighbor bonds with one endpoint in  $S$  and the other outside. We require  $I$  to lie strictly inside this cut. Let  $\mathcal{B}_{\text{int}}(S)$  be the bits on bonds with both endpoints in  $S$ . Every physical vertex in  $S$  is free. Once  $J = j$  is fixed, the conditional mass of the inside spins and bits is

$$K_S(\sigma_S, b_{\mathcal{B}_{\text{int}}(S)} \mid j) = \frac{1}{Z_S(j)} \prod_{v \in S} W_v(\sigma_v, b_{\mathcal{B}(v)}), \quad b_J = j.$$

Every bit incident to a vertex of  $S$  is either an interior bit or belongs to  $J$ ; all remaining vertex factors lie outside  $S$ . This proves the factorization and shows that  $K_S$  is identical in the domain and torus laws. In particular, their inside-bit marginals can differ only through their mixing distribution on  $j$ . The representation is the same for all the laws by Proposition 6.1; exterior pins in the spin model are retained exactly by Lemma 1.2.

Let  $\hat{\mu}$  denote the joint torus law of spins and bits, and write  $\hat{\mu}_j = \hat{\mu}(\cdot \mid J = j)$ . For a sign  $s$ , let  $C_r^s$  be the fixed spin test of a color- $s$  circuit in  $A_r$ , and let  $E_r^s$  be success of the exploration with that color in its wider band. These are different events. The geometric implication is

$$C_r^s \subseteq E_r^s. \quad (6.12)$$

Only the fixed tests  $C_r^s$  are used as inputs to bulk stability. By Theorem 5.1 and the pure simultaneous signed-circuit positivity of Proposition 2.13, for all sufficiently fine meshes

$$\hat{\mu}(C_1^+ \cap C_2^-) \geq c, \quad \hat{\mu}(C_1^- \cap C_2^+) \geq c \quad (6.13)$$

for a fixed  $c > 0$ . We may take  $c \leq 1$ .

We first show that, for *every* cut configuration  $j$ ,

$$\hat{\mu}_j(C_1^+) \geq c, \quad \hat{\mu}_j(C_1^-) \geq c. \quad (6.14)$$

Condition the unconditioned torus experiment on  $E_2^-$ , and then on a successful complete exploration transcript. All physical spins of  $A_1$  remain free. Let  $V$  be the unqueried nearest-neighbor component containing the protected inner disk, and hence all of  $A_1$ . It is a component of the common free set in the revealed experiment and in  $\hat{\mu}_j$ . Every neighbor outside  $V$  is a revealed minus spin in the first experiment and a free spin in the second. The queried outer boundary of the exploration separates  $V$  from the cut, so no bit of  $J$  touches  $V$ . The minus-pin/free case of Proposition 6.3 orders the conditional experiment below  $\hat{\mu}_j$  on the spins of  $A_1$ . Arbitrary further exterior bit conditions are allowed by that proposition. Average over the successful transcripts. Since  $C_1^+$  is increasing, this gives

$$\begin{aligned} \hat{\mu}_j(C_1^+) &\geq \hat{\mu}(C_1^+ \mid E_2^-) \\ &= \frac{\hat{\mu}(C_1^+ \cap E_2^-)}{\hat{\mu}(E_2^-)} \geq \hat{\mu}(C_1^+ \cap C_2^-) \geq c. \end{aligned}$$

The denominator is positive by (6.13). Interchanging the colors and applying the comparison to the decreasing event  $C_1^-$  proves the second inequality. This step uses no monotonicity in the cut configuration  $j$ .

Now take the two extreme cut configurations  $j_-, j_+$ , and denote the corresponding laws by  $\hat{\mu}_-$  and  $\hat{\mu}_+$ . Under  $\hat{\mu}_-$ , the event  $E_1^+$  has probability at least  $c$ , by (6.14) and (6.12). Conditional on any successful transcript, the component containing all endpoints of  $I$  has facing plus pins. Compare it to  $\hat{\mu}_+$  before any reveal. The plus-pin/free case of Proposition 6.3 makes the former law larger on the bits  $I$ . The cut bits are outside the comparison component in both experiments. The increasing event  $\{I = i_+\}$  therefore satisfies

$$\hat{\mu}_-(I = i_+) \geq \hat{\mu}_-(E_1^+) \hat{\mu}_+(I = i_+) \geq c \hat{\mu}_+(I = i_+). \quad (6.15)$$

The opposite construction, starting from  $\hat{\mu}_+$  and revealing a minus shield, gives

$$\hat{\mu}_+(I = i_-) \geq c \hat{\mu}_-(I = i_-). \quad (6.16)$$

These inequalities compare possibly very small probabilities; they do not require a lower bound for either all-equal configuration on  $I$ .

The joint torus bit marginal  $p(I, J)$  is positive and MTP<sub>2</sub>, by Lemma 6.2 and marginalization as proved above. Its extreme ratio satisfies

$$e^\Lambda = \frac{\hat{\mu}_+(I = i_+) \hat{\mu}_-(I = i_-)}{\hat{\mu}_-(I = i_+) \hat{\mu}_+(I = i_-)} \leq c^{-2},$$

by (6.15) and (6.16). Apply Lemma 6.5. It bounds the density ratio of every two mixtures of the laws of  $I$  given  $J$  between  $c^2$  and  $c^{-2}$ . One mixture is the torus law, and the other is the domain law, by the exact cut factorization.

Finally, conditional on  $I$ , the physical spins in  $Q$  have the same product of Bernoulli kernels with plus probabilities  $k_v(I)$  in both laws. Integrating the pointwise bit-density inequalities against this positive kernel gives (6.11), with  $C = c^{-2}$ . Summation gives the event version. Every geometric separation was fixed before taking the fine-mesh threshold, so the constant is independent of the domain and its exterior pins.  $\square$

The buffer in Proposition 6.6 stays fixed when a smaller test box shrinks inside  $Q$ . Consequently its constant can be used for all boxes in a fine grid inside a compact chart. Combining it with the torus four-passage estimate of Theorem 5.1 and Proposition 2.12 gives the compact-chart bottleneck bound used in Section 7. That section gives the grid argument with the order of the mesh and radius limits explicit.

## 7 Signed boundary barriers and convergence of the full curve

We now pass from the torus laws to the domains and exterior contours in Theorem 1.1. Theorem 5.1 compares finite vectors of polygonal spin tests in the pure and perturbed tori. Proposition 6.3 transfers colored crossings from those tori to the target domain, after suitable boundary paths have been revealed. These crossings will place the target trace between two pure comparison chords with the same limiting SLE<sub>3</sub> law.

Identifying a trace does not determine its traversal order. We use the other output of Section 6, the fixed-buffer density bound in Proposition 6.6, to transfer the bulk four-passage estimate to target interior events. It rules out reverse travel of positive diameter along the limiting chord and

gives the oriented curve metric (1.2). Thus density comparison controls exceptional events; the colored-barrier argument identifies the limiting law.

Fix a permitted  $J, U$ , and choose the perturbation interval small enough for Theorem 4.5, Theorem 5.1, and Proposition 6.3. No further reduction depending on a domain will be made. Throughout this section the perturbed model is at  $\beta_c(J, U, \lambda)$ , and  $\mu_{\lambda, N}$  denotes its torus spin law. The pure torus law is  $\mu_{0, N}$ . Write

$$D_n = \delta_n D_{\Omega_n}, \quad \gamma_n = \delta_n \gamma(P_n).$$

By Lemma 1.2, its spin law is the conditional law  $\nu_{\Omega_n, \beta_c, \lambda}^{\tau_n}$  with the actual exterior configuration  $\tau_n$  induced by  $P_{o, n}$ . We keep  $\gamma_n$  as the prescribed resolution of  $P_n$  and its two half-stubs. Spin paths and bulk passage tests use the fixed triangulation of Section 5. In a bulk face, triangular interpolation and the prescribed contour have the same connections and differ by  $O(\delta_n)$  with their traversal order preserved.

At the boundary one must distinguish the prescribed interface from the zero set obtained by interpolating the entire exterior spin field. Exterior-only disagreement edges can enter a boundary face and change its full-plane connections. We will instead redraw the target locally beside the signed strips, using the geometric filling proved in Lemma 7.2. This changes its drawing by  $O(\delta_n)$  and leaves the conditional spin law with exterior  $\tau_n$  intact.

## 7.1 Fixed displaced domains and their boundary incidences

Let  $I_+$  and  $I_-$  be the two open arcs of  $\partial D$  carrying the plus and minus exterior nearest neighbors. There are only two choices. It suffices to treat a subsequence on which this choice is fixed. Choose a Jordan–Schoenflies homeomorphism  $\Psi$  of the plane [6, Section 1 and Section 2(A), p. 860] such that

$$D = \Psi(\{\rho < 1\}), \quad a = \Psi(1), \quad b = \Psi(-1), \quad I_+ = \Psi(\{e^{i\theta} : 0 < \theta < \pi\}).$$

Only the topology of these coordinates is used. All spin tests will be fixed polygonal tests in the original Euclidean coordinates.

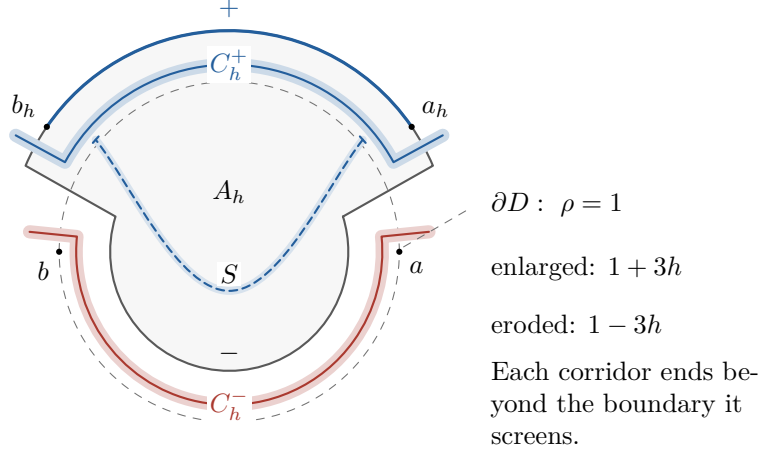
We first construct a domain favoring minus, from which plus crossings can be transferred to the target. Move its plus boundary outward and its minus boundary inward. A minus corridor will screen the target’s minus arc, while a disjoint plus corridor will screen the comparison domain’s plus arc. Conditioning a torus on both crossings then permits the two strict-incidence comparisons. The caps must extend beyond the boundaries they screen, so that a common free component cannot go around them.

For an explicit construction, take  $0 < h < \pi/30$ , also small enough that  $3h < 1/2$ , and let  $R_h$  be the continuous periodic function which is  $1 + 3h$  on  $[4h, \pi - 4h]$ , is  $1 - 3h$  outside  $(3h, \pi - 3h)$ , and is affine on the two remaining transition intervals. The preliminary comparison domain is

$$A_h^{\text{top}} = \Psi(\{\rho e^{i\theta} : 0 \leq \rho < R_h(\theta)\}). \quad (7.1)$$

Its marked points have angles  $6h$  and  $\pi - 6h$ , on the enlarged boundary. The arc between them through  $\theta = \pi/2$  is plus; the rest is minus.

Two disjoint topological rectangles implement the comparisons. The center of the minus rectangle  $C_h^-$  follows  $\rho = 1 - h$  along the long angular interval  $[\pi - h, 2\pi + h]$ . At its two ends it bends radially outward near angles  $\pi - h$  and  $h$ , crosses  $\rho = 1$ , and ends at gates with  $\rho > 1 + h$ . It is wholly outside  $\overline{A_h^{\text{top}}}$ . The center of the plus rectangle  $C_h^+$  follows  $\rho = 1 + h$  from angle  $5h$  to angle  $\pi - 5h$ . Its ends bend outward at those angles, cross  $\rho = 1 + 3h$  on the minus part of  $\partial A_h^{\text{top}}$ , and end at gates



**Figure 2:** The minus-favoring construction for a small displacement  $h > 0$ , in Schoenflies coordinates. Here  $A_h$  is a sufficiently close polygonal approximation, in the physical plane, of the preliminary displaced topological domain;  $a_h, b_h$  are its incoming and outgoing boundary marks. The dashed circle is  $\partial D$ . The preliminary boundary is enlarged along its middle plus arc and eroded elsewhere. The red minus corridor  $C_h^-$  is outside  $A_h$ , including its caps; the blue plus corridor  $C_h^+$  is outside  $D$ , and crosses the minus boundary of  $A_h$  before its end gates. Corridor widths and radial displacements are exaggerated. The dashed blue strip  $S$  illustrates an allowed plus strip: its gates lie just outside  $D$ , and the strip lies inside  $A_h$ , on the center side of both corridors. Such strips are selected from a finite deterministic list before coupling any crossing events. The diagram specifies topology and signs, not a metric distortion estimate for  $\Psi$ .

with  $\rho > 1 + 4h$ . This rectangle is wholly outside  $\bar{D}$ . Use sufficiently small positive widths. Include slightly larger rectangles on their radially exterior sides as the scopes in which a crossing will be revealed. The end caps of these larger rectangles still cross, and extend beyond, the boundary being cut. Compactness and the strict inequalities just specified allow the widths to be chosen so that

$$\overline{\text{scope}(C_h^-)} \cap \overline{A_h^{\text{top}}} = \emptyset, \quad \overline{\text{scope}(C_h^+)} \cap \bar{D} = \emptyset, \quad (7.2)$$

and the two scopes are disjoint. A lengthwise minus crossing cuts the minus boundary of  $D$  off from the center. A lengthwise plus crossing cuts the plus boundary of  $A_h^{\text{top}}$  off from the center. Figure 2 records these placements.

Replace the boundary, corridors, scopes, and gates by sufficiently close polygonal approximations in the physical plane. One way to do this is to subdivide each of the finitely many simple arcs into pieces lying in disjoint small neighborhoods of nonadjacent pieces, then join points in successive neighborhoods by polygonal arcs; small transverse polygonal gates give the rectangles. The rectangular neighborhoods of the arcs are transported from the coordinate plane by  $\Psi$ . Working within these neighborhoods preserves the order and sides in the polygonal approximations. All the displayed disjointness conditions, cap crossings, and gaps between designated arcs are open conditions with positive clearance. Hence they survive sufficiently close approximations. Denote the resulting polygonal domain by  $A_h$  and its marked points by  $a_h, b_h$ . Choose the approximation errors tending to zero with  $h$ , including in boundary parametrization. Ordinary lattice approximations of this fixed polygonal marked domain will always be used for its pure Dobrushin law.

For the comparison favoring plus use the radial function  $R_h(\theta - \pi)$  instead. Its incoming and outgoing marks have angles  $2\pi - 6h$  and  $\pi + 6h$ , respectively; the enlarged intervening arc through  $3\pi/2$  is minus, and its complement is plus. Rotate the two corridor placements by  $\pi$  and interchange their colors. Thus the inner plus corridor lies outside this comparison domain and the outer minus corridor lies outside  $D$ . Orient its pure chord from the mark near  $a$  to the mark near  $b$ , as before.

We spell out why these are valid comparisons for the given target approximations. If  $c$  parametrizes  $\partial D$ , then  $\phi_n \circ c$  parametrizes  $\partial D_n$  and converges uniformly to  $c$ . On any compact set disjoint from  $\partial D$ , winding numbers of these curves agree eventually; thus membership in  $D_n$  agrees there with membership in  $D$ . On a closed subarc separated from  $a, b$ , the order along the boundary is also preserved, since  $\phi_n$  is a homeomorphism taking the two marks to the prescribed marks. Consequently every fixed clearance, inclusion, and signed-arc placement above holds for  $D_n$  for all sufficiently large  $n$ . Fine teeth or fjords lie in the shrinking boundary collar. They do not replace any actual exterior pin, and they do not intersect a compact clearance chosen at fixed  $h$ .

An allowed *plus strip* is a polygonal topological rectangle  $S$  with two end gates, whose closure is in  $A_h$ , on the center side of both cutting corridors and outside both reveal scopes. Its two gates are outside  $\bar{D}$ , on the enlarged plus portion. We may require  $S \subset \Psi(\{\rho < 1 + h/2\})$ , with a slight decrease of this bound to absorb polygonal tolerances. Wherever  $S$  crosses  $\partial D$ , it does so on  $I_+$  with positive clearance from the marks. Only strips satisfying all these conditions are used. The relevant portion of  $A_h$  is on the center side of  $C_h^-$  because that entire corridor is outside  $A_h$ . The strips needed below exist around compact subarcs of a pure interface; their selection will be justified separately.

**Proposition 7.1** (Fixed-geometry boundary squeeze). *Fix the domain, signs, a displacement  $h$ , the preceding polygonal placements, and a finite list  $S_1, \dots, S_m$  of allowed plus strips. Let  $V_A \in \{0, 1\}^m$  record their plus crossings in the pure Dobrushin law in an ordinary mesh approximation of  $A_h$ . Let  $V_n^+$  record plus crossings in the target, filling the portions of the strips outside  $D_n$  virtually with plus. There is a coupling with*

$$\mathbb{P}(V_A \leq V_n^+) = 1 - o(1) \quad (n \rightarrow \infty), \quad (7.3)$$

where the inequality holds coordinatewise. The error is uniform over all admissible target exterior configurations. The sign-reversed construction gives the corresponding transfer of minus crossings from pure domains favoring plus. The smallness condition on  $\lambda$  is the one fixed at the beginning of the section; only the mesh threshold and the error bound depend on these fixed geometries.

*Proof.* Place all sets in a torus chart with a fixed positive external buffer. Let  $E_h$  be the event of simultaneous lengthwise minus and plus crossings of  $C_h^-$  and  $C_h^+$ . Proposition 2.13 gives  $\mu_{0,N}(E_h) \geq q_h > 0$  for small mesh. By Theorem 5.1,  $\mu_{\lambda,N}(E_h) \geq q_h/2$  eventually, and the joint laws of

$$(\mathbf{1}_{E_h}, V(S_1), \dots, V(S_m))$$

in the pure and perturbed tori have total variation distance  $\varepsilon_n \rightarrow 0$ . Here each  $V(S_i)$  is an ordinary bulk crossing test in the torus, without virtual filling. For any two laws  $p, q$  on a finite space, an event  $E$  with  $p(E), q(E) \geq q_* > 0$ , and a random vector  $V$  on that space, subtraction of the two conditional fractions gives

$$\|p(V \in \cdot \mid E) - q(V \in \cdot \mid E)\|_{\text{TV}} \leq \frac{2\|p - q\|_{\text{TV}}}{q_*}. \quad (7.4)$$

Thus the two middle vectors conditioned on  $E_h$  can be coupled to agree with probability  $1 - o(1)$ . The laws  $p, q$  in this application are the joint indicator laws just displayed; no total variation estimate on the underlying spin fields is used.

In each conditioned torus reveal the crossing closest to the radially exterior side of each corridor. Use the stopped reveal of Lemma 6.4. The transcript examines the exterior side up to the crossing and leaves only pins of its prescribed color facing the center. The crossing and the event of its

being the first such crossing are determined by this transcript. After both transcripts have been conditioned upon, no additional crossing-existence condition remains on the unrevealed center. The two scopes are disjoint, so the second reveal does not enter the core left by the first.

For precision, the free comparison components and all their boundary incidences are as follows. In the first comparison, the lower law is the pure law in  $A_h$ , and the upper law is the conditioned pure torus. Take the components of the common free region on the center side of its revealed plus path. This path is an upper plus pin against lower free spins. All accessible portions of  $\partial A_h$  are lower minus pins against upper free spins: the plus boundary is screened by the revealed path, whose caps continue outside  $A_h$ . At a meeting of a path and a boundary, the pair is upper plus against lower minus. The minus reveal is outside  $A_h$  by (7.2). These checks give a joint comparison on all spins used by the strips, and therefore

$$V_A \leq V_0^{E_h}.$$

In the second comparison the lower law is the conditioned perturbed torus and the upper law is the target. Take the common free components on the center side of the revealed minus path. On this path the lower pin is minus and the upper vertex is free. The target minus arc is screened by this path and its caps. Every remaining accessible target boundary neighbor is an upper plus pin against a lower free vertex. Where the two cuts meet, the pair is upper plus against lower minus. The plus reveal lies outside the target. The comparisons are precisely

$$(\text{upper}, \text{lower}) = (+, \text{free}), \quad (\text{free}, -), \quad (+, -).$$

No same-sign pinned pair, and no monotonicity of an arbitrary pinned perturbed factor, is being assumed. A diagonal segment of a separating triangulated path cannot be crossed by the interior of a nearest-neighbor edge. Thus each nearest-neighbor incidence leaving a comparison component actually meets one of the path or arc pins just enumerated. This remains true when a strip exits and reenters  $D_n$ . Condition on all other auxiliary bits as in Proposition 6.3; the strict inequalities are uniform in those bits and hence in the unprescribed spin pattern farther behind the arcs. The resulting spin comparison and the extremal virtual values outside  $D_n$  give

$$V_\lambda^{E_h} \leq V_n^+.$$

Finally glue the two order couplings with the equality coupling supplied by (7.4). This is elementary disintegration on finite spaces: first sample each common middle vector, then sample the remaining variables from the relevant conditional laws. Extend the outer vectors to their complete configurations in the same way. This proves (7.3). In particular the conclusion is about a finite vector of tests, not a total variation comparison of spin configurations. Interchanging plus and minus and exchanging the enlarged and eroded arcs proves the other assertion.  $\square$

The virtual filling in the proposition is a rule for evaluating a spin crossing. We now justify its use as a geometric barrier, including at corners of the target boundary.

**Lemma 7.2** (Local drawing beside a signed strip). *Fix finitely many allowed plus strips and, in the sign-reversed construction, finitely many allowed minus strips. For all sufficiently large  $n$ , the prescribed target curve has a drawing  $\tilde{\gamma}_n$  with the same traversal order and*

$$d_{\text{curv}}(\tilde{\gamma}_n, \gamma_n) = O(\delta_n)$$

*that is disjoint from every monochromatic crossing of its respective virtually filled strip. The two half-stubs and neighborhoods of the marks can retain their original drawings. The filling and redrawing change no spin in the conditional law and remove no interaction term.*

*Proof.* Consider first a plus strip. Its boundary contacts lie on a compact subarc of  $I_+$  separated from the marks. For large  $n$ , the same is true of its contacts with  $\partial D_n$ . Every exterior vertex sharing a nearest-neighbor bond with the domain at those contacts is already plus, by Lemma 1.2. Fill the remaining exterior vertices of the incident square faces with plus for the purpose of drawing only.

Inspect a square face of the primal lattice, or equivalently the four cells incident to one dual vertex. The Jordan condition allows one, two adjacent, or three interior cells at a boundary vertex; two opposite interior cells alone would give a diagonal pinch. With one interior cell, its two nearest exterior neighbors are already plus. Only the opposite exterior vertex might have a different actual spin, and filling it leaves every disagreement edge in  $E_{\Omega_n}$  unchanged. The resulting zero set joins the two occupied domain edges when there are two. With two adjacent or three interior cells, every exterior cell of the face is a nearest exterior neighbor, so its spin already has the filling value. In each case the interpolated zero segments have exactly the partial contour's connections: its unique connection at degree two, and the prescribed NE/SW pairing at degree four. Faces wholly inside the domain have the same property without filling.

Redraw the target portions in all faces meeting the strip, together with one adjacent lattice layer, as the corresponding zero segments. A monochromatic edge of the triangulation is disjoint from that zero set. Both the original drawing and each replacement meet a face side at the same disagreement-edge midpoint. The replacements therefore glue to one another and to the unchanged portions without changing a local connection. Matching the successive face crossings in their original order gives the displayed  $O(\delta_n)$  curve distance.

The minus construction is identical with the colors exchanged. The boundary-contact sets of the fixed plus and minus strip lists have positive separation from each other and from the marks. Therefore, for sufficiently fine mesh, no face receives conflicting filling instructions. In the bulk both instructions use the actual spins. The replacements can consequently be made simultaneously, leaving the marked neighborhoods and half-stubs untouched. This is a modification of the drawing, not an interpolation of the actual full-plane exterior configuration. In particular all farther exterior spins and every interaction across a narrow fjord remain in the Gibbs conditional law.  $\square$

## 7.2 From fixed strip tests to limiting barriers

We first record the domain-continuity fact needed for the displaced pure curves. If Jordan boundaries  $c_j : \partial \mathbb{D} \rightarrow \partial G_j$  are homeomorphic parametrizations converging uniformly to a Jordan parametrization  $c$ , normalized conformal maps  $f_j : \mathbb{D} \rightarrow G_j$  converge uniformly on  $\bar{\mathbb{D}}$  to the corresponding map onto  $G$ . Normalization can be by one fixed interior point and positive derivative, or by three convergent distinct boundary points. Here is the reason the boundary assumption matters. Uniform convergence to the injective  $c$  implies a uniform local connectedness modulus for the boundaries: nearby endpoints have one boundary subarc of small diameter. Indeed, otherwise inverse uniform continuity of  $c$  and uniform continuity of the parametrizations would give two separated parameter arcs with arbitrarily close images. Kernel convergence gives local uniform convergence of the normalized maps. The boundary crosscut proof of the Carathéodory extension theorem, applied with this common local connectedness modulus, gives equicontinuity on the closed disk, and hence upgrades local convergence to uniform convergence. This is Radó's Jordan-domain continuity theorem [21, Théorème, p. 182]. Its stated uniform convergence is on the closed disk. It requires more than kernel convergence alone.

For clarity, the crosscut argument is uniform here. The image areas are bounded by a common containing disk. Cauchy–Schwarz on circular arcs centered at a boundary point of  $\mathbb{D}$ , followed by integration in their radii, supplies a crosscut of image length tending to zero between any two

sufficiently small separated radius scales; the squared length bound is a constant divided by the logarithm of their ratio. Its image endpoints are close. The common local connectedness modulus bounds the diameter of the smaller boundary subarc joining them. The image crosscut together with that subarc cuts off a set of comparably small diameter. The component not containing the fixed normalized interior point is this small component, since that point has fixed positive distance from the boundary. This controls the images of the smaller boundary neighborhood uniformly. Finite covering of the unit circle then proves the required equicontinuity. Boundary normalization follows by precomposing with disk automorphisms; the preimages of the three distinct limiting marks converge by injectivity of the limiting boundary map.

To account explicitly for the moving marks, first normalize  $f_h, f$  by a common interior point and positive derivative. If  $\alpha_h, \beta_h$  are the boundary preimages of  $a_h, b_h$ , uniform convergence on the closed disk and injectivity of  $f$  imply  $\alpha_h \rightarrow \alpha = f^{-1}(a)$  and  $\beta_h \rightarrow \beta = f^{-1}(b)$ . Fix a third boundary point different from  $\alpha, \beta$ . The disk automorphism taking  $\alpha, \beta$  to  $\alpha_h, \beta_h$  and fixing that third point converges uniformly on the closed disk to the identity. Precomposition by these automorphisms therefore gives uniformly convergent maps with the required marked endpoints.

Applying this fact to  $A_h$ , and coupling by a single disk  $\text{SLE}_3$ , shows that its continuum pure interface law converges as  $h \downarrow 0$  to chordal  $\text{SLE}_3$  in  $D$  from  $a$  to  $b$ , in the oriented uniform curve metric. The same holds for the plus-favoring comparison domains. This uses the conformal definition of  $\text{SLE}_3$  and its continuous simple trace; uniform convergence of the conformal maps bounds the distance between the corresponding image curves. For fixed  $h$  the lattice convergence used here is only Theorem 2.10 in an ordinary polygonal approximation of  $A_h$ .

**Choosing a deterministic finite list.** Couple the continuum curves in  $A_h$  to their limit  $\Gamma$  so that their oriented distances tend to zero. The curve  $\Gamma$  meets  $\partial D$  only at  $a, b$ . For every fixed small parameter interval removed from each end, its remaining compact subarc is a positive distance inside  $D$ . Uniform convergence implies that, with probability tending to one as  $h \downarrow 0$ , a middle subarc of the curve in  $A_h$  lies below  $\rho = 1 + h/2$ , has endpoints outside  $\overline{D}$ , and the omitted terminal pieces have diameter tending to zero. To make this choice without a regularity assumption on level hits, choose a point of the curve deep inside  $D$ , take the last hit of  $\rho = 1 + h/3$  before that point and the first hit after it. The intervening subarc lies in  $\rho \leq 1 + h/3$ . Both hits lie outside  $D$  on the enlarged plus arc. Extend this subarc slightly at its two ends, within  $\rho < 1 + h/2$ , before choosing gates. Compact parts away from  $a, b$  cannot contain these hits in the limit. The omitted portions therefore shrink in diameter. Polygonal tolerances can be chosen smaller than the gaps between these levels.

For each such simple subarc, a rectangular neighborhood of a slightly extended arc gives a thin strip with transverse end gates lying outside  $D$ . Polygonal approximation of its sides and gates gives an allowed strip. One can prescribe a continuous parametrization of its cross-sections such that each section has small diameter and stays near the corresponding point of a simple polygonal prototype. A lengthwise traversal then stays near this prototype, even if it backtracks. The extended subarc crosses both gates with a margin, and stays away from the long sides. These properties persist in an open neighborhood in the oriented curve metric. On sufficiently fine lattices, a traversal by the pure interface with these margins supplies a plus spin crossing of the strip: follow its plus side in the fixed triangulation and crop at the gates. The discrepancy between that side and the interface is  $O(\delta_n)$ , smaller than the fixed margin.

The strip construction can be made from a countable collection, using rational polygonal vertices and rational positive margins. These open curve neighborhoods cover all the favorable continuum curves just described. The curve space is separable; equivalently use this explicit countable collection

and continuity from below of probability. A finite subcollection thus covers all but any prescribed positive probability. Fix that entire finite list before using Proposition 7.1. After its joint vector coupling has been made, the pure curve may select one strip from the list. The corresponding target crossing then exists outside an event of vanishing probability. No test chosen adaptively from an infinite family has been inserted into Theorem 5.1.

**Passing to moving geometries.** Choose errors  $\epsilon_j \downarrow 0$ . First choose  $h_j \downarrow 0$ , polygonal comparison domains of both signs, and favorable continuum events whose omitted end pieces have diameter at most  $\epsilon_j$ , with failure probability at most  $\epsilon_j$ . Then choose finite strip covers with widths and prototype errors at most  $\epsilon_j$  and additional failure probability at most  $\epsilon_j$ . Finally choose increasing integers  $n_j$  so that for every  $n \geq n_j$  the fixed placements, both pure interface approximations, and both finite-vector couplings have all their required errors at most  $\epsilon_j$ . Set

$$j(n) = \max\{j \leq n : n_j \leq n\}, \quad (7.5)$$

with an arbitrary definition before  $n_1$ . This index tends to infinity. The constants  $q_{h_j}$  in (7.4) may tend to zero; the threshold  $n_j$  has already absorbed their reciprocals. Couple each of the two comparison constructions with the target and glue over the target configuration. Their comparison curves need not be independent. Both marginal laws converge to the same chordal SLE<sub>3</sub> law, and, with probability tending to one, the target has both colored barriers in strips converging to those comparison curves.

### 7.3 Forbidden sides, order, and identification of the trace

For a simple chord  $C$  of  $D$  from  $a$  to  $b$ , with interior in  $D$ , let  $P(C)$  and  $M(C)$  be the open components of  $D \setminus C$  adjacent to  $I_+$  and  $I_-$ , respectively. Their closures below are in  $\overline{D}$ . All target traces lie in one fixed compact set for large  $n$ . The space of nonempty compact subsets of that set is compact in Hausdorff distance. Consequently, along any mesh subsequence we can pass further to a joint limit

$$(\text{tr } \gamma_n, L_n, U_n) \implies (K, L, U),$$

where  $L_n, U_n$  are the two comparison curves in the slow diagonal. Their marginal curve laws are tight by the preceding construction. The limit  $K$  is connected and contains  $a, b$ : separation of a compact limit into two sets at positive distance would separate every sufficiently close connected approximant. Also  $K \subset \overline{D}$ : winding stability excludes every fixed compact set outside  $\overline{D}$  from  $D_n$  eventually, and the drawing displacements vanish. The curves  $L, U$  have the chordal SLE<sub>3</sub> marginal law. Use a representation of this joint convergence in which the compact and curve distances tend to zero almost surely, and pass to a further subsequence so that barrier failures also occur only finitely often.

We claim that the barriers imply

$$K \subset \overline{M(L)} \cap \overline{P(U)}. \quad (7.6)$$

Choose once and for all  $\rho_* > 1$ , independently of  $h$  and  $n$ , and restrict to  $h$  with  $1 + h/2 < \rho_*$ . Consider the plus barrier associated with  $L_n$ . Erase loops from a successful strip crossing and connect its ends to the gate centers within exterior gate neighborhoods. In Schoenflies coordinates, close this path by radial segments from the gate centers to  $\rho = \rho_*$  and the plus-side circular arc between them. For each fixed  $h$  these added paths have positive clearance from  $\overline{D}$ , so they lie outside  $D_n$  for all sufficiently large  $n$ ; include this requirement in the choice of  $n_j$ . Together these paths form a closed barrier disjoint from the redrawn target  $\tilde{\gamma}_n$  of Lemma 7.2. We use this exact

disjointness here;  $O(\delta_n)$  closeness alone would not preserve disjointness from a lattice barrier. Both target marks are on its zero-winding exterior side: their outward escape paths lie beyond the ends of the extended plus arc and avoid the strip, whose ends are inside the target plus arc and separated from the marks. Thus the entire redrawn target interface has zero winding relative to the closed barrier.

On every compact subset of  $P(L)$ , the barrier has, for large  $n$ , the same nonzero winding as  $L$  closed around  $I_+$  outside  $D$ . To verify this without controlling the traversal order of the target crossing, use the rectangular chart of the strip. Any path connecting its two end gates is homotopic, relative to those gates, to its central prototype; its loops can be removed inside the strip. Its winding about a point outside that strip is therefore the prototype's winding. The widths vanish, the prototypes converge in curve distance to  $L$ , and the terminal errors shrink to  $a, b$ . The gate centers therefore tend to  $a, b$ , and their radial connectors tend to the endpoint radial segments. On every compact subarc of  $I_+$ , the fixed-radius part of the closure stays a positive distance beyond  $\partial D$ , while the connectors remain away from that subarc. The same winding conclusion consequently holds on small neighborhoods of every point of  $I_+$  other than  $a, b$ . The redrawn target cannot enter these neighborhoods. Its trace has the same Hausdorff limit  $K$  as the original target, by Lemma 7.2. This excludes both  $P(L)$  and the open plus boundary arc from  $K$ , leaving  $K \subset \overline{M(L)}$ . The minus barrier proves  $K \subset \overline{P(U)}$ . The half-stubs remain near the marks and therefore do not meet these fixed neighborhoods.

**Lemma 7.3** (Ordered chords with equal laws). *Let  $L, U$  be random simple chords from  $a$  to  $b$  in a Jordan domain, meeting its boundary only at their endpoints, and let  $K$  be a random compact connected subset of its closure containing  $a, b$ . Suppose (7.6) holds almost surely. Then  $P(L) \subset P(U)$  almost surely. If  $L$  and  $U$  have the same trace law, then  $L = U = K$  as compact sets almost surely.*

*Proof.* Work in the closed disk using a fixed Schoenflies map. If  $P(L) \cap M(U)$  were nonempty, connect a point of that intersection to an interior point of  $I_+$  through  $P(L)$  and to an interior point of  $I_-$  through  $M(U)$ . The two side domains are Jordan domains. Small endpoint arcs and interior polygonal paths, with loops removed, therefore give a simple crosscut between these opposite boundary arcs, contained in their union except for its two boundary endpoints. This crosscut is disjoint from  $K$  by (7.6). Its endpoints alternate with  $a, b$  around the circle, so it separates  $a$  from  $b$  in the closed disk. This contradicts connectedness of  $K$ . Hence  $P(L) \cap M(U) = \emptyset$ . It follows that  $P(L) \subset P(U) \cup U$ . No interior point of  $U$  can lie in the open set  $P(L)$ : every neighborhood of such a point meets  $M(U)$ . Thus  $P(L) \subset P(U)$ .

Let  $X, Y$  be the ordinary planar areas of the two plus sides after this fixed disk map. They are bounded random variables,  $X \leq Y$ , and their distributions agree when the trace laws agree. Thus  $\mathbb{E}(Y - X) = 0$  and  $X = Y$  almost surely. Proper inclusion of these particular side domains would give strictly different areas. Indeed a point of  $P(U) \setminus P(L)$  either belongs to  $M(L)$ , or lies on  $L$ ; in the latter case its open neighborhood in  $P(U)$  meets  $M(L)$ . In either case the difference contains a nonempty open subset of  $M(L)$ , of positive area. Consequently  $P(L) = P(U)$  and their interior boundaries give the same chord. This reasoning does not require the chords to have zero area. Finally (7.6) puts  $K$  inside that chord. The inverse of a simple parametrization sends  $K$  to a connected subset of  $[0, 1]$  containing both endpoints, which is all of  $[0, 1]$ .  $\square$

The lemma identifies every subsequential target trace limit as chordal SLE<sub>3</sub>. We have also excluded a boundary segment from that trace. We have not yet excluded repeated forward and backward travel in a thin neighborhood of the same chord. The next argument supplies that last distinction.

## 7.4 Interior passages and the orientation of the limit

For a compact  $F \subseteq D$  and a fixed  $R > 0$  small enough that all radius- $2R$  neighborhoods of  $F$  lie in  $D$ , let  $\mathcal{B}_n(F; r, R)$  be the event that the target contour has at least four passages across an annulus with some center in  $F$ , inner radius  $r$ , and outer radius  $R$ . Passages count disjoint parameter intervals, with endpoint hits allowed to coincide, as in Section 5. Square or round annuli can be used, after fixed-factor changes of radii. We claim

$$\lim_{r \downarrow 0} \limsup_{n \rightarrow \infty} \mathbb{P}(\mathcal{B}_n(F; r, R)) = 0. \quad (7.7)$$

Cover the fixed  $2R$ -neighborhood by finitely many interior boxes with fixed positive free buffers. Proposition 6.6 compares each of their target spin marginals pointwise with a torus marginal, with a constant independent of  $r, n, \tau_n$ . A grid of mesh comparable to  $r$  has  $O_F(r^{-2})$  centers. Four target passages about any center in  $F$  give four passages about a nearest grid center from radius  $Cr$  to radius  $R/C$ , by cropping the same paths at their hits. At each grid center enlarge this to the event  $F_4$  that *any* locally resolved contours supply four passages through that annulus. This enlarged event is determined by the spins in the annulus and a one-face margin. Identifying the marked target component is not local, but is unnecessary for this upper bound. The bulk drawing comparison absorbs its  $O(\delta_n)$  displacement into the fixed radius crops.

For each fixed  $r$  the enlarged events are finitely many fixed bulk tests of Section 5, and the density comparison applies to them. Theorem 5.1 and Proposition 2.12 bound their torus mesh upper limits by  $C_R(r/R)^{2+c}$  for some  $c > 0$  when  $r/R$  is small. A union bound and the fixed density constants give an upper bound  $C_{F,R}r^c$  after the mesh limit. This proves (7.7). The buffer in the density comparison was fixed before  $r$  decreased; there is no density constant at a shrinking buffer hidden in this bound. The finite grid union also gives a measurable upper event for arbitrary centers, so no measurable choice of a center is required below.

**Lemma 7.4** (Progress coordinates and the full curve metric). *Let  $\Gamma : [0, 1] \rightarrow \overline{D}$  be continuous and injective, with  $\Gamma(0) = a$ ,  $\Gamma(1) = b$ , and  $\Gamma((0, 1)) \subset D$ . Let  $g_n : [0, 1] \rightarrow \mathbb{C}$  be continuous simple curves whose endpoints tend to  $a, b$  and whose traces tend in Hausdorff distance to  $\Gamma([0, 1])$ . There are continuous functions  $u_n : [0, 1] \rightarrow [0, 1]$  with endpoint values tending to 0, 1 such that*

$$e_n := \sup_t |g_n(t) - \Gamma(u_n(t))| \rightarrow 0. \quad (7.8)$$

Put  $B_n = \sup_{s < t} (u_n(s) - u_n(t))$ . If  $B_n \rightarrow 0$ , then  $d_{\text{curv}}(g_n, \Gamma) \rightarrow 0$ . Conversely, for every fixed  $0 < \eta < 1$ , a backward drop  $B_n \geq \eta$ , with  $n$  sufficiently large, forces six disjoint passages from radius  $r$  to one fixed radius  $R_\eta > 0$ , for every  $e_n < r < R_\eta$ , about a center on  $\Gamma([ \eta/2, 1 - \eta/2 ]) \subseteq D$ .

For random curves, suppose the trace and endpoint convergence hold in probability in a coupling with such a random  $\Gamma$ , and suppose (7.7) holds for every deterministic compact interior  $F$  and sufficiently small fixed  $R$ . Then  $d_{\text{curv}}(g_n, \Gamma) \rightarrow 0$  in probability in that coupling.

*Proof.* The inverse coordinate  $\Gamma^{-1}$  is continuous on the compact trace. Extend it continuously, with values in  $[0, 1]$ , to a compact ambient neighborhood and call the extension  $p$ . Setting  $u_n = p \circ g_n$  gives (7.8) by uniform continuity: a point close to  $\Gamma(v)$  has  $p$ -value close to  $v$ , and hence is close to  $\Gamma(p(\cdot))$ . The endpoints have the asserted coordinates.

Here is the matching when drops vanish. The running maximum

$$m_n(t) = \max_{s \leq t} u_n(s)$$

is continuous, nondecreasing, and satisfies  $0 \leq m_n(t) - u_n(t) \leq B_n$ . Its endpoint values tend to 0, 1. For all large  $n$  let

$$v_n(t) = \frac{m_n(t) - m_n(0)}{m_n(1) - m_n(0)}, \quad h_n(t) = (1 - \kappa_n)v_n(t) + \kappa_n t,$$

where  $0 < \kappa_n \downarrow 0$ . Then  $h_n$  is an increasing homeomorphism of  $[0, 1]$  and  $\|h_n - u_n\|_\infty \rightarrow 0$ . If  $\omega_\Gamma$  denotes the modulus of continuity of  $\Gamma$ , then

$$d_{\text{curv}}(g_n, \Gamma) \leq e_n + \omega_\Gamma(\|h_n - u_n\|_\infty) \rightarrow 0.$$

This uses exactly the allowed matching in (1.2), with the identity on  $g_n$  and  $h_n$  on  $\Gamma$ .

For the converse, the continuous function  $u_n(s) - u_n(t)$  attains its maximum on  $0 \leq s \leq t \leq 1$ . If  $B_n \geq \eta > 0$ , a maximizing pair has  $s < t$  and  $u_n(s) - u_n(t) \geq \eta$ . Set  $c_n = (u_n(s) + u_n(t))/2 \in [\eta/2, 1 - \eta/2]$ . When the endpoint coordinate errors are less than  $\eta/4$ , each of the intervals  $[0, s]$ ,  $[s, t]$ , and  $[t, 1]$  has endpoints on opposite sides of  $c_n$ , at coordinate distance at least  $\eta/4$ . Each therefore contains a visit to  $c_n$ . Injectivity and compactness imply

$$q_\eta = \min_{|v-w| \geq \eta/4} |\Gamma(v) - \Gamma(w)| > 0, \quad d_\eta = \text{dist}(\Gamma([\eta/2, 1 - \eta/2]), \partial D) > 0.$$

Choose  $R_\eta < \min(q_\eta, d_\eta)/4$ . Once  $e_n < R_\eta$ , the four anchor points  $g_n(0), g_n(s), g_n(t), g_n(1)$  are outside the radius- $R_\eta$  disk about  $\Gamma(c_n)$ , whereas each of the three level visits is within  $e_n$  of its center. In each time interval crop the incoming and outgoing portions at the last outer and first inner hits, and at the last inner and first outer hits. For every  $e_n < r < R_\eta$  this supplies two annular passages. The six parameter interiors are disjoint, and simplicity gives disjoint spatial interiors. They are thus six passages in the stated convention, and in particular at least four. Fixed-factor cropping gives the same assertion for square annuli.

We give the measurability and order of limits for a random  $\Gamma$ . Take a measurable continuous representative, for example the image of a disk SLE representative in the coupling above. We need not choose a random extension  $p$ . Write  $\Delta_n = d_H(\text{tr } g_n, \text{tr } \Gamma)$ ,  $z_n = 2\Delta_n + 1/n$ , and define instead

$$u_n(t) = \frac{\int_0^1 v(z_n - |g_n(t) - \Gamma(v)|)_+ dv}{\int_0^1 (z_n - |g_n(t) - \Gamma(v)|)_+ dv}. \quad (7.9)$$

The denominator is positive since every  $g_n(t)$  is within  $\Delta_n$  of the trace, and continuity gives a parameter interval with positive weight. This coordinate is continuous in  $t$  and jointly measurable. Let

$$\iota_\Gamma(r) = \sup\{|v - w| : |\Gamma(v) - \Gamma(w)| \leq r\}.$$

It tends to zero as  $r \downarrow 0$ . All parameters receiving positive weight in (7.9) have diameter at most  $\iota_\Gamma(2z_n)$ , and their weighted average lies between their extreme values. Consequently

$$e_n \leq z_n + \omega_\Gamma(\iota_\Gamma(2z_n)) \rightarrow 0$$

in probability; the endpoint coordinate errors also tend to zero. These assertions follow first almost surely along any subsequence on which  $\Delta_n$  and the endpoint errors converge almost surely, and hence hold in probability for the full sequence.

Fix  $0 < \eta < 1$  and an exceptional probability  $\zeta > 0$ . Since  $q_\eta, d_\eta$  are strictly positive random variables, choose deterministic  $q, d > 0$  such that

$$\mathbb{P}(q_\eta < q \text{ or } d_\eta < d) < \zeta.$$

The random centers then lie in the deterministic compact set  $F_d = \{z \in \overline{D} : \text{dist}(z, \partial D) \geq d\}$ , and one deterministic  $R < \min(q, d)/8$  works. For each fixed  $0 < r < R$ , the preceding six-passage argument, outside an event of probability  $\zeta + o(1)$ , bounds the probability of a drop of size  $\eta$  by  $\mathbb{P}(\mathcal{B}_n(F_d; r, R))$ . First let  $n \rightarrow \infty$ , then  $r \downarrow 0$  using (7.7), and finally  $\zeta \downarrow 0$ . Thus  $B_n \rightarrow 0$  in probability. To deduce curve distance convergence, first choose a deterministic coordinate tolerance on which  $\omega_\Gamma$  is small outside probability  $\zeta$ , then use the running-maximum bound and let  $n \rightarrow \infty$ . Finally remove  $\zeta$ . This localizes the continuity modulus as well as the injectivity and boundary-distance moduli. All annulus radii and compact charts are deterministic before the mesh limit is taken.  $\square$

The simplicity used in this lemma holds for the prescribed locally resolved interface: the resolved embedded graph has degree two except at its two degree-one stub midpoints, so its connecting component is an embedded arc. Distinct visits to a small disk along that arc count as distinct passages. No estimate for path length is required. The lemma also explains why a separate boundary passage estimate is unnecessary at this stage: the limiting chord avoids the boundary away from its endpoints, and its short terminal parameter intervals have arbitrarily small diameter by continuity.

## 7.5 Completion of the universality theorem

*Proof of Theorem 1.1.* For the fixed finite-range real potential  $U$ , Lemma 1.2 converts the exact contour weight into a finite-range, even, square-symmetric spin perturbation, with all interactions meeting the interior retained. Theorem 3.1 and Theorem 4.5 give an analytic bulk choice  $\beta_c(J, U, \lambda)$  near  $\beta_0 = \log(1 + \sqrt{2})/(2J)$ , positive for sufficiently small real  $\lambda$  of either sign and equal to  $\beta_0$  at zero. Shrink this interval once to satisfy the independent-edit stability threshold and the finite-table strict-incidence margins. This gives a number  $\lambda_0(J, U) > 0$  before any choice of  $D, a, b$ , approximation, exterior contours, corridor, or strip. The universal fixed-shape bootstrap in Theorem 5.1 is what permits all later fixed geometries to use this same interval.

Fix any  $|\lambda| < \lambda_0$ , any bounded Jordan marked domain, any sequence of meshes and uniformly boundary-parametrized Jordan lattice domains in the theorem, and any admissible sequence  $P_{o,n}$ . On each subsequence pass further so that the choice of plus boundary arc is fixed. The exact conditional-law reduction permits the target to be placed in a buffered torus chart while retaining all its exterior pins. Finite range permits any continuation beyond that chart without changing the conditional law. Proposition 7.1, the finite strip construction, and the slow diagonal produce a joint limit of every further Hausdorff-convergent subsequence with two comparison chords of the same SLE<sub>3</sub> law. Their forbidden-side inclusions and Lemma 7.3 identify the target trace with that chord. Thus the target traces converge in law in Hausdorff distance to chordal SLE<sub>3</sub>.

Lift the subsequential trace coupling to the target paths using their conditional distributions given their traces. At each mesh the spin space is finite, so this disintegration is immediate and preserves every target passage probability. No tightness of these path variables is needed to perform the lift. The limiting chord already has a measurable continuous representative from its SLE marginal. The interior density comparison and bulk four-passage estimate give (7.7), uniformly in the permitted exterior configurations. Lemma 7.4 therefore gives convergence in probability in the oriented curve metric in this representation, and hence convergence in distribution in that metric. Since every subsequence has a further subsequence with this same limit, the full sequence converges. This argument also gives tightness in the requested curve topology; it did not assume it for the perturbed paths.

The conclusion concerns the prescribed  $\gamma_n$ . The local redrawings used for its barriers preserve traversal order and have  $O(\delta_n)$  curve distance from it; they were used only to identify its trace limit.

The half-stubs remained part of the original curve, with endpoints tending to  $a, b$ , and the orientation is always from the incoming midpoint to the outgoing midpoint. The unit-lattice potential  $U$  was fixed throughout; only coordinates were multiplied by  $\delta_n$ . No sign condition on  $U$  or  $\lambda$ , boundary regularity beyond the stipulated Jordan parametrizations, or restriction on the additional exterior contours was imposed. The value of  $\beta_c$  and its interval of validity are consequently independent of every domain and boundary choice required in the theorem.  $\square$

The public mathematical results used in the argument are credited at their points of use.

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